

IDENTIFICATION

PRODUCT CODE: MAINDEC-8E-DBAC-D

PRODUCT NAME: DK8E CLOCKS DIAGNOSTIC

DATE CREATED: OCTOBER 8, 1971

MAINTAINER: DIAGNOSTIC PROGRAMMING GROUP

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1. ABSTRACT -----

THE DK8E CLOCKS DIAGNOSTIC IS DESIGNED TO VERIFY CORRECT OPERATION OF THE DK8-EA, DK8-EC, DK8-ES, AND DK8-EP REAL TIME CLOCK OPTIONS. THE PROGRAM UTILIZES AND TESTS IOT'S ASSOCIATED WITH THE DK8-EA LINE, DK8-EC CRYSTAL, AND THE DK8-EP/DK8-ES PROGRAMMABLE REAL TIME CLOCKS.

2. REQUIREMENTS -----

2.1 EQUIPMENT -----

A PDP-8E WITH THE DK8-EA, DK8-EC, DK8-ES, OR THE DK8-EP OPTION INSTALLED AND AN ASR-33 TELETYPE OR EQUIVALENT.

A SPECIAL TEST CABLE IS NECESSARY TO CONNECT THE CLOCK FRONT PANEL TO THE PDP8/E POWER SUPPLY FOR THE DK8-ES CLOCK OPTION.

A SPECIAL CABLE IS NECESSARY TO CONNECT THE DK8-EA CLOCK MODULE TO THE PDP8/E POWER SUPPLY FOR THE DK8-EA CLOCK OPTION.

2.2 STORAGE -----

THE PROGRAM OCCUPIES LOCATIONS 0000-6600.

2.3 PRELIMINARY PROGRAMS -----

ALL PROGRAMS FOR THE BASIC PDP-8E MUST HAVE BEEN RUN SUCCESSFULLY.

3. LOADING PROCEDURE -----

3.1 METHOD -----

THE PROGRAM IS LOADED INTO BANK 0, USING THE STANDARD BINARY LOADER TECHNIQUE.

4. STARTING PROCEDURE -----

4.1 CONTROL SWITCH SETTINGS -----

SWR0=1	FOR DK8-EP/DK8-ES REGISTER TEST
SWR1=1	FOR DK8-ES SCHMITT TRIGGER LOGIC TEST
SWR2=1	FOR INHIBIT ERROR PRINT OUT
SWR3=1	FOR INHIBIT ERROR BELL
SWR4=1	FOR INHIBIT ERROR HALT
SWR5=1	FOR ENTER SCOPE LOOP ON ERROR
SWR6=1	FOR LOOP ON NON-FAILING TEST
SWR7=1	FOR DK8-EP/DK8-ES EXTERNAL PULSE SCOPE LOOP TEST

SWR8=1

FOR DK8-ES EXTERNAL CLOCK SCOPE LOOP TEST

4.1.1 FREQUENCY SWITCH SETTINGS FOR DK8-EA/DK8-EC TEST

SWR9-11=0 TEST 1 CPS CRYSTAL CLOCK
SWR9-11=1 TEST 50 CPS CRYSTAL CLOCK
SWR9-11=2 TEST 50 CPS LINE CLOCK
SWR9-11=3 TEST 60 CPS LINE CLOCK
SWR9-11=4 TEST 500 CPS CRYSTAL CLOCK
SWR9-11=5 TEST 5000 CPS CRYSTAL CLOCK

4.2 STARTING ADDRESS

THE STARTING ADDRESS IS 0200 OCTAL.

4.3 OPERATOR ACTION

4.3.1 DK8-EA/DK8-EC TEST

WITH THE PROGRAM IN BANK 0, SET SWITCH REGISTER TO 0200.
PRESS ADDRESS LOAD.

SET THE SWITCH REGISTER TO 0000.

SET SWITCH REGISTER TO INDICATE FREQUENCY OF DK8-EA
OR DK8-EC CLOCK UNDER TEST.

PRESS CLEAR AND THEN PRESS CONTINUE.

THE PROGRAM SHOULD RUN UNTIL AN ERROR OCCURES OR UNTIL
STOPPED BY THE OPERATOR.

THE TTY WILL SIGNAL "DK8E PASS COMPLETE" AT
THE COMPLETION OF EVERY PASS.

4.3.2 DK8-EP/DK8-ES REGISTER TEST

WITH THE PROGRAM IN BANK 0, SET SWITCH REGISTER TO 0200.
PRESS ADDRESS LOAD.

SET SWITCH REGISTER TO 0000.

SET SWITCH REGISTER TO INDICATE DK8-EP/DK8-ES REGISTER TEST.

PRESS CLEAR AND THEN PRESS CONTINUE.

THE PROGRAM SHOULD RUN UNTIL AN ERROR OCCURES OR UNTIL
STOPPED BY THE OPERATOR.

THE TTY WILL SIGNAL "DK8E PASS COMPLETE" AT
THE COMPLETION OF EVERY PASS.

4.3.3 DK8-ES SCHMITT TRIGGER INPUT LOGIC TEST

WITH THE PROGRAM IN BANK 0, SET THE SWITCH REGISTER TO 0200.
PRESS ADDRESS LOAD.

SET SWITCH REGISTER TO 0000.

SET THE SWITCH REGISTER TO INDICATE DK8-ES SCHMITT TRIGGER
INPUT LOGIC TEST.

PRESS CLEAR AND THEN CONTINUE.

THE PROGRAM SHOULD RUN UNTIL AN ERROR OCCURES OR UNTIL
STOPPED BY THE OPERATOR.

THE TTY WILL SIGNAL "DK8E PASS COMPLETE" AT
THE COMPLETION OF EVERY PASS.

4.3.4 DK8-EP/DK8-ES EXTERNAL PULSE SCOPE LOOP TEST

WITH THE PROGRAM IN MEMORY, SET THE SWITCH REGISTER TO 0200.
PRESS ADDRESS LOAD.

SET SWITCH REGISTER TO 0000.

SET SWITCH REGISTER TO INDICATE EXTERNAL PULSE SCOPE LOOP TEST.

PRESS CLEAR AND THEN PRESS CONTINUE.

USE OSCILLOSCOPE TO VERIFY 40 MICRO SECOND PULSE RATE AT
FJ2, FJ1, HM1, AND HM2 ON THE DK8-EP/DK8-ES MODULES.

USE OSCILLOSCOPE TO VERIFY 40 MICRO SECOND PULSE RATE AT
OVERFLOW ON DK8-ES CLOCK FRONT PANEL. (DK8-ES ONLY)

4.3.5 DK8-ES EXTERNAL CLOCK SCOPE LOOP TEST

WITH THE PROGRAM IN MEMORY, SET THE SWITCH REGISTER TO 0200.
PRESS ADDRESS LOAD.

SET SWITCH REGISTER TO 0000.

SET SWITCH REGISTER TO INDICATE EXTERNAL CLOCK SCOPE LOOP TEST.

PRESS CLEAR AND THEN PRESS CONTINUE.

GROUND CLOCK IN ON DK8-ES CLOCK FRONT PANEL.

THE TTY BELL WILL SIGNAL, IF AN EXTERNAL CLOCK IN WAS
RECEIVED.

5. OPERATING PROCEDURE

5.1 OPERATIONAL SWITCH SETTINGS

NONE

5.2 SUBROUTINE ABSTRACTS

NONE

5.3 OPERATOR TEST SELECTION

5.3.1 DK8-EA OR DK8-EC CLOCK OPTION

INSTALL DK8-EA OR DK8-EC CLOCK OPTION

RUN DK8-EA/DK8-EC TEST 4.3.1.

5.3.2 DK8-EP CLOCK OPTION

INSTALL DK8-EP CLOCK OPTION.

RUN DK8-EP/DK8-ES REGISTER TEST 4.3.2.

RUN DK8-EP/DK8-ES EXTERNAL PULSE SCOPE LOOP TEST 4.3.4.

5.3.3. DK8-ES CLOCK OPTION

INSTALL DK8-ES CLOCK OPTION.

RUN DK8-EP/DK8-ES REGISTER TEST 4.3.2.

CONNECT EXTERNAL SOURCE FREQUENCY LOCATED AT J5 ON THE PDP8/E POWER SUPPLY TO THE EXTERNAL SCHMITT TRIGGER INPUT LOGIC VIA THE DK8-ES CLOCK FRONT PANEL WITH THE SPECIAL TEST CABLE.

SET THE THREE SLOPE SELECTION SWITCHES ON DK8-ES CLOCK FRONT PANEL TO THE POSITIVE POSITION.

ADJUST THE THREE INPUT THRESHOLD POTENTIOMETERS ON DK8-ES CLOCK FRONT PANEL TO THE CENTER POSITION.

RUN THE DK8-ES SCHMITT TRIGGER INPUT LOGIC TEST 4.3.3.

RUN THE DK8-EP/DK8-ES EXTERNAL PULSE SCOPE LOOP TEST 4.3.4.

RUN THE DK8-ES EXTERNAL CLOCK SCOPE LOOP TEST 4.3.5.

6. ERRORS

ALL RECOVERABLE ERRORS ENCOUNTERED IN THE PROGRAM WILL RESULT IN AN ERROR HALT OR AN ERROR TYPEOUT AND THEN AN ERROR HALT.

6.1 ERRORS AND DISCRPTION -----

6.1.1 ERROR HALTS -----

ERROR HALTS IN PROGRAM ARE AS FOLLOWS:

EHLT1: MONITOR ERROR HALT, READ ERROR TYPE-OUT.

EHLT2: SKIP TRAP, CLZE

EHLT3: SKIP TRAP, CLOE

EHLT4: SKIP TRAP, CLOE

EHLT5: SKIP TRAP, CLAB

EHLT6: SKIP TRAP, CLEN

EHLT7: SKIP TRAP, CLSA

EHLT10: SKIP TRAP, CLBA

EHLT11: SKIP TRAP, CLCA

6.1.2 ERROR TYPECUTS -----

ERROR TYPECUTS IN PROGRAM ARE AS FOLLOWS:

TEST XXXX FAILED, STARTING ADDRESS XXXX

THE GOOD AC = XXXX AND BAD AC = XXXX

CLOCK BUFFER REGISTER AND AC TRANSFER FAILED

CLOCK COUNTER REGISTER AND AC TRANSFER FAILED

CLOCK ENABLE REGISTER AND AC TRANSFER FAILED

THE AC WAS CHANGED BY A CLOCK IOT

PROGRAM INTERRUPT FAILED, NO INTERRUPT EXPECTED

PROGRAM INTERRUPT FAILED, INTERRUPT EXPECTED

CLOCK SKIP FAILED, NO SKIP EXPECTED

CLOCK SKIP FAILED, SKIP EXPECTED

CLOCK OUTPUT FAILED, CLOCK FREQUENCY FAST

CLOCK OUTPUT FAILED, CLOCK FREQUENCY SLOW

6.2 ERROR RECOVERY -----

ALL ERRORS ENCOUNTERED MUST BE CORRECTED BEFORE PROCEEDING
ON IN THE PROGRAM. IN ALL CASES ACCESS THE LISTING FOR

FURTHER INFORMATION.

6.2.1 SCOPE LOOPS

A SCOPE LOOP IS AVAILABLE FOR ALL MONITOR ERROR HALTS. THE OPERATOR MAY ENTER A SCOPE LOOP AFTER A MONITOR ERROR HALT BY DOING THE FOLLOWING.

SET SWR4=1 TO INDICATE INHIBIT ERROR HALT.

SET SWR5=1 TO INDICATE ENTER SCOPE LOOP.

SET SWR6=1 TO INDICATE LOOP ON THIS TEST.

PRESS CLEAR AND THEN PRESS CONTINUE.

SET SWR2=1 TO INHIBIT ERROR TYPEOUT.

SET SWR3=1 TO INHIBIT ERROR BELL.

7. RESTRICTIONS

7.1 STARTING RESTRICTIONS

NONE

7.2 OPERATING RESTRICTIONS

THE PROGRAM MUST RESIDE IN BANK 0 .

PDP-8E WITH THE DK8-EA, DK8-EC, DK8-ES, OR THE DK8-EP CLOCK OPTION INSTALLED.

THE EXT. CPS SOURCE USED IN THE DK8-ES EXTERNAL SCHMITT TRIGGER INPUT LOGIC TEST MUST BE DISCONNECTED WHEN RUNNING THE DK8-EP/DK8-ES REGISTER TEST.

THE PDP-8E MUST BE RUNNING FAST CYCLE "1.2" MICRO. SECONDS.

ALL CLOCK OUTPUTS SHOULD BE VERIFIED WITH AN OSCILLOSCOPE TO INSURE CORRECT OPERATION.

8. MISCELLANECUS

8.1 SPECIFICATIONS

THE DK8-EA CLOCK COUNTS AT INTERVALS OF TIME AT 100 OR 120 TIMES A SECOND. THE FREQUENCY IS DETERMINED BY THE FULL WAVE RECTIFIER NETWORK WHICH OPERATES OFF THE 50 OR 60 CPS LINE WHICH EVER IT MAY BE. THIS MAKES THE CLOCK CAPABLE OF SUPPLYING PROGRAM INTERRUPT REQUESTS AT A RATE OF 100 OR 120 TIMES A SECOND.

8.2 EXECUTION TIME

DK8-EA/DK8-EC TEST, APPROXIMATIVELY 2.5 MINUTES PER PASS.

DK8-EP/DK8-ES REGISTER TEST, APPROXIMATIVELY 3.5 MINUTES
PER PASS.

DK8-ES SCHMITT TRIGGER INPUT LOGIC TEST, APPROXIMATIVELY
2 MINUTES PER PASS.

9. PROGRAM DISCRIPTION

9.1 DK8-EA OR DK8-EC CLOCK

THE PROGRAM EXERCISES AND TESTS THE FOLLOWING IOT'S FOR CORRECT
OPERATION AND FUNCTION.

SKIP ON A CLOCK FLAG AND CLEAR THE FLAG (CLSK)

OCTAL CODE: 6133

OPERATION: SENSES THE CLOCK FLAG, WHICH IS SET WITH
EACH CLOCK PULSE. IF IT IS SET, THE NEXT
SEQUENTIAL INSTRUCTION IS SKIPPED AND THE
FLAG IS THEN CLEARED.

ENABLE CLOCK INTERRUPT (CLEI)

OCTAL CODE: 6131

OPERATION: ENABLES THE CLOCK FLAG, WHICH IS SET WITH
EACH CLOCK PULSE, TO CAUSE A PROGRAM
INTERRUPT REQUEST. THE FLAG WILL REMAIN
SET UNTIL CLEARED WITH CLSK.

DISABLE CLOCK INTERRUPT (CLED)

OCTAL CODE: 6132

OPERATION: DISABLES THE CLOCK FLAG FROM CAUSING
AN INTERRUPT REQUEST, THE FLAG IS NOT AFFECTED.

9.2 DK8-EP/DK8-ES CLOCK

THE PROGRAM EXERCISES AND TESTS THE FOLLOWING IOT'S FOR CORRECT
OPERATION AND FUNCTION.

CLEAR THE CLOCK ENABLE REGISTER PER AC (CLZE)

OCTAL CODE: 6130

OPERATION: CLEARS THE BITS IN THE CLOCK ENABLE
REGISTER CORRESPONDING TO THOSE BITS
SET IN THE AC. THE AC IS NOT AFFECTED.

SKIP ON A CLOCK INTERRUPT (CLSK)

OCTAL CODE: 6131

OPERATION: SENSES FOR INTERRUPT CONDITIONS, IF THE
CONDITIONS ARE PRESENT THE NEXT SEQUENTIAL
INSTRUCTION IS SKIPPED. THE CONDITIONS
ARE AS FOLLOWS:

A. ENABLE EVENT INTERRUPT 1 AND INPUT 4

B. ENABLE EVENT INTERRUPT 2 AND INPUT 2

C. ENABLE EVENT INTERRUPT 3 AND INPUT 1

D. ENABLE OVERFLOW INTERRUPT AND OVERFLOW

AC TO CLOCK ENABLE REGISTER (CLOE)

OCTEL CODE:

6132

OPERATION:

CAUSES THE CONTENTS OF THE AC TO BE LOADED INTO THE CLOCK ENABLE REGISTER CORRESPONDING TO THOSE BITS SET IN THE AC, THE AC IS NOT AFFECTED. CLOCK ENABLE REGISTER FUNCTIONS ARE AS FOLLOWS.

AC BIT

FUNCTION

0

ENABLE CLOCK OVERFLOW

1 & 2

MODE CONTROL

00 COUNTER RUNS AT SELECTED RATE, OVERFLOW OCCURS EVERY 4096 COUNTS. OVERFLOW REMAINS SET UNTIL CLEARED BY (CLSA) IOT 6135.

01 COUNTER RUNS AT SELECTED RATE, OVERFLOW CAUSES THE CLOCK BUFFER REGISTER TO BE TRANSFERRED TO THE CLOCK COUNTER REGISTER WHICH WILL CONTINUE TO RUN AFTER TRANSFER. OVERFLOW WILL REMAIN SET UNTIL CLEARED BY (CLSA) IOT 6135.

10 COUNTER RUNS AT SELECTED RATE. AN EXTERNAL SCHMITT TRIGGER SIGNAL, IF ENABLED, CAUSES THE CLOCK COUNTER REGISTER TO BE TRANSFERRED TO THE CLOCK BUFFER REGISTER AND THE CLOCK COUNTER CONTINUES TO RUN.

11 COUNTER RUNS AT SELECTED RATE. AN EXTERNAL SCHMITT TRIGGER SIGNAL, IF ENABLED, CAUSES THE CLOCK COUNTER REGISTER TO BE TRANSFERRED TO THE CLOCK BUFFER REGISTER AND THE CLOCK COUNTER WILL CONTINUE TO RUN FROM 0.

3, 4 & 5

COUNT RATE

000 STOP

001 EXTERNAL CLOCK SOURCE

010 100 CPS

011 1000 CPS

100 10000 CPS

101 100000 CPS

110 1000000 CPS

111 STOP

6

WHEN SET TO A 1, OVERFLOW CAUSES AN EXTERNAL PULSE.

7 WHEN SET TO A 1, THE CLOCK COUNTER IS INHIBITED FROM COUNTING.

8 WHEN SET TO A 1, ENABLES EXTERNAL SCHMITT TRIGGER SIGNALS AND THE OVERFLOW FLOP TO CAUSE AN INTERRUPT REQUEST IF THEY ARE ENABLED.

9,10 & 11 ENABLE SCHMITT TRIGGER EVENTS

100 INPUT 4
010 INPUT 2
001 INPUT 1

AC TO CLOCK BUFFER REGISTER (CLAB)
OCTAL CODE: 6133
OPERATION: CAUSES THE CONTENTS OF THE AC TO BE TRANSFERRED INTO THE CLOCK BUFFER REGISTER; THE CONTENTS OF BUFFER REGISTER IS THEN TRANSFERRED TO THE CLOCK COUNTER REGISTER. THE AC IS NOT AFFECTED.

CLOCK ENABLE REGISTER TO AC (CLEN)
OCTAL CODE: 6134
OPERATION: CAUSES THE CONTENTS OF THE CLOCK ENABLE REGISTER TO BE TRANSFERRED TO THE AC. THE ENABLE REGISTER IS NOT AFFECTED.

CLOCK STATUS TO AC (CLSA)
OCTAL CODE: 6135
OPERATION: CAUSES THE CONTENTS OF THE CLOCK STATUS REGISTER TO BE TRANSFERRED INTO THE AC. THE STATUS BITS ARE THEN CLEARED CORRESPONDING TO THOSE BITS THAT WERE SET IN THE AC. THE STATUS REGISTER FUNCTIONS ARE AS FOLLOWS.

AC BIT -----	STATUS CONDITION -----
0	OVERFLOW
1-8	NOT USED
9	INPUT 4
10	INPUT 2
11	INPUT 1

CLOCK BUFFER REGISTER TO AC (CLBA)
OCTAL CODE: 6136
OPERATION: CAUSES THE CONTENTS OF THE CLOCK BUFFER REGISTER TO BE TRANSFERRED INTO THE AC. THE BUFFER REGISTER IS NOT AFFECTED.

CLOCK COUNTER REGISTER TO AC (CLCA)
OCTAL CODE: 6137

OPERATION:

CAUSES THE CONTENTS OF THE CLOCK
COUNTER TO BE TRANSFERED INTO THE
CLOCK BUFFER REGISTER. THE BUFFER
REGISTER IS THEN TRANSFERED INTO
THE AC. THE COUNTER REGISTER
IS NOT AFFECTED.

10.

LISTING

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/

/DK8E CLOCKS DIAGNOSTIC

/

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/

/THE STARTING ADDRESS 0200 OCTAL.

/

/PLEASE READ DOCUMENT FOR FURTHER INFORMATION.

/

0000	0000	*0000
0001	0000	/
0002	5001	0000
0003	0002	0000
0004	0003	0000
0005	0000	0000
0006	0207	0000
0007	0007	0207, 0207
0010	0000	K0207, 0007
0011	0000	AUTO10, 0000
0012	7700	SAVAC, 0000
0013	0100	K7700, 7700
0014	4000	K0100, 0100
0015	0200	K4000, 4000
0016	2525	K0200, 0200
0017	5252	K2525, 2525
0020	5102	K5252, 5252
0021	5107	X10TA, 10TA
0022	5114	X10TB, 10TB
0023	5121	X10TC, 10TC
0024	5127	X10TD, 10TD
0025	5134	X10TE, 10TE
0026	5142	X10TF, 10TF
0027	5146	X10TF1, 10TF1
0030	5154	X10TG, 10TG
0031	5163	X10TH, 10TH
0032	5200	X10TI, 10TI
0033	5207	X10TJ, 10TJ
0034	5350	X10TK, 10TK
0035	5360	X10TS, 10TS
0036	5370	X10TS1, 10TS1
0037	5400	X10TS2, 10TS2
0040	0000	X10TS3, 10TS3
0041	0000	REGA, 0000
0042	0000	REGB, 0000
0043	0000	REGC, 0000
0044	0000	REGD, 0000
0045	0000	REGE, 0000
0046	5642	REGF, 0000
0047	5255	SKPWAT, XWA1T
0050	5270	XPIG01, PIG01
0051	5323	XPIG02, PIG02
0052	5336	XPIG03, PIG03
0053	5234	XPIG04, PIG04
		XPIG05, PIG05

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PAL10

0054	0055	0056	0057	0060	0061	0062	0063	0064	0065	0066	0067	0070	0071	0072	0073	0074	0075	0076	0077	0100	0101	0102	0103	0104	0105	0106	0107	0110	0111	0112	0113	0114	0115	0116	0117	0120	0121	0122	0123	0124	0125	0126	0127	0130	0131	0132	0133	0134	0135	0136	0137	0140	0141	0142
5310	5224	5216	5302	5065	0215	0217	0570	3561	3556	5660	5746	0000	0000	5000	5020	5413	0000	0000	0000	5402	5441	5563	5471	5420	5542	5624	5056	5046	7730	7400	0000	6007	0026	0400	6000	3000	5000	7770	0260	4100	3740	0240	0017	7774	7773	7772	0077	0215	0212	0377	0040	0020	7000	0010
XISE,	RANDY,	XSNDRV,	XSYNC,	XCLREG,	OVER2,	OVER2A,	XOK0EP,	XMITT,	XMITT1,	XLAS,	XGTAD,	SEND,	RECEV,	NERROR,	ERROR,	XCLOCK,	CLOCKS,	KREGC,	LOOP,	JMP12,	XCRLF,	XREG,	XSORT,	XOCTEL,	XMESS,	XPRINT,	XTYPE,	XBELL,	KPRMTI,	K7400,	KT1CPS,	K6007,	K0006,	K0400,	K6000,	K3000,	K5000,	K7770,	K0260,	K4100,	K3740,	K0240,	K0017,	K7774,	K7773,	K7772,	K0077,	K0215,	K0212,	K0377,	K0040,	K0020,	K7000,	K0010,
ISZLOP	RANDOM	SNDRV	SYNC	CLRREG	BCNEAC	BCNEAC +2	TST30	TST202	TST202 -3	SWLAS	GTAD	0000	0000	NERRO	ERRO	CLOCK	0000	0000	0000	JMP 1 2	CRLF	PREG	SORT	OCTEL	MESS	PRINT	TYPE	BELL	7730	7400	0000	6007	0006	0400	6000	3000	5000	7770	0260	4100	3740	0240	0017	7774	7773	7772	0077	0215	0212	0377	0040	0020	7000	0010

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0143 2000, 2000
0144 1000, 1000
0145 0300, 0300
0146 0500, 0500
0147 0600, 0600
0150 0700, 0700
0151 2725, 2725
0152 2650, 2650
0153 7425, 7425
0154 7350, 7350
0155 7753, 7753
0156 0225, 0225
0157 0150, 0150
0160 1450, 1450
0161 1425, 1425
0162 6575, 6575
0163 6525, 6525
0164 5600, 5600
0165 5450, 5450
0166 0070, 0070
0167 5771, 5771
0170 5740, 5740
0171 1775, 1775
0172 2200, 2200
0173 2603, 2603
0174 2505, 2505
0175 4003, 4003

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0200 0200

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0200 7300
0201 6007
0202 4501
0203 4506
0204 6000
0205 4501
0206 4460
0207 4504
0210 4466
0211 5465
0212 5463
0213 4474
0214 4565
0215 4567
0216 3077
0217 4460
0220 3040

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/ CLEAR THE AC AND LINK
/ CAF OR CLEAR THE WORLD
/ CRLF
/ PRINT DK8E CLOCKS DIAGNOSTIC
/ MESSAGE POINTER
/ CRLF
/ CLEAR ALL MY REGISTERS
/ SET UP FOR PI RETURN
/ GET HIS SWITCHES
/ TEST SCHMITT
/ TEST DK8EP CLOCK
/ TEST DK8EA OR DK8EC
/ SORT AND PRINT FREQ. SELECTFD
/ GET TIME LENGTH
/ SET LOOP COUNTER
/ CLEAR ALL REGISTERS

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/ DOES IOT CLEI CHANGE AC ?
/ CHECK ALL COMBINATIONS
/
TST0, TAD REGA /GET AC NUMBER
DCA SEND /SAVE OUTPUT FOR ERROR PRINTER
TAD SEND
JMS I XIOTA /IOT 6131, CLEI

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0225      3071      DCA RECEV      /SAVE INPUT FOR ERROR PRINTER
0226      1071      TAD RECEV
0227      4456      JMS I XSNDRV      /CHECK SEND AND RECEV REGISTERS
0230      4472      JMS I NERROR      /CHECK NON-ERROR HANDLER.
0231      4473      JMS I ERROR      /ERROR: CLEI CHANGED AC.
0232      3000      /TST0 ERROR MESSAGE.
0233      0221      TST0
0234      3040      DCA REGA      /SCOPE LOOP.

/DOES IOT CLED CHANGE AC ?
/CHECK ALL COMBINATIONS
/
TST1,
0235      1040      TAD REGA      /GET AC NUMBER
0236      3070      DCA SEND      /SAVE OUTPUT FOR ERROR PRINTER
0237      1070      TAD SEND
0240      4421      JMS I XIOTB      /IOT 6132, CLED
0241      3071      DCA RECEV      /SAVE INPUT FOR ERROR PRINTER
0242      1071      TAD RECEV
0243      4456      JMS I XSNDRV      /CHECK SEND AND RECEV REGISTERS
0244      4472      JMS I NERROR      /CHECK NON-ERROR HANDLER.
0245      4473      JMS I ERROR      /ERROR: CLED CHANGED AC.
0246      3001      /TST1 ERROR MESSAGE.
0247      0235      TST1
0250      3040      DCA REGA      /SCOPE LOOP.

/DOES IOT CLSK CHANGE AC ?
/CHECK ALL COMBINATIONS
/
TST2,
0251      1040      TAD REGA      /GET AC NUMBER
0252      3070      DCA SEND      /SAVE OUTPUT FOR ERROR PRINTER
0253      1070      TAD SEND
0254      4422      JMS I XIOTC      /IOT 6133, CLSK
0255      7000      NOP      /WAIT JUST IN CASE !
0256      3071      DCA RECEV      /SAVE INPUT FOR ERROR PRINTER
0257      1071      TAD RECEV
0260      4456      JMS I XSNDRV      /CHECK SEND AND RECEV REGISTERS
0261      4472      JMS I NERROR      /CHECK NON-ERROR HANDLER.
0262      4473      JMS I ERROR      /ERROR: CLSK CHANGED AC.
0263      3002      /TST2 ERROR MESSAGE.
0264      0251      TST2      /SCOPE LOOP.

/TEST FOR NO INTERRUPT RQST.
/
TST3,
0265      6007      /CAF OR CLEAR THE WORLD
0266      4447      JMS I XPIG01      /GO TO PI, NO PI EXPECTED
0267      4472      JMS I NERROR      /CHECK NON-ERROR HANDLER
0270      4473      JMS I ERROR      /ERROR:PI OR INT. RQST. FAILED
0271      1003      /TST3 ERROR MESSAGE
0272      0265      TST3      /SCOPE LOOP

/DOES CLSK SKIP ON A CLOCK FLAG
/
TST4,
0273      1113      TAD KT1CPS
0274      3045      DCA REGF      /SET UP TIMER
0275      4422      JMS I XIOTC      /IOT 6133, CLSK

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0276      7020      NOP
0277      4422      JMS I XIOTC      /IOT 6133, CLSK
0300      4446      JMS I SKPWAT      /GO WAIT FOR FLAG
0301      4472      JMS I NERROR      /CHECK NON-ERROR HANDLER
0302      4473      JMS I ERROR      /ERROR: CLSK OR FLAG FAILED
0303      0404      /TST4 ERROR MESSAGE
0304      0273      TST4      /SCOPE LOOP

/DOES CLSK CLEAR THE FLAG ?
/
TST5,      TAD KTICPS      /SET UP TIMER
DCA REGF      /IOT 6133, CLSK
JMS I XIOTC
NOP
JMS I XIOTC      /IOT 6133, CLSK
JMS I SKPWAT      /GO WAIT FOR FLAG
SKP      /GOT THE FLAG
JMP I .-10      /GO BACK TO TEST 4
JMS I XIOTC      /IOT 6133, CLSK
JMS I NERROR      /CHECK NON-ERROR HANDLER
JMS I ERROR      /ERROR: CLSK CLEAR THE FLAG FAILED
0005
TST5      /TST5 ERROR MESSAGE
/SCOPE LOOP

/DOES CLEI ENABLE CLOCK INTERRUPT ?
/
TST6,      JMS I XIOTA      /IOT 6131, CLEI
JMS I XPIG02      /GO TO PI, PI EXPECTED
JMS I NERROR      /CHECK NON-ERROR HANDLER,
JMS I ERROR      /ERROR: DID CLEI ENABLE CLOCK INTERRUPT ?
1406      /TST6 ERROR MESSAGE
TST6      /SCOPE LOOP.

/DOES CLED DISABLE CLOCK INTERRUPT ?
/
TST7,      JMS I XIOTA      /IOT 6131, CLEI
JMS I XIOTB      /IOT 6132, CLED
JMS I XPIG01      /GO TO PI, NO PI EXPECTED
JMS I NERROR      /CHECK NON-ERROR HANDLER,
JMS I ERROR      /ERROR: DID CLED DISABLE CLOCK INTERRUPT?
1007      /TST7 ERROR MESSAGE
TST7      /SCOPE LOOP.

/DOES CAF DISABLE CLOCK INTERRUPT ?
/
TST10,      JMS I XIOTA      /IOT 6131, CLEI
6007      /CAF OR CLEAR THE WORLD
JMS I XPIG01      /GO TO PI, NO PI EXPECTED
JMS I NERROR      /CHECK NON-ERROR HANDLER,
JMS I ERROR      /ERROR: DID CAF DISABLE CLOCK INTERRUPT ?
1010      /TST10 ERROR MESSAGE
TST10      /SCOPE LOOP.

/DOES CLEI ENABLE CLOCK INTERRUPT ?
/
0322      4420
0323      4420
0324      4472
0325      4473
0326      1406
0327      0322

0330      4420
0331      4421
0332      4447
0333      4472
0334      4473
0335      1007
0336      0330

0337      4420
0340      6027
0341      4447
0342      4472
0343      4473
0344      1010
0345      0337

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0346	4420	TST11,	JMS I XIOTA	/IOT 6131, CLEI
0347	4447		JMS I XPIG01	/GO TO PI, PI EXPECTED
0350	5354		JMP T11A	
0351	4420		JMS I XIOTA	/IOT 6131, CLEI
0352	4450		JMS I XPIG02	/GO TO PI, PI EXPECTED
0353	4472		JMS I NERROR	/CHECK NON-ERROR HANDLER.
0354	4473	T11A,	JMS I ERROR	/ERROR: CLEI AND CLED FAST TOGGLE
0355	1411		1411	/TS11 ERROR MESSAGE
0356	0346		TST11	/SCOPE.
		/	/DOES CLED DISABLE CLOCK INTERRUPT ?	
		/		
0357	4420	TST12,	JMS I XIOTA	/IOT 6131, CLEI
0360	4421		JMS I XIOTB	/IOT 6132, CLED
0361	4450		JMS I XPIG02	/GO TO PI, NO PI EXPECTED
0362	5366		JMP T12A	
0363	4421		JMS I XIOTB	/IOT 6132, CLED
0364	4447		JMS I XPIG01	/GO TO PI, NO PI EXPECTED
0365	4472		JMS I NERROR	/CHECK NON-ERROR HANDLER.
0366	4473	T12A,	JMS I ERROR	/ERROR: CLEI AND CLED FAST TOGGLE
0367	1012		1012	/TS12 ERROR MESSAGE
0370	0357		TST12	/SCOPE LOOP.
		/	/TEST DECODER FOR 6135, NOT CLEI	
		/		
0371	4421	TST13,	JMS I XIOTB	/IOT 6132, CLED
0372	4431		JMS I XIOTI	/IOT 6135, NOT AN IOT 6131
0373	4447		JMS I XPIG01	/GO TO PI, NO PI EXPECTED
0374	4472		JMS I NERROR	/CHECK NON-ERROR HANDLER.
0375	4473		JMS I ERROR	/ERROR: DID DECODER WORK
0376	1013		1013	/TS13 ERROR MESSAGE
0377	0371		TST13	/SCOPE LOOP.
		/	/TEST DECODER FOR A 6136, NOT CLED	
		/		
0400	4420	TST14,	JMS I XIOTA	/IOT 6131, CLEI
0401	4432		JMS I XIOTJ	/IOT 6136, NOT AN IOT 6132.
0402	4450		JMS I XPIG02	/GO TO PI, PI EXPECTED
0403	4472		JMS I NERROR	/CHECK NON-ERROR HANDLER.
0404	4473		JMS I ERROR	/ERROR: DID DECODER WORK
0405	1414		1414	/TS14 ERROR MESSAGE
0406	0400		TST14	/SCOPE LOOP.
		/	/TEST DECODER FOR 6137, NOT CLSK	
		/		
0407	1113	TST15,	TAD KT1CPS	/SET UP TIMER
0410	3045		DCA REGF	/IOT 6132, CLED
0411	4422		JMS I XIOTC	
0412	7000		NOP	
0413	4433		JMS I XIOTK	/IOT 6137, NOT AN IOT 6133
0414	4446		JMS I SKPWAT	/GO WAIT FOR FLAG
0415	7410		SKP	/ERROR, SKIP OCCURRED
0416	4472		JMS I NERROR	/CHECK NON-ERROR HANDLER.
0417	4473		JMS I ERROR	/ERROR: DID DECODER WORK
0420	0015		0015	/TS15 ERROR MESSAGE

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/
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0421 0407
TST15 /SCOPE LOOP.
/DOES CLSK ENABLE CLOCK INTERRUPT ?
TST16. JMS I XIOTC /IOT 6133, CLSK
NOP
JMS I XPIG01 /GO TO PI, NO PI EXPECTED
JMS I NERROR /CHECK NON-ERROR HANDLER,
JMS I ERROR /ERROR: DID CLSK CAUSE INTERRUPT
1016 /TST16 ERROR MESSAGE
TST16 /SCOPE LOOP.
/DOES CLSK DISABLE CLOCK INTERRUPT ?
TST17. JMS I XIOTA /IOT 6131, CLEI
JMS I XIOTC /IOT 6133, CLSK
NOP
JMS I XPIG02 /GO TO PI, PI EXPECTED
JMS I NERROR /CHECK NON-ERROR HANDLER,
JMS I ERROR /ERROR: CLSK DISABLED CLOCK INTERRUPT
1417 /TST17 ERROR MESSAGE
TST17 /SCOPE LOOP.
/DOES CLEI CAUSE A SKIP ON FLAG ?
TST20. TAD KI1CPS
DCA REGF /SET UP TIMER
JMS I XIOTA /IOT 6131, CLEI
JMS I SKPWAT /GO WAIT FOR FLAG
SKP /ERROR, SKIP OCCURRED
JMS I NERROR /CHECK NON-ERROR HANDLER,
JMS I ERROR /ERROR: DID CLEI CAUSE A SKIP
0020 /TST20 ERROR MESSAGE
TST20 /SCOPE LOOP.
/DOES CLED CAUSE A SKIP ON FLAG ?
TST21. TAD KI1CPS
DCA REGF /SET UP TIMER
JMS I XIOTB /IOT 6132, CLED
JMS I SKPWAT /GO WAIT FOR FLAG
SKP /ERROR, SKIP OCCURRED
JMS I NERROR /CHECK NON-ERROR HANDLER,
JMS I ERROR /ERROR: DID CLED CAUSE A SKIP ON FLAG
0021 /TST21 ERROR MESSAGE
TST21 /SCOPE LOOP.
/DOES INT. ROST STAY DOWN ?
TST22. JMS I XSYNC /SYNC WITH CLOCK
JMS I XIOTA /IOT 6131, CLEI
JMS I XPIG01 /GO TO PI, PI EXPECTED
JMP T22A /ERROR, PI FAILED
ISZ REGB
JMP .-1 /WAIT 15.5 MS
0422 4422
0423 7000
0424 4447
0425 4472
0426 4473
0427 1016
0430 0422
0431 4420
0432 4422
0433 7000
0434 4450
0435 4472
0436 4473
0437 1417
0440 0431
0441 1113
0442 3045
0443 4420
0444 4446
0445 7410
0446 4472
0447 4473
0450 0020
0451 0441
0452 1113
0453 3045
0454 4421
0455 4446
0456 7410
0457 4472
0460 4473
0461 0021
0462 0452
0463 4457
0464 4420
0465 4447
0466 5273
0467 2041
0470 5267

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0471	4452	JMS I XPIG04	/GO TO PI, PI EXPECTED	
0472	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
0473	4473	JMS I ERROR	/ERROR:DID RQST. LAST ?	
0474	1422	1422	/TST21 ERROR MESSAGE	
0475	0463	TST22	/SCOPE LOOP	
		/		
		/DOES CLSK CLEAR RQST. LINE ?		
		/		
0476	4420	TST23,	JMS I XIOTA	/IOT 6131, CLEI
0477	4457	JMS I XSYNC	/SYNC WITH CLOCK FLAG	
0500	4451	JMS I XPIG03	/GO TO PI, NO PI EXPECTED	
0501	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
0502	4473	JMS I ERROR	/ERROR: DID CLSK CLEAR RQST. FLAG	
0503	1023	1023	/TST23 ERROR MESSAGE	
0504	0476	TST23	/SCOPE LOOP	
		/		
		/SYNC WITH CLOCK AND		
		/CHECK FOR FAST OUTPUT		
		/		
0505	4467	TST24,	JMS I XGTAD	/GET TIME CONSTANTS
0506	0000	0000	/MODIFIED BY TEST	
0507	1706	TAD I .-1		
0510	3043	DCA REGD		
0511	4420	JMS I XIOTA	/IOT 6131, CLEI	
0512	4457	JMS I XSYNC	/SYNC WITH CLOCK	
0513	4447	JMS I XPIG01	/GO TO PI, NO PI EXPECTED	
0514	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER.	
0515	4473	JMS I ERROR	/ERROR: CLOCK FREQUENCY FAST.	
0516	2024	2024	/TST24 ERROR MESSAGE.	
0517	0505	TST24	/SCOPE LOOP.	
		/		
		/SYNC WITH CLOCK AND		
		/CHECK FOR SLOW OUTPUT		
		/		
0520	1115	TST25,	TAD K0006	/SETUP FOR SLOW CLOCK
0521	4467	JMS I XGTAD	/GET TIME CONSTANTS	
0522	0000	0000	/MODIFIED BY TEST	
0523	1722	TAD I .-1		
0524	3043	DCA REGD		
0525	4420	JMS I XIOTA	/IOT 6131, CLEI	
0526	4457	JMS I XSYNC	/SYNC WITH CLOCK	
0527	4450	JMS I XPIG02	/GO TO PI, PI EXPECTED	
0530	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER.	
0531	4473	JMS I ERROR	/ERROR: CLOCK FREQUENCY SLOW.	
0532	2425	2425	/TST25 ERROR MESSAGE.	
0533	0520	TST25	/SCOPE LOOP.	
		/		
		/CHECK FOR FAST CLOCK AND		
		/BAD CLOCK FLAG WITH CLSK.		
		/		
0534	4467	TST26,	JMS I XGTAD	/GET TIME CONSTANTS
0535	0000	0000	/MODIFIED BY TEST	
0536	1735	TAD I .-1		
0537	3043	DCA REGD		
0540	4457	JMS I XSYNC	/SYNC WITH CLOCK	

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0541      4454      JMS I XISZ      /WAIT
0542      4422      JMS I XIOTC      /IOT 6133, CLSK
0543      4472      JMS I NERRR      /CHECK NON-ERROR HANDLER
0544      4473      JMS I ERROR      /ERROR: CLOCK FAILED
0545      2026      /TST26 ERROR MESSAGE
0546      0534      TST26      /SCOPE LOOP

/
/CHECK FOR SLOW CLOCK AND
/BAD CLOCK FLAG WITH CLSK
/
TST27,      TAD K0006      /SET UP FOR SLOW CLOCK
JMS I XGTAD      /GET TIME CONSTANTS
0000      /MODIFIED BY TEST

TAD I --1
DCA REGD
JMS I XSYNC
JMS I XISZ
JMS I XIOTC
SKP
JMS I NERRR
JMS I ERROR
2427
TST27
ISZ LOOP
JMP I OVER2A
JMS I XPASS
JMP I OVER2

/DOES IOT CLZE CHANGE AC?
/CHECK ALL COMBINATIONS.
/
TST30,      TAD REGA      /GET AC NUMBER
JMS I XIOTD      /IOT 6130,CLZE
DCA RECEV      /SAVE INPUT FOR ERROR PRINTER
TAD RECEV
JMS I XSNDRV      /CHECK SEND AND RECEV REGISTERS
JMS I NERRR      /CHECK NON-ERROR HANDLER
JMS I ERROR      /ERROR:CLZE CHANGED AC
3030      /TST30 ERROR MESSAGE
TST30      /SCOPE LOOP

/DOES IOT CLSK CHANGE AC?
/CHECK ALL COMBINATIONS
/
TST31,      TAD REGA      /GET AC NUMBER
DCA SEND      /SAVE OUTPUT FOR ERROR PRINTER
TAD SEND
JMS I XIOTE      /IOT 6131, CLSK
NOP
DCA RECEV      /SAVE INPUT FOR ERROR PRINTER
TAD RECEV
JMS I XSNDRV      /CHECK SEND AND RECEV REGISTERS
JMS I NERRR      /CHECK NON-ERROR HANDLER
JMS I ERROR      /ERROR:CLSK CHANGED AC.
0601      1040
0602      3070
0603      1070
0604      4424
0605      7020
0606      3071
0607      1071
0610      4426
0611      4472
0612      4473

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0613	3031	3031		/TST31 ERROR MESSAGE
0614	0601	TST31		/SCOPE LOOP
				/
				/DOES IOT CLOE CHANGE AC?
				/CHECK ALL COMBINATIONS
				/
0615	1040	TST32,		/GET AC NUMBER
0616	4425	JMS I XIOTF		/IOT 6132, CLOE
0617	3071	DCA RECEV		/SAVE INPUT FOR ERROR PRINTER
0620	1071	TAD RECEV		
0621	4456	JMS I XSNDRV		/CHECK SEND AND RECEV REGISTERS
0622	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
0623	4473	JMS I ERROR		/ERROR:CLDE CHANGED AC
0624	3032	3032		/TST32 ERROR MESSAGE
0625	0615	TST32		/SCOPE LOOP
				/
				/DOES IOT CLAB CHANGE AC?
				/CHECK ALL COMBINATIONS
				/
0626	1040	TST33,		/GET AC NUMBER
0627	4427	JMS I XIOTG		/IOT 6133, CLAB
0630	3071	DCA RECEV		/SAVE INPUT FOR ERROR PRINTER
0631	1071	TAD RECEV		
0632	4456	JMS I XSNDRV		/CHECK SEND AND RECEV REGISTERS
0633	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
0634	4473	JMS I ERROR		/ERROR:CLAB CHANGED AC
0635	3033	3033		/TST33 ERROR MESSAGE
0636	0626	TST33		/SCOPE LOOP
				/
				/DOES CAF CLEAR BUFFER REGISTER?
				/CHECK FOR JAM TO AC, CLBA.
				/
0637	6027	TST34,		/CAF OR CLEAR T.C WORLD
0640	7340	CLA CLL CMA		/AC TO 7777
0641	4432	JMS I XIOTJ		/IOT 6136, CLJA
0642	7650	SNA CLA		/WAS BUFFER ALL 'S?
0643	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
0644	4473	JMS I ERROR		/ERROR:CAF OR CLBA FAILED.
0645	3434	3434		/TST34 ERROR MESSAGE
0646	0637	TST34		/SCOPE LOOP
				/
				/DOES CAF CLEAR ENABLE REGISTER?
				/CHECK FOR JAM TO AC, CLEN.
				/
0647	6027	TST35,		/CAF OR 'LEAR THE WORLD
0650	7340	CLA CLL CMA		/AC TO 7777
0651	4430	JMS I XIOTH		/IOT 6134, CLEN
0652	7650	SNA CLA		/WAS ENABLE REGISTER ALL 0'S?
0653	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
0654	4473	JMS I ERROR		/ERROR:CAL OR CLEN FAILED.
0655	4435	4435		/TST35 ERROR MESSAGE
0656	0647	TST35		/SCOPE LOOP

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/
/DOES CAF CLEAR STATUS REGISTER ?
/CHECK JAM TO AC CLSA
/
TST36, 6007 /CAF OR THE CLEAR THE WORLD
        CLA ALL CMA /AC TO 7777
        JMS I XIOTI /IOT 6135, CLSA
        SNA CLA /WAS STATUS REGISTER ALL 0'S ?
        JMS I NERROR /CHECK NON-ERROR HANDLER
        JMS I ERROR /CAF OR CLSA FAILED
        5036 /TST36 ERROR MESSAGE
        TST36 /SCOPE LOOP
/
/DOES AC LOAD BUFFER REGISTER?
/CHECK ALL 0'S TRANSFER
/CHECK JAM TO AC, CLBA
/
TST37, JMS I XIOTG /IOT 6133, CLAB
        CLA CLL CMA /AC TO 7777
        JMS I XIOTJ /IOT 6136, CLBA
        SNA CLA /WAS BUFFER ALL 0'S?
        JMS I NERROR /CHECK NON-ERROR HANDLER
        JMS I ERROR /ERROR:CLAB OR CLBA FAILED
        3437 /TST37 ERROR MESSAGE
        TST37 /SCOPE LOOP
/
/DOES AC LOAD BUFFER REGISTER ?
/CHECK ALL 1'S TRANSFER
/CHECK JAM TO AC, CLBA
/
TST40, CLA CLL CMA /AC TO 7777
        JMS I XIOTG /IOT 6133, CLAB
        CLA CLL /CLEAR THE AC AND LINK
        JMS I XIOTJ /IOT 6136, CLBA
        CMA /COMPLEMENT THE AC
        SNA CLA /WAS BUFFER ALL 1'S?
        JMS I NERROR /CHECK NON-ERROR HANDLER
        JMS I ERROR /ERROR:CLAB OR CLBA FAILED
        3440 /TST40 ERROR MESSAGE
        TST40 /SCOPE LOOP
/
/DOES BUFFER SURVIVE PATTERN 2525 ?
/
TST41, TAD K2525 /GET AC NUMBER
        JMS I XIOTG /IOT 6133, CLAB
        CMA /COMPLEMENT AC
        JMS I XIOTJ /IOT 6136, CLBA
        JMS I XSNDV /CHECK SEND AND RECEV REGISTERS
        JMS I NERROR /CHECK NON-ERROR HANDLER
        JMS I ERROR /ERROR: BUFFER OR AC FAILED
        3441 /TST41 ERROR MESSAGE
        TST41 /SCOPE LOOP
/
/DOES BUFFER SURVIVE PATTERN 5252 ?
/
TST41, 1016
        0711 4427 /GET AC NUMBER
        0712 7040 /IOT 6133, CLAB
        0713 4432 /COMPLEMENT AC
        0714 4436 /IOT 6136, CLBA
        0715 4472 /CHECK SEND AND RECEV REGISTERS
        0716 4473 /CHECK NON-ERROR HANDLER
        0717 3441 /ERROR: BUFFER OR AC FAILED
        0720 0711 /TST41 ERROR MESSAGE
        0721 0711 /SCOPE LOOP
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/
TST42,      TAD K5252      /GET AC NUMBER
            JMS I XIOTG    /IOT 6133, CLXB
            CMA             /COMPLEMENT AC
            JMS I XIOTJ    /IOT 6136, CLBA
            JMS I XSNDRV   /CHECK SEND AND RECEV REGISTERS
            JMS I NERROR   /CHECK NON-ERROR HANDLER
            JMS I ERROR    /ERROR: BUFFER OR AC FAILED
            3442           /TST42 ERROR MESSAGE
            TST42         /SCOPE LOOP

/
/DOES CAF REALLY CLEAR BUFFER ?
/
TST43,      CLA CLA CMA    /AC TO ALL 7777
            JMS I XIOTG    /IOT 6133, CLAB
            6007          /CAF OR CLEAR THE WORLD
            DCA SEND       /SAVE OUTPUT FOR ERROR PRINTER
            CLA CLL CMA    /IOT 6136, CLBA
            JMS I XIOTJ    /WAS BUFFER ALL 0'S ?
            SNA CLA        /CHECK NON-ERROR HANDLER
            JMS I NERROR   /ERROR: CAF OR BUFFER FAILED
            JMS I ERROR    /TST43 ERROR MESSAGE
            3443           /SCOPE LOOP
            TST43

/
/DOES CAF REALLY CLEAR BUFFER ?
/DO ALL COMBINATIONS
/
TST44,      TAD REGA      /GET AC NUMBER
            JMS I XIOTG    /IOT 6133, CLAB
            6007          /CAF OR CLEAR THE WORLD
            DCA SEND       /SAVE OUTPUT FOR ERROR PRINTER
            CLA CLL CMA    /IOT 6136, CLBA
            JMS I XIOTJ    /WAS BUFFER ALL 0'S ?
            SNA CLA        /CHECK NON-ERROR HANDLER
            JMS I NERROR   /ERROR: CAF OR BUFFER FAILED
            JMS I ERROR    /TST44 ERROR MESSAGE
            3444           /SCOPE LOOP
            TST44

/
/CHECK AC TO BUFFER REGISTER AND
/BUFFER REGISTER TO AC TRANSFERS.
/CHECK ALL COMBINATIONS.
/CHECK LOAD ON BUFFER REGISTER.
/
TST45,      CLA CLL CMA    /GET AC NUMBER
            DCA REGA      /IOT 6133, CLAB
            TAD REGB      /COMPLEMENT THE AC
            JMS I XIOTG    /IOT 6136, CLBA
            CMA             /CHECK SEND RECEV REGISTERS
            JMS I XIOTJ
            JMS I XSNDRV
            SKP CLA
            JMP T45A
            15Z REGB      /UPDATE AC NUMBER
            0772 2041

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/
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0773 5363 JMP T45B /CHECK NON-ERROR HANDLER
0774 4472 JMS I NERRR /ERROR: AC OR BUFFER FAILED.
0775 4473 JMS I ERROR /TST45 ERROR MESSAGE
0776 3445 3445 /SCOPE LOOP
0777 0761 TST45

/DOES READING BUFFER CHANGE ITS CONTENTS ?
/
TST46, CLA CLL CMA /AC TO 7777
DCA REGA
TAD K2525 /GET AC NUMBER
JMS I XIOTG /IOT 6133, CLAB
CMA /COMPLEMENT AC
T46B, JMS I XIOTJ /IOT 6136, CLBA
JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
SKP CLA
JMP T46A
ISZ REGB /UPDATE COUNTER
JMP T46B /DO 4096 TIMES
JMS I NERRR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: BUFFER FAILED
3446 /TST46 ERROR MESSAGE
TST46 /SCOPE LOOP

/DOES READING BUFFER CHANGE ITS CONTENTS ?
/
TST47, CLA CLL CMA /AC TO 7777
DCA REGA
TAD K5252 /GET AC NUMBER
JMS I XIOTG /IOT 6133, CLAB
CMA /COMPLEMENT AC
T47B, JMS I XIOTJ /IOT 6136, CLBA
JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
SKP CLA
JMP T47A
ISZ REGB /UPDATE COUNTER
JMP T47B /DO 4096 TIMES
JMS I NERRR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: BUFFER FAILED
3447 /TST47 ERROR MESSAGE
TST47 /SCOPELOOP

/DOES BUFFER SURVIVE RANDOM PATTERNS ?
/
TST50, CLA CLL CMA /AC TO 7777
DCA REGA
T50B, JMS I RANDY /GET RANDOM NUMBER
JMS I XIOTG /IOT 6133, CLAB
CMA /COMPLEMENT AC
T47A, JMS I XIOTJ /IOT 6136, CLBA
JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
SKP CLA
JMP T50A
ISZ REGB /UPDATE COUNTER
JMP T50B /DO 4096 TIMES

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1051      4472      JMS I NERROR      /CHECK NON-ERROR HANDLER
1052      4473      JMS I ERROR      /ERROR: BUFFER FAILED
1053      3450      TST50      /TST50 ERROR MESSAGE
1054      1036      TST50      /SCOPE LOOP

/DOES BUFFER SURVIVE FAST TOGGLE ?
/
TST51,      TAD REGA      /GET AC NUMBER
DCA SEND      /SAVE OUTPUT FOR ERROR PRINTER
TAD REGA
JMS I XIOTS1      /IOT'S 6133 AND 6136
DCA RECEV      /SAVE INPUT FOR ERROR PRINTER
TAD RECEV
JMS I XSNDRV      /CHECK SEND RECEV REGISTERS
JMS I NERROR      /CHECK NON-ERROR HANDLER
JMS I ERROR      /ERROR: BUFFER FAILED
3451      /TST51 ERROR MESSAGE
TST51      /SCOPE LOOP

/DOES AC SET ENABLE REGISTER?
/CHECK ALL 1'S TRANSFER.
/CHECK JAM TO AC, CLEN
/
TST52,      CLA CLL CMA      /AC TO 7777
JMS I XIOTF      /IOT 6132, CLOE
CMA      /COMPLEMENT AC
JMS I XIOTH      /IOT 6134, CLEN
CMA      /COMPLEMENT AC
SNA CLA      /WAS ENABLE REGISTER ALL 1'S ?
JMS I NERROR      /CHECK NON-ERROR HANDLER
JMS I ERROR      /ERROR: CLOE OR CLEN FAILED.
4452      /TST52 ERROR MESSAGE
TST52      /SCOPE LOOP

/DOES AC SET ENABLE REGISTER?
/CHECK ALL 0'S TRANSFER.
/
TST53,      CLA CLL CMA      /AC TO 7777
JMS I XIOTF      /IOT 6132, CLOE
CLA CLL      /CLEAR THE AC AND LINK
JMS I XIOTF1      /IOT 6132, CLOE
JMS I XIOTH      /IOT 6134, CLEN
CMA      /COMPLEMENT THE AC
SNA CLA      /WAS ENABLE REGISTER ALL 1'S ?
JMS I NERROR      /CHECK NON-ERROR HANDLER
JMS I ERROR      /ERROR: CLOE OR CLEN FAILED
4453      /TST53 ERROR MESSAGE
TST53      /SCOPE LOOP

/DOES CAF REALLY CLEAR ENABLE REGISTER?
/
TST54,      CLA CLL CMA      /AC TO 7777
JMS I XIOTF      /IOT 6132, CLOE

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1117	6007	6007		/CAF OR CLEAR THE WORLD
1120	3070	DCA SEND		/SAVE OUTPUT FOR ERROR PRINTER
1121	7340	CLA CLL CMA		/AC TO 7777
1122	4430	JMS I X10TH		/IOT 6134, CLEN
1123	7650	SNA CLA		/WAS REGISTER ALL 0'S
1124	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
1125	4473	JMS I ERROR		/ERROR:CAF,CLOE,OR CLEN FAILED
1126	4454	4454		/TST54 ERROR MESSAGE
1127	1115	TST54		/SCOPE LOOP
/DOES CAF REALLY CLEAR ENABLE REGISTER ?				
/DO ALL COMBINATIONS				
/TST55,				
1130	1040	TAD REGA		/GET AC NUMBER
1131	4426	JMS I X10TF1		/IOT 6132, CLOE
1132	6007	6007		/CAF OR CLEAR THE WORLD
1133	7340	CLA CLL CMA		/AC TO 7777
1134	4430	JMS I X10TH		/IOT 6134, CLEN
1135	7650	SNA CLA		/WAS ENABLE REGISTER ALL 0'S ?
1136	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
1137	4473	JMS I ERROR		/ERROR: ENABLE REGISTER FAILED
1140	4455	4455		/TST55 ERROR MESSAGE
1141	1130	TST55		/SCOPE LOOP
/DOES ENABLE REGISTER SURVIVE PATTERN 2525 ?				
/TST56,				
1142	1016	TAD K2525		/GET AC NUMBER
1143	4425	JMS I X10TF		/IOT 6132, CLOE
1144	7040	CMA		/COMPLEMENT AC
1145	4430	JMS I X10TH		/IOT 6134, CLEN
1146	4456	JMS I XSNDRV		/CHECK SEND AND RECEV REGISTERS
1147	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
1150	4473	JMS I ERROR		/ERROR: EBABLE REGISTER FAILED
1151	4456	4456		/TST56 ERROR MESSAGE
1152	1142	TST56		/SCOPE LOOP
/DOES ENABLE REGISTER SURVIVE PATTERN 2522 ?				
/TST57,				
1153	1017	TAD K5252		/GET AC NUMBER
1154	4425	JMS I X10TF		/IOT 6132, CLOE
1155	7040	CMA		/COMPLEMENT AC
1156	4430	JMS I X10TH		/IOT 6134, CLEN
1157	4456	JMS I XSNDRV		/CHECK SEND AND RECEV REGISTERS
1160	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
1161	4473	JMS I ERROR		/ERROR: ENABLE REGISTER FAILED
1162	4457	4457		/TST57 ERROR MESSAGE
1163	1153	TST57		/SCOPE LOOP
/DOES ENABLE REGISTER SURVIVE PATTERN 2525 ?				
/TST60,				
1164	1016	TAD K2525		/GET AC NUMBER
1165	4425	JMS I X10TF		/IOT 6132, CLOE
1166	7300	CLA CLL		/CLEAR THE AC AND LINK
1167	4426	JMS I X10TF1		/IOT 6132, CLOE
1170	7340	CLA CLL CMA		/AC TO 7777

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/
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1171      4430      JMS I X10TH      /IOT 6134, CLEN
1172      4426      JMS I XSNDRV      /CHECK SEND AND RECEV REGISTERS
1173      4472      JMS I NERROR      /CHECK NON-ERROR HANDLER
1174      4473      JMS I ERROR      /ERROR: ENABLE RFGISTER FAILED
1175      4460      4460      /TST60 ERROR MESSAGE
1176      1164      TST60      /SCOPE LOOP

/DOES ENABLE REGISTER SURVIVE PATTERN 5252 ?
/
TST61,      TAD K5252      /GET AC NUMBER
1177      1017      JMS I X10TF      /IOT 6132, CLOE
1200      4425      CLA CLL      /CLEAR THE AC AND LINK
1201      7300      JMS I X10TF1      /IOT 6132, CLOE
1202      4426      CLA CLL CMA      /AC TO 7777
1203      7340      JMS I X10TH      /IOT 6134, CLEN
1204      4430      JMS I XSNDRV      /CHECK SEND AND RECEV REGISTERS
1205      4456      JMS I NERROR      /CHECK NON-ERROR HANDLER
1206      4472      JMS I ERROR      /ERROR: ENABLE REGISTER FAILED
1207      4473      4461      /TST61 ERROR MESSAGE
1210      4461      TST61      /SCOPE LOOP
1211      1177

/DOES ENABLE REGISTER SURVIVE COMPLEMENT PATTERN ?
/
TST62,      CLA CLL CMA      /AC TO 7777
1212      7340      DCA SEND      /SAVE OUTPUT FOR ERROR PRINTER
1213      3070      TAD K5252      /GET AC NUMBER
1214      1016      JMS I X10TF1      /IOT 6132, CLOE
1215      4426      CMA      /COMPLEMENT AC
1216      7040      JMS I X10TF1      /IOT 6132, CLOE
1217      4426      CLA CLL      /CLEAR THE AC AND LINK
1220      7300      JMS I X10TH      /IOT 6134, CLEN
1221      4430      JMS I XSNDRV      /CHECK SEND RECEV REGISTERS
1222      4456      JMS I NERROR      /CHECK NON-ERROR HANDLER
1223      4472      JMS I ERROR      /ERROR: ENABLE REGISTER FAILED
1224      4473      4462      /TST62 ERROR MESSAGE
1225      4462      TST62      /SCOPE LOOP
1226      1212

/DOES ENABLE REGISTER SURVIVE COMPLEMENT PATTERN ?
/
TST63,      CLA CLL CMA      /AC TO 7777
1227      7340      DCA SEND      /SAVE OUTPUT FOR ERROR PRINTER
1230      3070      TAD K5252      /GET AC NUMBER
1231      1017      JMS I X10TF1      /IOT 6132, CLOE
1232      4426      CMA      /COMPLEMENT AC
1233      7040      JMS I X10TF1      /IOT 6132, CLOE
1234      4426      CLA CLL      /IOT 6134, CLEN
1235      7300      JMS I X10TH      /CHECK SEND AND RECEV REGISTERS
1236      4430      JMS I XSNDRV      /CHECK NON-ERROR HANDLER
1237      4456      JMS I NERROR      /ERROR: ENABLE REGISTERS
1240      4472      JMS I ERROR      /TST63 ERROR MESSAGE
1241      4473      4463      /SCOPE LOOP
1242      4463      TST63
1243      1227

/DO AC TO ENABLE REGISTER AND
/ENABLE REGISTER TO AC TRANSFERS

```

/CHECK ALL COMBINATIONS

```
1244 1040      TAD REGA      /GET AC NUMBER
1245 4425      JMS I XIOTF    /IOT 6132, CLOE
1246 7340      CLA CLL CMA    /AC TO 7777
1247 4430      JMS I XIOTH    /IOT 6134, CLEN
1250 4456      JMS I XSNDRV    /CHECK SEND AND RECEV REGISTERS
1251 4472      JMS I NERROR    /CHECK NON-ERROR HANDLER
1252 4473      JMS I NERROR    /ERROR: AC OR ENABLE REGISTER FAILED.
1253 4464      JMS I NERROR    /TST64 ERROR MESSAGE
1254 1244      TST64          /SCOPE LOOP
```

/DOES ENABLE REGISTER SURVIVE COMPLEMENT PATTERN.
/DO ALL COMBINATIONS.

```
1255 7340      CLA CLL CMA    /AC TO 7777
1256 3070      DCA SEND      /SAVE OUTPUT FOR ERROR PRINTER
1257 1040      TAD REGA      /GET AC NUMBER
1260 4426      JMS I XIOTF1    /IOT 6132, CLOE
1261 7040      CMA          /COMPLEMENT THE AC
1262 4426      JMS I XIOTF1    /IOT 6132, CLOE
1263 4430      JMS I XIOTH    /IOT 6134, CLEN
1264 4456      JMS I XSNDRV    /CHECK SEND AND RECEV REGISTERS
1265 4472      JMS I NERROR    /CHECK NON-ERROR HANDLER
1266 4473      JMS I NERROR    /ERROR: AC OR ENABLE REGISTER FAILED.
1267 4465      JMS I NERROR    /TST65 ERROR MESSAGE
1270 1255      TST65          /SCOPE LOOP
```

/DOES ENABLE REGISTER SURVIVE RANDOM PATTERN ?

```
1271 4455      JMS I RANDY     /GET RANDOM NUMBER
1272 4425      JMS I XIOTF    /IOT 6132, CLOE
1273 7300      CLA CLL        /CLEAR THE AC AND LINK
1274 4430      JMS I XIOTH    /IOT 6134, CLEN
1275 4456      JMS I XSNDRV    /CHECK SEND AND RECEV REGISTERS
1276 4472      JMS I NERROR    /CHECK NON-ERROR HANDLER
1277 4473      JMS I NERROR    /ERROR: ENABLE REGISTER FAILED
1300 4466      JMS I NERROR    /TST66 ERROR MESSAGE
1301 1271      TST66          /SCOPE LOOP
```

/DOES ENABLE REGISTER SURVIVE RANDOM COMPLEMENT PATTNRN ?

```
1302 7340      CLA CLL CMA    /AC TO 7777
1303 3070      DCA SEND      /SAVE OUTPUT FOR ERROR PRINTER
1304 4455      JMS I RANDY     /GET RANDOM NUMBER
1305 4426      JMS I XIOTF1    /COMPLEMENT AC
1306 7040      CMA          /IOT 6132, CLOE
1307 4426      JMS I XIOTF1    /IOT 6134, CLEN
1310 4430      JMS I XSNDRV    /CHECK SEND AND RECEV REGISTERS
1311 4456      JMS I NERROR    /CHECK NON-ERROR HANDLER
1312 4472      JMS I NERROR    /ERROR: ENABLE REGISTER FAILED
1313 4473      JMS I NERROR    /TST67 ERROR MESSAGE
1314 4467      JMS I NERROR    /SCOPE LOOP
1315 1302      TST67
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/DOES READING ENABLE REGISTER CHANGE ITS CONTENTS ?
/
TST70,      CLA CLL CMA      /AC TO 7777
DCA REGA
TAD K2525      /GET AC NUMBER
JMS I XIOTF      /IOT 6132, CLOE
CLA CLL CMA      /AC TO 7777
JMS I XIOTW      /IOT 6134, CLEN
JMS I XSNDRV      /CHECK SEND AND RECEV REGISTERS
SKP CLA
JMP T70A
IS2 REGB      /UPDATE COUNTER
JMP T70B      /DO 4096 TIMES
JMS I NERROR      /CHECK NON-ERROR HANDLER
JMS I ERROR      /ERROR: ENABLE REGISTER FAILED
4470      /TST70 ERROR MESSAGE
TST70      /SCOPE LOOP

/DOES READING ENABLE REGISTER CHANGE TIS CONTENTS ?
/
TST71,      CLA CLL CMA      /AC TO 7777
DCA REGA
TAD K2525      /GET AC NUMBER
JMS I XIOTF      /IOT 6132, CLOE
CLA CLL CMA      /CLEAR THE AC AND LINK
JMS I XIOTW      /IOT 6134, CLEN
JMS I XSNDRV      /CHECK SEND RECEV REGISTERS
SKP CLA
JMP T71A
IS2 REGB      /UPDATE COUNTER
JMP T71B      /DO 4096 TIMES
JMS I NERROR      /CHECK NON-ERROR HANDLER
JMS I ERROR      /ERROR: ENABLE REGISTER FAILED
4471      /TST71 ERROR MESSAGE
TST71      /SCOPE LOOP

/DOES ENABLE REGISTER SURVIVE FAST TOGGLE ?
/
TST72,      TAD REGA      /GET AC NUMBER
DCA SEND      /SAVE OUTPUT FOR ERROR PRINTER
TAD REGA
JMS I XIOTS      /IOT'S 6132 AND 6134
DCA RECEV      /SAVE INPUT FOR ERROR PRINTER
TAD RECEV
JMS I XSNDRV      /CHECK SEND RECEV REGISTERS
JMS I NERROR      /CHECK NON-ERROR HANDLER
JMS I ERROR      /ERROR: ENABLE REGISTER FAILED
4472      /TST72 ERROR MESSAGE
TST72      /SCOPE LOOP

/DOES CLZE CLEAR ENABLE REGISTER?
/
TST73,      CLA CLL CMA      /AC TO 7777
JMS I XIOTF1      /IOT 6132, CLOE
CLA CLL CMA

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1372	4423	JMS I XIOTD	/IOT 6130, CLZE
1373	7300	CLA CLL	/CLEAR THE AC AND LINK
1374	3070	DCA SEND	/SAVE OUTPUT FOR ERROR PRINTER
1375	7340	CLA CLL CMA	/AC TO 7777
1376	4430	JMS I XIOTH	/IOT 6134, CLEN
1377	7650	SNA CLA	/WAS REGISTER ALL 0'S
1400	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER
1401	4473	JMS I ERROR	/ERROR:CLZE OR CLEN FAILED.
1402	4473	4473	/TST73 ERROR MESSAGE
1403	1367	TST73	/SCOPE LOOP

/DOES CLZE CLEAR ENABLE REGISTER?			
/			
1404	7340	TST74,	CLA CLL CMA
1405	4425	JMS I XIOTF	/AC TO 7777
1406	7300	CLA CLL	/IOT 6132, CLOE
1407	4423	JMS I XIOTD	/IOT 6130, CLZE
1410	7340	CLA CLL CMA	/AC TO 7777
1411	3070	DCA SEND	/SAVE OUTPUT FOR ERROR PRINTER
1412	4430	JMS I XIOTH	/IOT 6134, CLEN
1413	7040	CMA	/COMPLEMENT AC
1414	7650	SNA CLA	/WAS REGISTER ALL 0'S?
1415	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER
1416	4473	JMS I ERROR	/ERROR:CLZE OR CLEN FAILED.
1417	4474	4474	/TST74 ERROR MESSAGE
1420	1424	TST74	/SCOPE LOOP

/DOES CLZE CLEAR ENABLE REGISTER?			
/			
1421	1016	TST75,	TAD K2525
1422	4425	JMS I XIOTF	/IOT 6132,CLOE
1423	7040	CMA	/COMPLEMENT THE AC
1424	4423	JMS I XIOTD	/IOT 6130, CLZE
1425	7040	CMA	/COMPLEMENT AC
1426	3070	DCA SEND	/SAVE OUTPUT FOR ERROR PRINTER
1427	4430	JMS I XIOTH	/IOT 6134, CLEN
1430	4456	JMS I XSNDV	/CHECK SEND AND RECEV REGISTERS
1431	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER
1432	4473	JMS I ERROR	/ERROR:CLZE,CLOE, OR CLEN FAILED
1433	4475	4475	/TST75 ERROR MESSAGE
1434	1421	TST75	/SCOPE LOOP

/DOES CLZE CLEAR ENABLE REGISTER ?			
/			
1435	1017	TST76,	TAD K5252
1436	4425	JMS I XIOTF	/GET AC NUMBER
1437	7040	CMA	/IOT 6132, CLOE
1440	4423	JMS I XIOTD	/COMPLEMENT AC
1441	7040	CMA	/IOT 6130, CLZE
1442	3070	DCA SEND	/COMPLEMENT AC
1443	4430	JMS I XIOTH	/SAVE OUTPUT FOR ERROR PRINTER
1444	4456	JMS I XSNDV	/IOT 6134, CLEN
1445	4472	JMS I NERROR	/CHECK SEND AND RECEV REGISTERS
1446	4473	JMS I ERROR	/CHECK NON-ERROR HANDLER
1447	4476	4476	/ERROR: ENABLE REGISTER FAILED
			/TST76 ERROR MESSAGE

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1450  1435      /
1450  1435      /DOES CLZE CLEAR ENABLE REGISTER?
1450  1435      /CHECK ALL COMBINATIONS
1450  1435      /
1450  1435      TST77,
1450  1435      /GET AC NUMBER
1450  1435      /IOT 6132, CLOE
1450  1435      /IOT 6130, CLZE
1450  1435      /CLEAR THE AC AND LINK
1450  1435      /SAVE OUTPUT FOR ERROR PRINTER
1450  1435      /AC TO ALL 1'S
1450  1435      /IOT 6134, CLEN
1450  1435      /WAS REGISTER ALL 1'S?
1450  1435      /CHECK NON-ERROR HANDLER
1450  1435      /ERROR:CLZE,CLOE, OR CLEN FAILED
1450  1435      /TST77 ERROR MESSAGE
1450  1435      /SCOPE LOOP
1450  1435      TST77
1450  1435
1450  1435      /DOES CLZE CLEAR ENABLE REGISTER?
1450  1435      /DO ALL COMBINATIONS
1450  1435      /
1450  1435      TST100,
1450  1435      /GET AC NUMBER
1450  1435      /IOT 6132, CLOE
1450  1435      /COMPLEMENT THE AC
1450  1435      /IOT 6130, CLZE
1450  1435      /COMPLEMENT THE AC
1450  1435      /SAVE OUTPUT FOR ERROR PRINTER
1450  1435      /IOT 6134, CLEN
1450  1435      /CHECK SEND AND RECEV REGISTERS
1450  1435      /CHECK NON-ERROR HANDLER
1450  1435      /ERROR:CLZE, CLOE, OR CLEN FAILED
1450  1435      /TST100 ERROR MESSAGE
1450  1435      /SCOPE LOOP
1450  1435      TST100
1450  1435
1450  1435      /DOES CLZE SURVIVE RANDOM PATTERN ?
1450  1435      /
1450  1435      TST101,
1450  1435      /GET RANDOM NUMBER
1450  1435      /IOT 6132, CLOE
1450  1435      /IOT 6130, CLZE
1450  1435      /CLEAR THE AC AND LINK
1450  1435      /SAVE OUTPUT FOR ERROR PRINTER
1450  1435      /IOT 6134, CLEN
1450  1435      /CHECK SEND AND RECEV REGISTERS
1450  1435      /CHECK NON-ERROR HANDLER
1450  1435      /ERROR: ENABLE REGISTER FAILED
1450  1435      /TST101 ERROR MESSAGE
1450  1435      /SCOPE LOOP
1450  1435      TST101
1450  1435
1450  1435      /DOES CLZE SURVIVE RANDOM COMPLEMENT PATTERN ?
1450  1435      /
1450  1435      TST102,
1450  1435      /GET RANDOM NUMBER
1450  1435      /IOT 6132, CLOE
1450  1435      /COMPLEMENT AC
1450  1435      CMA
1450  1435

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/
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1517 4423 JMS I XIOTD /IOT 6130, CLCA
1520 7040 CMA /COMPLEMENT AC
1521 3070 DCA SEND /SAVE OUTPUT FOR ERROR PRINTER
1522 4430 JMS I XIOTH /IOT 6134, CLEN
1523 4456 JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
1524 4472 JMS I NERROR /CHECK NON-ERROR HANDLER
1525 4473 JMS I ERROR /ERROR: ENABLE REGISTER FAILED
1526 4502 4502 /TST102 ERROR MESSAGE
1527 1514 TST102 /SCOPE LOOP
/
/DOES CLZE SURVIVE FAST TOGGLE ?
/
TST103, TAD REGA /GET AC NUMBER
1530 1040 JMS I XIOTF /IOT 6132, CLOE
1531 4425 JMS I XIOTS3 /IOT'S 6130 AND 6134
1532 4437 DCA RECEV /SAVE INPUT FOR ERROR PRINTER
1533 3071 TAD RECEV
1534 1071 JMS I XSNDRV /CHECK SEND RECEV REGISTERS
1535 4456 JMS I NERROR /CHECK NON-ERROR HANDLER
1536 4472 JMS I ERROR /ERROR: ENABLE REGISTER FAILED
1537 4473 4503 /TST103 ERROR MESSAGE
1540 4523 TST103 /SCOPE LOOP
1541 1530
/
/DOES AC TRANSFER TO BUFFER THEN TO COUNTER ?
/
TST104, JMS I XIOTG /IOT 6133, CLAH
1542 4427 CLA CLL CMA /AC TO ALL 1'S
1543 7340 JMS I XIOTK /IOT 6137, CLCA
1544 4433 SNA CLA /WAS COUNTER ALL 0'S?
1545 7050 JMS I NERROR /CHECK NON-ERROR HANDLER
1546 4472 JMS I ERROR /ERROR: CLAB OR CLCA FAILED
1547 4473 4104 /TST104 ERROR MESSAGE
1550 4124 TST104 /SCOPE LOOP
1551 1542
/
/DOES AC TRANSFER TO BUFFER THEN TO COUNTER?
/
TST105, CLA CLL CMA /IOT 6133, CLAB
1552 7340 JMS I XIOTG /IOT 6137, CLCA
1553 4427 CMA /COMPLEMENT THE AC
1554 4433 SNA CLA /WAS COUNTER ALL 1'S?
1555 7040 JMS I NERROR /CHECK NON-ERROR HANDLER
1556 7050 JMS I ERROR /ERROR: CLAB OR CLCA FAILED
1557 4472 4105 /TST105 ERROR MESSAGE
1560 4473 TST105 /SCOPE LOOP
1561 4125
1562 1552
/
/DOES COUNTER SURVIVE PATTERN 2525 ?
/
TST106, TAD K2525 /GET AC NUMBER
1563 1016 JMS I XIOTG /IOT 6133, CLAB
1564 4427 CLA CLL /CLEAR THE AC AND LINK
1565 7300 JMS I XIOTK /IOT 6137, CLCA
1566 4433 JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
1567 4456 JMS I NERROR /CHECK NON-ERROR HANDLER
1570 4472

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1571 4473 JMS I ERROR /ERROR: COUNTER FAILED
1572 4126 4106 /TST106 ERROR MESSAGE
1573 1563 TST106 /SCOPE LOOP
/
/DOES COUNTER SURVIVE PATTERN 5252 ?
/
TST107, TAD K5252 /GET AC NUMBER
1574 1017 JMS I XIOTG /IOT 6133, CLAB
1575 4427 CLA CLL CMA /AC TO ALL 7777
1576 7340 JMS I XIOTK /IOT 6137, CLCA
1577 4433 JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
1600 4436 JMS I NERRR /CHECK NON-ERROR HANDLER
1601 4472 JMS I ERROR /ERROR: COUNTER FAILED
1602 4473 4107 /TST107 ERROR MESSAGE
1603 4127 TST107 /SCOPE LOOP
1604 1574
/
/DOES AC TRANSFER TO BUFFER THEN TO COUNTER?
/CHECK ALL COMBINATIONS
/
TST110, TAD REGA /IOT 6133, CLAB
1605 1040 JMS I XIOTG /COMPLEMENT THE AC
1606 4427 CMA /IOT 6137, CLCA
1607 7040 JMS I XIOTK /CHECK SEND AND RECEV REGISTERS
1610 4433 JMS I XSNDRV /CHECK NON-ERROR HANDLER
1611 4436 JMS I NERRR /ERROR: CLAB OR CLCA FAILED
1612 4472 JMS I ERROR /TST110 ERROR MESSAGE
1613 4473 4110 /SCOPE LOOP
1614 4110 TST110
1615 1605
/
/DOES COUNTER SURVIVE FAST TOGGLE?
/
TST111, TAD REGA /GET AC NUMBER
1616 1040 DCA SEND /SAVE OUTPUT FOR ERROR PRINTF
1617 3070 TAD SEND /IOT 6133 AND 6137
1620 1070 JMS I XIOTS2 /SAVE INPUT FOR ERROR PRINTF
1621 4436 DCA RECEV /CHECK SEND AND RECEV REGISTERS
1622 3071 TAD RECEV /CHECK NON-ERROR HANDLER
1623 1071 JMS I XSNDRV /ERROR:CLAB OR CLCA FAILED
1624 4436 JMS I NERRR /TST111 ERROR MESSAGE
1625 4472 JMS I ERROR /SCOPE LOOP
1626 4473 4111
1627 4111 TST111
1630 1616
/
/DOES CAF AFFECT COUNTER ?
/
TST112, TAD REGA /GET AC NUMBER
1631 1040 JMS I XIOTG /IOT 6133, CLAB
1632 4427 6007 /CAF OR CLEAR THE WORLD
1633 6007 JMS I XIOTK /IOT 6137, CLCA
1634 4433 JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
1635 4436 JMS I NERRR /CHECK NON-ERROR HANDLER
1636 4472 JMS I ERROR /ERROR: CLAB OR CLCA FAILED.
1637 4473 4112 /TST112 ERROR MESSAGE
1640 4112
```

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1641 1631

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TST112 /SCOPE LOOP

```
/
/DOES READING COUNTER CHANGE ITS CONTENTS?
/PATTERN 2525,
/
TST113, CLA CLL CMA /AC TO 7777
DCA REGA
TAD K2525
JMS I XIOTG /IOT 6133, CLAB
T113B, JMS I XIOTK /IOT 6137, CLCA
JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
SKP T113A
JMP T113A
ISZ REGB
JMP T113B
JMS I NERROR /CHECK NON-ERROR
T113A, JMS I ERROR /ERROR: CLAR OR CLCA FAILED
4113 /TST113 ERROR MESSAGE
TST113 /SCOPE LOOP
```

```
/DOES READING COUNTER CHANGE ITS CONTENTS?
/PATTERN 5252
/
TST114, CLA CLL CMA /AC TO 7777
DCA REGA
TAD K2525
JMS I XIOTG /IOT 6133, CLAB
T114R, JMS I XIOTK /IOT 6137, CLCA
JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
SKP
JMP T114A
ISZ REGB
JMP T114B
JMS I NERROR /CHECK NON-ERROR HANDLER
T114A, JMS I ERROR /ERROR: COUNTER FAILED
4114 /TST114 ERROR MESSAGE
TST114 /SCOPE LOOP
```

```
/DOES COUNTER SURVIVE RANDOM PATTERN ?
/
TST115, JMS I RANDY /GET RANDOM NUMBER
JMS I XIOTG /IOT 6133, CLAB
CLA CLL CMA
JMS I XIOTK /IOT 6137, CLCA
T115, JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
JMS I NERROR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: COUNTER FAILED
4115 /TST115 ERROR MESSAGE
TST115 /SCOPE LOOP
```

/TEST FOR NO INT. RQST.

1707 7340
TST116, CLA CLL CMA /AC TO 7777

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1710	4427	JMS I XIOTG	/IOT 6133, CLAB	
1711	3040	DCA REGA		
1712	1142	TAD K0010		
1713	1147	TAD K0600	/GET ENABLES	
1714	4425	JMS I XIOTF	/IOT 6132, CLOE	
1715	4447	JMS I XPIG01	/GO TO PI, NO PI EXPECTED	
1716	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
1717	4473	JMS I ERROR	/ERROR: INT. ROST. FAILED	
1720	1116	1116	/TST116, ERROR MESSAGE	
1721	1707	TST116	/SCOPE LOOP	
		/DOES CLSK SKIP ON CLOCK OVERFLOW?		
		/SKIP EXPECTED, MODE 0, RATE 6		
		/		
1722	7340	TST117, CLA CLL CMA	/AC TO 7777	
1723	4427	JMS I XIOTG	/IOT 6133, CLAB	
1724	7300	CLA CLL	/CLEAR THE AC AND LINK	
1725	1147	TAD K0600	/GET RATE 6	
1726	4425	JMS I XIOTF	/IOT 6132, CLOE	
1727	4424	JMS I XIOTE	/IOT 6131, CLSK	
1730	7410	SKP		
1731	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
1732	4473	JMS I ERROR	/ERROR: CLSK OR OVERFLOW FAILED	
1733	0517	0517	/TST117 ERROR MESSAGE	
1734	1722	TST117	/SCOPE LOOP	
		/DOES OVERFLOW REMAIN SET ?		
		/		
1735	7340	TST120, CLA CLL CMA	/IOT 6133, CLAB	
1736	4427	JMS I XIOTG		
1737	3040	DCA REGA	/GET ENABLES	
1740	1147	TAD K0600	/IOT 6132, CLOE	
1741	4425	JMS I XIOTF	/IOT 6131, CLSK	
1742	4424	JMS I XIOTE		
1743	9391	JMP T120A		
1744	2041	ISZ REGB		
1745	5344	JMP -1	/WAIT ABOUT 15 MS	
1746	4424	JMS I XIOTE	/IOT 6131, CLSK	
1747	7410	SKP		
1750	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
1751	4473	JMS I ERROR	/ERROR: CLSK OR OVERFLOW FAILED	
1752	0520	0520	/TST120 ERROR MESSAGE	
1753	1735	TST120	/SCOPE LOOP	
		/DOES CAF CLEAR THAT FLAG ?		
		/		
1754	7340	TST121, CLA CLL CMA	/IOT 6133, CLAB	
1755	4427	JMS I XIOTG		
1756	3040	DCA REGA	/GET ENABLES	
1757	1147	TAD K0600	/IOT 6132, CLOE	
1760	4425	JMS I XIOTF	/IOT 6131, CLSK	
1761	4424	JMS I XIOTE		
1762	5361	JMP -1	/CAF OR CLEAR THE WORLD	
1763	6007	6007	/IOT 6131, CLSK	
1764	4424	JMS I XIOTE		

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1765	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
1766	4473	T121A, JMS I ERROR	/ERROR: CAF OR OVERFLOW FAILED	
1767	0121	0121	/TST121 ERROR MESSAGE	
1770	1754	TST121	/SCOPE LOOP	
/DOES CLSK SKIP ON OVERFLOW ?				
/SKIP EXPECTED, RATE 2-6, MODE 0				
1771	1131	TST122, TAD K7773		
1772	3041	DCA REG8		
1773	1015	TAD K0200		
1774	3044	DCA REGE		
1775	7340	CLA CLL CMA	/AC TO 7777	
1776	4427	JMS I XIOTG	/IOT 6133, CLAB	
1777	3040	DCA REGA		
2000	1044	TAD REGE	/GET ENABLES	
2001	4425	JMS I XIOTF	/IOT 6132, CLOE	
2002	2043	ISZ REGD		
2003	5202	JMP -1	/WAIT	
2004	4424	JMS I XIOTE	/IOT 6131, CLSK	
2005	5214	JMP T122A	/NO OVERFLOW FOUND	
2006	1013	TAD K0100		
2007	3044	DCA REGE	/UPDATE CLOCK RATE	
2010	0007	6007	/CAF OR CLEAR THE WORLD	
2011	2041	ISZ REG8		
2012	5571	JMP I XCRS1		
2013	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
2014	4473	JMS I ERROR	/ERROR: CLSK OR OVERFLOW FAILED	
2015	0522	0522	/TST122 ERROR MESSAGE	
2016	1771	TST122	/SCOPE LOOP	
/DOES CLSK SKIP ON OVERFLOW ?				
/SKIP EXPECTED, RATE 2-6, MODE 1				
2017	1131	TST123, TAD K7773		
2020	3041	DCA REG8		
2021	1144	TAD K1000		
2022	1015	TAD K0200		
2023	3044	DCA REGE		
2024	7340	CLA CLL CMA	/AC TO 7777	
2025	4427	JMS I XIOTG	/IOT 6133, CLAB	
2026	3040	DCA REGA		
2027	1044	TAD REGE	/GET ENABLES	
2030	4425	JMS I XIOTF	/IOT 6132, CLOE	
2031	2043	ISZ REGD		
2032	5231	JMP -1	/WAIT	
2033	4424	JMS I XIOTE	/IOT 6131, CLSK	
2034	5243	JMP T123A	/NO OVERFLOW FOUND	
2035	1013	TAD K0100		
2036	3044	DCA REGE	/UPDATE CLCOK RATE	
2037	0007	6007	/CAF OR CLEAR THE WORLD	
2040	2041	ISZ REG8		
2041	5224	JMP T123B	/DO RATES 2-6	
2042	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
2043	4473	JMS I ERROR	/ERROR: CLSK OR OVERFLOW FAILED	
T123A,				

2044 0523
2045 2017

0523 /TST123 ERROR MESSAGE
TST123 /SCOPE LOOP

/DOES CLSK SKIP ON OVERFLOW ?
/SKIP EXPECTED, MODE 2, RATE 2-6
/

2046 1131
2047 3041
2048 1143
2049 1019
2050 3044
2051 7340
2052 4427
2053 3040
2054 1044
2055 4425
2056 4023
2057 5260
2058 4424
2059 5272
2060 1013
2061 3044
2062 6007
2063 2041
2064 5253
2065 4472
2066 4473
2067 0524
2068 2046

TST124, TAD K7773
DCA REGB
TAD K2000
TAD K0200
DCA REGE
CLA CLL CMA
JMS I X10TG
DCA REGA
TAD REGE
JMS I X10TF
ISE REGD
JMP -1
JMS I X10TE
JMP T124A
TAD K0100
DCA REGE
6007
ISE REGB
JMP T124B
JMS I NERRR
JMS I ERROR
0524
TST124

T124B, /MAKE ENABLES
/IOT 6133, CLAB
/GET ENABLES
/IOT 6132, CLOE
/WAIT ABOUT 15 MS
/IOT 6131, CLSK
/UPDATE RATE
/CAF OR CLEAR THE WORLD
/DO RATES 2-6
/CHECK NON-ERROR HANDLER
/ERROR! CLSK OR OVERFLOW FAILED
/TST124 ERROR MESSAGE
/SCOPE LOOP

/DOES CLSK SKIP ON OVERFLOW ?
/SKIP EXPECTED, RATE 2-6, MODE 3
/

2075 1131
2076 3041
2077 1120
2100 1019
2101 3044
2102 7340
2103 4427
2104 3040
2105 1044
2106 4425
2107 2043
2110 5307
2111 4424
2112 5320
2113 1013
2114 3044
2115 2041
2116 5302
2117 4472
2120 4473
2121 0525
2122 2075

TST125, TAD K7773
DCA REGB
TAD K3000
TAD K0200
DCA REGE
CLA CLL CMA
JMS I X10TG
DCA REGA
TAD REGE
JMS I X10TF
ISE REGD
JMP -1
JMS I X10TE
JMP T125A
TAD K0100
DCA REGE
ISE REGB
JMP T125B
JMS I NERRR
JMS I ERROR
0525
TST125

T125B, /MAKE ENABLES
/SAVE ENABLES
/IOT 6133, CLAB
/GET ENABLES
/IOT 6132, CLOE
/WAIT ABOUT 15 MS
/IOT 6131, CLSK
/UPDATE RATE
/DO RATES 2-6
/CHECK NON-ERROR HANDLER
/ERROR! CLSK OR OVERFLOW FAILED
/TST125 ERROR MESSAGE
/SCOPE LOOP

/DOES CLK SKIP ON OVERFLOW ?
/NO SKIP EXPECTED, RATE 0-7, MODE 0, DISABLE BIT 7

2123	1122
2124	3043
2125	7340
2126	4427
2127	3040
2130	1140
2131	1041
2132	4425
2133	2042
2134	5333
2135	4424
2136	7410
2137	5347
2140	0150
2141	1013
2142	3041
2143	6007
2144	2043
2145	5325
2146	4472
2147	4473
2150	0126
2151	2123

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TST126,      TAD K770
DCA REGD
CLA CLL CMA
JMS I XIOTG
DCA REGA
TAD K0020
TAD REGB
JMS I XIOTF
ISZ REGC
JMP ,-1
JMS I XIOTE
SKP
JMP T126A
AND K0700
TAD K0100
DCA REGB
6007
ISZ REGD
JMP T126B
JMS I NERROR
JMS I ERROR
0126
TST126
T126A,

```

```

/UPDATE RATE
/CAF OR CLEAR THE WORLD

/DO RATES 0-7
/CHECK NON-ERROR HANDLER
/ERROR: CLK OR OVERFLOW FAILED
/TST126 ERROR MESSAGE
/SCOPE LOOP

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/DOES CLSK SKIP ON OVERFLOW ?
/NO SKIP EXPECTED, RATE 0,1,7

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2192	7340
2193	4427
2194	3040
2195	4425
2196	2041
2197	9356
2198	4422
2199	7410
2200	5572
2201	1013
2202	4426
2203	2042
2204	5365
2205	4424
2206	7410
2207	5572
2208	1147
2209	4426
2210	2043
2211	5374
2212	4424
2213	7472
2214	4473
2215	0127
2216	
2217	
2218	
2219	
2220	
2221	

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TST127,          CLA  CLL CMA
JMS  I XIOTG
DCA REGA
JMS  I XIOTF
ISZ REGB
JMP  -1
JMS  I XIOTE
SKP  I XCRS2
JMP  I XCRS2
TAD K0100
JMS  I XIOTF1
ISZ REGC
JMP  -1
JMS  I XIOTE
SKP  I XCRS2
JMP  I XCRS2
TAD K0600
JMS  I XIOTF1
ISZ REGD
JMP  -1
JMS  I XIOTE
JMS  I XIOTE
JMS  I ERROR
JMS  I ERROR
T127A,          TST127 ERROR MS

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/UPDATE ENABLE
/!OT 6132, CLOC

/!WAIT ABOUT 15 MS
/!OT 6131, CLSK
/!CHECK NON-ERROR HANDLER
/!ERROR! CLSK OR OVERFLOW FAILED
/!TST127 ERROR MESSAGE

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2202 2152

TST127

/SCOPE LOOP

/DOES CLSA READ OVERFLOW BIT ?
/

2203 7340 CLA CLL CMA
2204 4427 JMS I XIOTG /IOT 6132, CLOE
2205 7330 CLA CLL CML RAR /AC TO 4000
2206 3070 DCA SEND /SAVE OUTPUT FOR ERROR PRINTER
2207 7313 CLA CLL IAC RTR /AC TO 4000
2210 1147 TAD K0600 /GET ENABLE
2211 4426 JMS I XIOTF1
2212 4424 JMS I XIOTE /IOT 6131, CLSK
2213 5212 JMP -1
2214 7350 CLA CLL CMA RAR /AC TO 3777
2215 4431 JMS I XIOTI /IOT 6135, CLSA
2216 4456 JMS I XSNDV /CHECK SEND AND RECEV REGISTERS
2217 4472 JMS I NERRR /CHECK NON-ERROR HANDLER
2220 4473 JMS I ERROR /ERROR! CLSI OR OVERFLOW FAILED
2221 5130 5130 /TST130 ERROR MESSAGE
2222 2203 TST130 /SCOPE LOOP

T130A,

/DOES CLSA CLEAR OVERFLOW FLOP ?
/

2223 7340 CLA CLL CMA /AC TO 7777
2224 4427 JMS I XIOTG /IOT 6133, CLAB
2225 7313 CLA CLL IAC RTR /AC TO 4000
2226 1147 TAD K0600 /GET ENABLE
2227 4426 JMS I XIOTF1 /IOT 6132, CLOE
2230 4424 JMS I XIOTE /IOT 6131, CLSK
2231 5230 JMP -1
2232 7350 CLA CLL CMA RAR /AC TO 3777
2233 4431 JMS I XIOTI /IOT 6135, CLSA
2234 7300 CLA CLL /CLEAR AC AND LINK
2235 3070 DCA SEND /SAVE OUTPUT FOR ERROR PRINTER
2236 7340 CLA CLL CMA /AC TO 7777
2237 4431 JMS I XIOTI /IOT 6135, CLSA
2240 7650 SNA CLA /WAS STATUS REGISTER ALL 0'S ?
2241 4472 JMS I NERRR /CHECK NON-ERROR HANDLER
2242 4473 JMS I ERROR /ERROR! CLSA OR OVERFLOW FAILED
2243 5131 5131 /TST131 ERROR MESSAGE
2244 2223 TST131 /SCOPE LOOP

/DOES CLSA READ OVERFLOW BIT ?
/

2245 7340 CLA CLL CMA
2246 4427 JMS I XIOTG /IOT 6133, CLAB
2247 7300 CLA CLL
2250 3070 DCA SEND /SAVE OUTPUT FOR ERROR PRINTER
2251 1147 TAD K0600 /GET ENABLES
2252 4426 JMS I XIOTF1 /IOT 6132, CLOE
2253 4424 JMS I XIOTE /IOT 6131, CLSK
2254 5253 JMP -1
2255 7340 CLA CLL CMA RAR /AC TO 3777
2256 4431 JMS I XIOTI /IOT 6135, CLSA
2257 7650 SNA CLA /WAS STATUS 0 ?

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2260	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
2261	4473	JMS I ERROR	/ERROR: CLSA OR STATUS FAILED	
2262	5132	5132	/TST132 ERROR MESSAGE	
2263	2245	TST132	/SCOPE LOOP	
/DOES BUFFER TO COUNTER ON OVERFLOW ?				
/MODE 1, RATE 2				
2264	7340	TST133, CLA CLL CMA	/IOT 6133, CLAB	
2265	4427	JMS I XIOTG	/AC TO 4000	
2266	3040	DCA REGA	/GET ENABLES	
2267	7313	CLA CLL IAC RTR	/IOT 6132, CLOE	
2270	1116	TAD K0400	/IOT 6131, CLSK	
2271	1144	TAD K1000	/WAIT FOR FLAG	
2272	4426	JMS I XIOTF1	/CLEAR THE AC AND LINK	
2273	4424	JMS I XIOTE	/IOT 6137, CLCA	
2274	5273	JMP .-1	/FOR TESTING	
2275	7300	CLA CLL	/WAS COUNTER ALL 1'S ?	
2276	4433	JMS I XIOTK	/IOT 6135, CLSA	
2277	7040	CMA	/DO TEST 4096 TIMES	
2300	7440	SEA T133A	/CHECK NON-ERROR HANDLER	
2301	5306	JMP T133A	/ERROR: COUNTER FAILED	
2302	4431	JMS I XIOTI	/TST133 ERROR MESSAGE	
2303	2041	IS2 REGB	/SCOPE LOOP	
2304	5273	JMP T133B		
2305	4472	JMS I NERROR		
2306	4473	JMS I ERROR		
2307	4133	4133		
2310	2264	TST133		
/DOES BUFFER TO COUNTER ON OVERFLOW ?				
/MODE 1, RATE 4				
2311	1017	TST134, TAD K5252	/GET AC NUMBER	
2312	4427	JMS I XIOTG	/IOT 6133, CLAB	
2313	7340	CLA CLL CMA	/AC TO 7777	
2314	3040	DCA REGA	/GET ENABLES	
2315	1144	TAD K1000	/IOT 6132, CLOE	
2316	1116	TAD K0400	/IOT 6131, CLSK	
2317	4426	JMS I XIOTF1	/WAIT FOR FLAG	
2320	4424	JMS I XIOTE	/AC TO 7777	
2321	5320	JMP .-1	/IOT 6137, CLCA	
2322	7340	CLA CLL CMA	/CHECK SEND AND RECEV REGISTERS	
2323	4433	JMS I XIOTK	/CHECK NON-ERROR HANDLER	
2324	4456	JMS I XSNDV	/ERROR: COUNTER FAILED	
2325	4472	JMS I NERROR	/TST134 ERROR MESSAGE	
2326	4473	JMS I ERROR	/SCOPE LOOP	
2327	4134	4134		
2330	2311	TST134		
/DOES BUFFER TO COUNTER ON OVERFLOW ?				
/MODE 1, RATE 4				
2331	1016	TST135, TAD K5255	/GET AC NUMBER	
2332	4427	JMS I XIOTG	/IOT 6133, CLAB	

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2333	7340	CLA CLL CMA	/AC TO 7777
2334	3040	DCA REGA	
2335	1144	TAD K1000	
2336	1116	TAD K0400	/GET ENABLES
2337	4426	JMS I XIOTF1	/IOT 6132, CLOE
2340	4424	JMS I XIOTE	/IOT 6131, CLSK
2341	5340	JMP -1	/WAIT FOR OVERFLOW
2342	4433	JMS I XIOTK	/IOT 6137, CLCA
2343	4456	JMS I XSNDV	/CHECK SEND AND RECEV REGISTERS
2344	4472	JMS I NERRR	/CHECK NON-ERROR HANDLER
2345	4473	JMS I ERROR	/ERROR: COUNTER FAILED
2346	4135	4135	/TST135 ERROR MESSAGE
2347	2331	TST135	/SCOPE LOOP

/DOES BUFFER TO COUNTER ON OVERFLOW ?			
/RATE 4, MODE 2			
/			
2350	7340	TST136, CLA CLL CMA	/AC TO 7777
2351	4427	JMS I XIOTG	/IOT 6133, CLAB
2352	3040	DCA REGA	
2353	3070	DCA SEND	/SAVE OUTPUT FOR ERROR PRINTER
2354	1116	TAD K0400	
2355	1143	TAD K2000	/GET ENABLES
2356	4426	JMS I XIOTF1	/IOT 6132, CLOE
2357	4424	JMS I XIOTE	/IOT 6131, CLSK
2360	5357	JMP -1	/WAIT FOR FLAG
2361	4433	JMS I XIOTK	/IOT 6137, CLCA
2362	7650	SNA CLA	/WAS COUNTER ALL 0'S ?
2363	4472	JMS I NERRR	/CHECK NON-ERROR HANDLER
2364	4473	JMS I ERROR	/ERROR: COUNTER FAILED
2365	4136	4136	/TST136 ERROR MESSAGE
2366	2350	TST136	/SCOPE LOOP

/DOES BUFFER TO COUNTER ON OVERFLOW ?			
/MODE 3, RATE 4			
/			
2367	7340	TST137, CLA CLL CMA	/IOT 6133, CLAB
2370	4427	JMS I XIOTG	
2371	3040	DCA REGA	/SAVE OUTPUT FOR ERROR PRINTER
2372	3070	DCA SEND	
2373	1116	TAD K0400	/GET ENABLES
2374	1120	TAD K3000	/IOT 6132, CLOE
2375	4426	JMS I XIOTF1	/IOT 6131, CLSK
2376	4424	JMS I XIOTE	/WAIT FOR OVERFLOW
2377	5376	JMP -1	
2400	7340	CLA CLL CMA	/IOT 6137, CLCA
2401	4433	JMS I XIOTK	/WAS COUNTER ALL 0'S ?
2402	7650	SNA CLA	/CHECK NON-ERROR HANDLER
2403	4472	JMS I NERRR	/ERROR: COUNTER FAILED
2404	4473	JMS I ERROR	/TST137 ERROR MESSAGE
2405	4137	4137	/SCOPE LOOP
2406	2367	TST137	

/DOES INT. WITHOUT BIT 0 ?			
/			

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2407	7340	TST140,	CLA CLL CMA	
2410	4427	JMS I XIOTG		/IOT 6133, CLAB
2411	3040	DCA REGA		
2412	7313	CLA CLL IAC RTR	/AC TO 4000	
2413	1007	TAD K0007		/GET ENABLES
2414	1147	TAD K0600		/IOT 6132, CLOE
2415	4425	JMS I XIOTF		/GO TO PI, NO PI EXPECTED
2416	4447	JMS I XPIG01		/CHECK NON-ERROR HANDLER
2417	4472	JMS I NERROR		/ERROR: INT. RQST. OR ENA 0 FAILED
2420	4473	JMS I ERROR		/TST140 ERROR MESSAGE
2421	1140	1140		/SCOPE LOOP
2422	2407	TST140		
		/		
		/DOES OVERFLOW CAUSE INT. RQST. ?		
		/RATE 6, MODE 0		
		/		
2423	7340	TST141,	CLA CLL CMA	/AC TO 7777
2424	4427	JMS I XIOTG		/IOT 6133, CLAB
2425	7300	CLA CLL		/CLEAR THE AC AND LINK
2426	1014	TAD K4000		
2427	1142	TAD K0010		/GET RATE * MODE
2430	1147	TAD K0600		/IOT 6132, CLOE
2431	4425	JMS I XIOTF		/GO TO PI, PI EXPECTED
2432	4452	JMS I XPIG04		/CHECK NON-ERROR HANDLER
2433	4472	JMS I NERROR		/ERROR: OVERFLOW OR ENA 0 FAILED
2434	4473	JMS I ERROR		/TST141 ERROR MESSAGE
2435	1541	1541		/SCOPE LOOP
2436	2423	TST141		
		/		
		/DOES INT. RQST. WITHOUT ENA 0 ?		
		/RATE 6, MODE 0		
		/		
2437	7340	TST142,	CLA CLL CMA	/AC TO 7777
2440	4427	JMS I XIOTG		/IOT 6133, CLAB
2441	7300	CLA CLL		/CLEAR THE AC AND LINK
2442	1142	TAD K0010		
2443	1147	TAD K0600		/GET RATE * MODE
2444	4425	JMS I XIOTF		/IOT 6132, CLOE
2445	4451	JMS I XPIG03		/GO TO PI, NO PI EXPECTED
2446	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
2447	4473	JMS I ERROR		/ERROR: ENA 0 FAILED
2450	1142	1142		/TST142 ERROR MESSAGE
2451	2437	TST142		/SCOPE LOOP
		/		
		/DOES COUNTER COUNT ?		
		/RATE 6, MODE 0		
		/		
2452	7340	TST143,	CLA CLL CMA	/AC TO 7777
2453	3040	DCA REGA		
2454	4427	JMS I XIOTG		/IOT 6133, CLAB
2455	1014	TAD K4000		
2456	1142	TAD K0010		/GET RATE * MODE
2457	1147	TAD K0600		/IOT 6132, CLOE
2460	4425	JMS I XIOTF		/GO TO PI
2461	4450	JMS I XPIG02		

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2462	4472	JMS I ERROR	/CHECK NON-ERROR HANDLER	
2463	4473	JMS I ERROR	/ERROR: OVERFLOW OR COUNTER FAILED	
2464	1543	TST143	/TST143 ERROR MESSAGE	
2465	2492		/SCOPE LOOP	
/				
		/DOES COUNTER COUNT ?		
		/RATE 6, MODE 1		
/				
2466	7340	TST144, CLA CLL		
2467	3040	DCA REGA		
2470	4427	JMS I XIOTG	/IOT 6133, CLAB	
2471	1121	TAD K5000		
2472	1142	TAD K0010		
2473	1147	TAD K0600	/GET RATE * MODE	
2474	4425	JMS I XIOTF	/IOT 6132, CLOE	
2475	4490	JMS I XPIG02	/GO TO PI	
2476	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
2477	4473	JMS I ERROR	/ERROR: OVERFLOW OR COUNTER FAILED	
2500	1544	TST144	/TST144 ERROR MESSAGE	
2501	2466		/SCOPE LOOP	
/				
		/DOES COUNTER COUNT ?		
		/RATE 6, MODE 2		
/				
2502	7340	TST145, CLA CLL CMA	/AC TO 7777	
2503	3040	DCA REGA		
2504	4427	JMS I XIOTG	/IOT 6133, CLAB	
2505	1117	TAD K6000		
2506	1142	TAD K0010	/GET ENABLES	
2507	1147	TAD K0600	/IOT 6132, CLOE	
2510	4425	JMS I XIOTF	/GO TO PI, PI EXPECTED	
2511	4490	JMS I XPIG02	/CHECK NON-ERROR HANDLER	
2512	4472	JMS I NERROR	/ERROR: OVERFLOW OR COUNTER FAILED	
2513	4473	JMS I ERROR	/TST145 ERROR MESSAGE	
2514	1545	TST145	/SCOPE LOOP	
2515	2502			
/				
		/DOES COUNTER COUNT ?		
		/RATE 6, MODE 3		
/				
2516	7340	TST146, CLA CLL CMA	/AC TO 7777	
2517	3040	DCA REGA		
2520	4427	JMS I XIOTG	/IOT 6133, CLAB	
2521	1141	TAD K7000		
2522	1142	TAD K0010	/GET ENABLES	
2523	1147	TAD K0600	/IOT 6132, CLOE	
2524	4425	JMS I XIOTF	/GO TO PI, PI EXPECTED	
2525	4490	JMS I XPIG02	/CHECK NON-ERROR HANDLER	
2526	4472	JMS I NERROR	/ERROR: COUNTER OR MODE 3 FAILED	
2527	4473	JMS I ERROR	/TST146 ERROR MESSAGE	
2530	1546	TST146	/SCOPE LOOP	
2531	2516			
/				
		/DOES OVERFLOW CAUSE ROST. ?		
		/RATE 2-6, MODE 0		

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2532 1131
2533 3041
2534 1014
2535 1142
2536 1015
2537 3044
2540 7340
2541 4427
2542 3040
2543 1044
2544 4425
2545 4447
2546 5355
2547 6007
2550 1013
2551 1044
2552 2041
2553 5337
2554 4472
2555 4473
2556 1547
2557 2532

/
TST147, TAD K7773
DCA REG8
TAD K4000
TAD K0010
TAD K0200
DCA REGE
CLA CLL CMA
JMS I XIOTG
DCA REGA
TAD REGE
JMS I XIOTF
JMS I XPIG01
JMP T147A
6007
TAD K0100
TAD REGE
ISZ REG8
JMP T147B
JMS I NERR0R
JMS I ERROR
1547
TST147
/DOES OVERFLOW CAUSE ROST. ?
/RATE 2-6, MODE 1
/
TST150, TAD K7773
DCA REG8
TAD K5020
TAD K0010
TAD K0200
DCA REGE
CLA CLL CMA
JMS I XIOTG
DCA REGA
TAD REGE
JMS I XIOTF
JMS I XPIG01
JMP I XCRS3
6007
TAD K0100
TAD REGE
ISZ REG8
JMP I XCRS4
JMS I NERR0R
JMS I ERROR
1550
TST150
/DOES OVERFLOW CAUSE ROST. ?
/RATE 2-6, MODE 2
/
TST151, TAD K7773
DCA REG8

2560 1131
2561 3041
2562 1121
2563 1142
2564 1015
2565 3044
2566 7340
2567 4427
2570 3040
2571 1044
2572 4425
2573 4447
2574 5573
2575 6007
2576 1013
2577 1044
2600 2041
2601 5574
2602 4472
2603 4473
2604 1550
2605 2560

2560 1131
2567 3041

/SET UP ENABLES
/AC TO 7777
/IOT 6133, CLAB
/GET ENABLES
/IOT 6132, CLOE
/GO TO PI, PI EXPECTED
/CAF OR CLEAR THE WORLD

/DO RATES 2-4
/CHECK NON-ERROR HANDLER
/ERROR: OVERFLOW OR MODE FAILED
/TST147 ERROR MESSAGE
/SCOPE LOOP

/MAKE ENABLES
/AC TO 7777
/IOT 6133, CLAB
/GET ENABLES
/IOT 6132, CLOE
/GO TO PI, PI EXPECTED
/CAF OR CLEAR THE WORLD

/CHECK NON-ERROR HANDLER
/ERROR: OVERFLOW OR MODE FAILED
/TST150 ERROR MESSAGE
/SCOPE LOOP

2606 1131
2607 3041
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PAL10	V141	9-OCT-71	15144	PAGE 1-33
2610	1117		TAD K0000	
2611	1142		TAD K0010	
2612	1015		TAD K0200	
2613	3044	T1518,	DCA REG	/MAKE ENABLES
2614	7340		CLA CLL CMA	/AC TO 7777
2615	4427		JMS I XIOTG	/IOT 6133, CLAB
2616	3040		DCA REG	
2617	1044		TAD REG	/GET ENABLES
2620	4425		JMS I XIOTF	/IOT 6132, CLDE
2621	4447		JMS I XPI001	/GO TO PI, PI EXPECTED
2622	5231		JMP T151A	
2623	6007		6007	/CAF OR CLEAR THE WORLD
2624	1013		TAD K0100	
2625	1044		TAD REG	
2626	2041		IS2 REG	
2627	5213		JMP T151B	
2630	4472		JMS I NERROR	/CHECK NON-ERROR HANDLER
2631	4473	T151A,	JMS I ERROR	/ERROR: OVERFLOW OR MODE FAILED
2632	1551		1551	/TST151 ERROR MESSAGE
2633	2606		TST151	/SCOPE LOOP
			/DOES OVERFLOW CAUSE ROST, ?	
			/RATE 2-6, MODE 3	
			/	
2634	1131	TST152,	TAD K7773	
2635	3041		DCA REG	
2636	1141		TAD K7000	
2637	1142		TAD K0010	
2640	1015		TAD K0200	/MAKE ENABLES
2641	3044	T1528,	DCA REG	
2642	7340		CLA CLL CMA	/AC TO 7777
2643	4427		JMS I XIOTG	/IOT 6133, CLAB
2644	3040		DCA REG	
2645	1044		TAD REG	/GET ENABLES
2646	4425		JMS I XIOTF	/IOT 6132, CLDE
2647	4447		JMS I XPI001	/GO TO PI, PI EXPECTED
2650	5257		JMP T152A	
2651	6007		6007	/CAF OR CLEAR THE WORLD
2652	1013		TAD K0100	
2653	1044		TAD REG	
2654	2041		IS2 REG	
2655	5241		JMP T152B	/DO RATES 2-6
2656	4472		JMS I NERROR	/CHECK NON-ERROR HANDLER
2657	4473	T152A,	JMS I ERROR	/ERROR: OVERFLOW OR MODE FAILED
2660	1552		1552	/TST152 ERROR MESSAGE
2661	2634		TST152	/SCOPE LOOP
			/DOES OVERFLOW CAUSE ROST, ?	
			/RATE 0-7, MODE 1, DISABLE BIT 7	
			/	
			/	
2662	1142	TST153,	TAD K7770	
2663	3041		DCA REG	
2664	1141		TAD K5000	
2665	1142		TAD K0010	
2666	1140		TAD K0020	

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/
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2667 3044 T153B, DCA REGE /MAKE ENABLES
2670 7340 CLA CLL CMA /AC TO 7777
2671 4427 JMS I XIOTG /IOT 6133, CLAB
2672 3040 DCA REGA
2673 1044 TAD REGE /GET ENABLES
2674 4425 JMS I XIOTF /IOT 6132, CLOE
2675 4450 JMS I XPIG02 /GO TO PI, NO PI EXPECTED
2676 5305 JMP T153A /CAF OR CLEAR THE WORLD
2677 6027 6007
2700 1013 TAD K0100
2701 1044 TAD REGE
2702 2041 IS2 REG8
2703 5267 JMP T153B /DO RATE 0-7
2704 4472 JMS I NERROR /CHECK NON-ERROR HANDLER
2705 4473 JMS I ERROR /ERROR: OVERFLOW OR CLK ENA FAILED
2706 1153 1153 /TST153 ERROR MESSAGE
2707 2662 TST153 /SCOPE LOOP

/DOES OVERFLOW CAUSE RQST. ?
/RATE 0-7, MODE 2, DISABLE INT. RQST. BIT
/

TST154, TAD K7770
DCA REG8
TAD K0000
TAD K0010
TAD K0020
DCA REGE /MAKE ENABLES
CLA CLL CMA /AC TO 7777
JMS I XIOTG /IOT 6133, CLAB
DCA REGA
TAD REGE /GET ENABLES
JMS I XIOTF /IOT 6132, CLOE
JMS I XPIG02 /GO TO PI, NO PI EXPECTED
JMP T154A /CAF OR CLEAR THE WORLD
6007
TAD K0100
TAD REGE
IS2 REG8
JMP T154B /DO RATE 0-7
JMS I NERROR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: OVERFLOW OR CLK ENA FAILED
1154 1154 /TST154 ERROR MESSAGE
TST154 /SCOPE LOOP

/DOES OVERFLOW CAUSE INT. RQST. ?
/MODE 0, RATE 6
/

TST155, CLA CLL CMA /AC TO 7777
JMS I XIOTG /IOT 6133, CLAB
CLA CLL CML RAR /AC TO 4000
TAD K0600
TAD K0010 /GET ENABLES
JMS I XIOTF /IOT 6132, CLOE
JMS I XPIG04 /GO TO PI, PI EXPECTED
JMS I NERROR /CHECK NON-ERROR HANDLER
2736 7340
2737 4427
2740 7330
2741 1147
2742 1142
2743 4425
2744 4422
2745 4472

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2746 4473 JMS I ERROR /ERROR: OVERFLOW OR COUNTER FAILED
2747 1555 1555 /TST155 ERROR MESSAGE
2750 2736 TST155 /SCOPE LOOP

/DOES CLSK SKIP THEN INTERRUPT ?
/RATE 6, MODE 0

TST156, CLA CLL CMA /AC TO 7777
JMS I XIOTG /IOT 6133, CLAB
CLA CLL CML RAR
TAD K0010
TAD K0600 /MAKE ENABLES
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP -1 /WAIT FOR OVERFLOW
JMS I XPIG04 /GO TO PI, PI EXPECTED
JMS I NERRR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: CLSK ON PI FAILED
1556 /TST156 ERROR MESSAGE
TST156 /SCOPE LOOP

/CHECK FOR NO INT, ROST.
/MODE 0, RATE 6, DISABLE WITH CLSA

TST157, CLA CLL CMA /AC TO 7777
JMS I XIOTG /IOT 6133, CLAB
CLA CLL CML RAR /AC TO 4000
TAD K0600
TAD K0010
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP -1 /WAIT FOR OVERFLOW
JMS I XIOTI /IOT 6135, CLSA
JMS I XPIG03 /GO TO PI, NO PI EXPECTED
JMS I NERRR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: INT, ROST, FAILED
1157 /TST157 ERROR MESSAGE
TST157 /SCOPE LOOP

/DOES CLOCK FREQUENCY TIME OUT ?
/RATE 2, MODE 0

TST160, CLA CLL CMA /AC TO 7777
DCA REGA
TAD KTA
DCA KREGC
JMS I XIOTG /IOT 6133, CLAB
TAD K4000
TAD K0010
TAD K0200 /MAKE ENABLES
JMS I XIOTF /IOT 6132, CLOE
JMS I XPIG05
SKP CLA
JMS I NERRR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: CLOCK FREQUENCY FAST

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PAL10	V141	9-OCT-71	15144	PAGE 1-36
3021	2160	2160		/TST160 ERROR MESSAGE
3022	3024	TST160		/SCOPE LOOP
/				
/DOES CLOCK FREQUENCY TIME OUT ?				
/RATE 2, MODE 0				
/				
3023	7340	TST161, CLA CLL CMA	/AC TO 7777	
3024	3040	DCA REGA		
3025	1152	TAD KTA1		
3026	3076	DCA KREGC		
3027	4427	JMS I XIOTG	/IOT 6133, CLAB	
3030	1014	TAD K4000		
3031	1142	TAD K0010		
3032	1015	TAD K0200		
3033	4425	JMS I XIOTF	/MAKE ENABLES	
3034	4453	JMS I XPIG05	/IOT 6132, CLOE	
3035	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
3036	4473	JMS I ERROR	/ERROR: CLOCK FREQUENCY SLOW	
3037	2561	2561	/TST161 ERROR MESSAGE	
3040	3023	TST161	/SCOPE LOOP	
/				
/DOES CLOCK FREQUENCY TIME OUT ?				
/RATE 3, MODE 0				
/				
3041	7340	TST162, CLA CLL CMA	/AC TO 7777	
3042	3040	DCA REGA		
3043	1153	TAD KTB		
3044	3076	DCA KREGC		
3045	4427	JMS I XIOTG	/IOT 6133, CLAB	
3046	1014	TAD K4000		
3047	1142	TAD K0010		
3050	1145	TAD K0300	/MAKE ENABLES	
3051	4425	JMS I XIOTF	/IOT 6132, CLOE	
3052	4453	JMS I XPIG05		
3053	7610	SKP CLA		
3054	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
3055	4473	JMS I ERROR	/ERROR: CLOCK FREQUENCY FAST	
3056	2162	2162	/TST162 ERROR MESSAGE	
3057	3041	TST162	/SCOPE LOOP	
/				
/DOES CLOCK FREQUENCY TIME OUT ?				
/RATE 3, MODE 0				
/				
3060	7340	TST163, CLA CLL CMA	/AC TO 7777	
3061	3040	DCA REGA		
3062	1154	TAD KTB1		
3063	3076	DCA KREGC		
3064	4427	JMS I XIOTG	/IOT 6133, CLAB	
3065	1014	TAD K4000		
3066	1142	TAD K0010		
3067	1145	TAD K0300	/MAKE ENABLES	
3070	4425	JMS I XIOTF	/IOT 6132, CLOE	
3071	4453	JMS I XPIG05		
3072	4472	JMS I NERROR	/CHECK NON-ERROR HANDLER	
3073	4473	JMS I ERROR	/ERROR: CLOCK FREQUENCY SLOW	

PAL10	V141	9-OCT-71	15144	PAGE 1-37
3074	2563	2563		/TST163 ERROR MESSAGE
3075	3060	TST163		/SCOPE LOOP
		/		/DOES CLOCK FREQUENCY TIME OUT ?
		/RATE 4, MODE 0		/
3076	7340	TST164,	CLA CLL CMA	/AC TO 7777
3077	3040		DCA REGA	
3100	1155		TAD KTC	
3101	3076		DCA KREGC	
3102	1156		TAD KTC1	
3103	3043		DCA REGD	
3104	4427		JMS I XIOTG	/SET TIMER FOR 10000 CPS CLOCK
3105	1014		TAD K4000	/IOT 6133, CLAB
3106	1142		TAD K0010	
3107	1116		TAD K0400	/MAKE ENABLES
3110	4425		JMS I XIOTF	/IOT 6132, CLOE
3111	4453		JMS I XPIG05	
3112	7610		SKP CLA	/CHECK NON-ERROR HANDLER
3113	4472		JMS I NERROR	/ERROR: CLOCK FREQUENCY FAST
3114	4473		JMS I ERROR	/TST164 ERROR MESSAGE
3115	2164		2164	/SCOPE LOOP
3116	3076		TST164	
		/		/DOES CLOCK FREQUENCY TIME OUT ?
		/RATE 4, MODE 0		/
3117	7340	TST165,	CLA CLL CMA	/AC TO 7777
3120	3040		DCA REGA	
3121	1155		TAD KTC	
3122	3076		DCA KREGC	
3123	1157		TAD KTC2	
3124	3043		DCA REGD	
3125	4427		JMS I XIOTG	/SET TIMER FOR 10000 CLOCK
3126	1014		TAD K4000	/IOT 6133, CLAB
3127	1142		TAD K0010	
3130	1116		TAD K0400	/MAKE ENABLES
3131	4425		JMS I XIOTF	/IOT 6132, CLOE
3132	4493		JMS I XPIG05	
3133	4472		JMS I NERROR	/CHECK NON-ERROR HANDLER
3134	4473		JMS I ERROR	/ERROR: CLOCK FREQUENCY SLOW
3135	2565		2565	/TST165 ERROR MESSAGE
3136	3117		TST165	/SCOPE LOOP
		/		/DOES CLOCK FREQUENCY TIME OUT ?
		/RATE 5, MODE 0		/
3137	7340	TST166,	CLA CLL CMA	/AC TO 7777
3140	3040		DCA REGA	
3141	7350		CLA CLL CMA RAR	
3142	4427		JMS I XIOTG	/IOT 6133, CLAB
3143	7300		CLA CLL	/CLEAR THE AC AND LINK
3144	1160		TAD KTD	
3145	3043		DCA REGD	/SET TIMER FOR 10000 CPS CLOCK
3146	1014		TAD K4000	

PAL10	V141	9-OCT-71	15:44	PAGE 1-38
3147	1142			
3150	1146			
3151	4425			/MAKE ENABLES
3152	4447			/IOT 6132, CLOE
3153	4472			/CHECK NON-ERROR HANDLER
3154	4473			/ERROR: CLOCK FREQUENCY FAST
3155	2166			/TST166 ERROR MESSAGE
3156	3137			/SCOPE LOOP
				/DOES CLOCK FREQUENCY TIME OUT ?
				/RATE 5, MODE 0
				/
3157	7340			TST167, CLA CLL CMA /AC TO 7777
3160	3040			DCA REGA
3161	7350			CLA CLL CMA RAR
3162	4427			JMS I XIOTG
3163	7300			CLA CLL
3164	1161			TAD KTD1
3165	3043			DCA REGD
3166	1014			TAD K4000
3167	1142			TAD K0010
3170	1146			TAD K0500
3171	4425			JMS I XIOTF
3172	4450			JMS I XPIG02
3173	4472			JMS I NERRR
3174	4473			JMS I ERROR
3175	2567			2567
3176	3157			TST167
				/DOES CLOCK FREQUENCY TIME OUT ?
				/RATE 6, MODE 0
				/
3177	7340			TST170, CLA CLL CMA /AC TO 7777
3200	3040			DCA REGA
3201	1162			TAD KTE
3202	3043			DCA REGD
3203	4427			JMS I XIOTG
3204	1014			TAD K4000
3205	1142			TAD K0010
3206	1147			TAD K0600
3207	4425			JMS I XIOTF
3210	4447			JMS I XPIG01
3211	4472			JMS I NERRR
3212	4473			JMS I ERROR
3213	2170			2170
3214	3177			TST170
				/DOES CLOCK FREQUENCY TIME OUT ?
				/RATE 6, MODE 0
				/
3215	7340			TST171, CLA CLL CMA /AC TO 7777
3216	3040			DCA REGA
3217	1163			TAD KTE1
3220	3043			DCA REGD
3221	4427			JMS I XIOTG
				/SET TIMER FOR 1000000 CPS CLOCK
				/IOT 6133, CLAB

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3222 1014      TAD K4000
3223 1142      TAD K0010
3224 1147      TAD K0600      /MAKE ENABLES
3225 4425      JMS I XIOTF      /IOT 6132, CLOE
3226 4450      JMS I XPIG02
3227 4472      JMS I NERR0R
3230 4473      JMS I ERROR
3231 2571      2571
3232 3215      TST171
/
/DOES COUNTER REALLY COUNT ?
/RATE 2, MODE 0
/
TST172, CLA CLL CMA      /AC TO 7777
JMS I XIOTG      /IOT 6133, CLAB
DCA REGA
TAD K0200      /GET RATE + MODE
JMS I XIOTF1      /IOT 6132, CLOE
CLA CLL      /CLEAR THE AC AND LINK
DCA REGC
TAD REGB
DCA SEND
JMS I XIOTK
CIA
TAD REGB
SNA CLA
JMP T172A
ISZ REGC
JMP T172B
JMP T172A1
ISZ REGB
JMP T172B1
JMS I NERR0R
JMS I ERROR
T172A1, JMS I ERROR
4172
TST172
/
/DOES COUNTER REALLY COUNT ?
/RATE 3, MODE 0
/
TST173, CLA CLL CMA      /AC TO 7777
JMS I XIOTG      /IOT 6133, CLAB
DCA REGA
TAD K0300      /GET RATE + MODE
JMS I XIOTF1      /IOT 6132, CLOE
CLA CLL      /CLEAR THE AC AND LINK
DCA REGC
TAD REGB
DCA SEND
JMS I XIOTK
CIA
TAD REGB
SNA CLA
JMP T173A
ISZ REGC
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3301      JMP T173B
3302      JMP T173A1
3303      ISZ REGB
3304      JMP T173B1
3305      JMS I NERRR
3306      JMS I ERROR
3307      4173
3310      TST173
          /DOES COUNTER REALLY COUNT ?
          /RATE 2, MODE 1
          /
TST174,   CLA CLL CMA
          JMS I XIOTG
          DCA REGA
          TAD K0200
          TAD K1020
          JMS I XIOTF1
          JMS I XIOTE
          JMP -1
          CLA CLL
          JMS I XIOTG
          DCA REGC
          TAD REGB
          T174B1, DCA SEND
          DCA SEND
          JMS I XIOTK
          CIA
          TAD REGB
          SNA CLA
          JMP T174A
          ISZ REGC
          JMP T174B
          JMP T174A1
          ISZ REGB
          T174A,   JMS I NERRR
          JMS I ERROR
          4174
          TST174
          /DOES COUNTER REALLY COUNT ?
          /RATE 4, MODE 1
          /
TST175,   CLA CLL CMA
          JMS I XIOTG
          DCA REGA
          TAD K0400
          TAD K1000
          JMS I XIOTF1
          JMS I XIOTE
          JMP -1
          CLA CLL
          JMS I XIOTG
          DCA REGC
          T175B1, DCA REGC
          TAD REGB
3311      7340
3312      4427
3313      3040
3314      1015
3315      1144
3316      4426
3317      4424
3320      5317
3321      7300
3322      4427
3323      3042
3324      1041
3325      3070
3326      4433
3327      7041
3330      1041
3331      7650
3332      5336
3333      2042
3334      5326
3335      5341
3336      2041
3337      5323
3340      4472
3341      4473
3342      4174
3343      3311

3344      7340
3345      4427
3346      3040
3347      1116
3350      1144
3351      4426
3352      4424
3353      5352
3354      7300
3355      4427
3356      3042
3357      1041

JMP T173B      /WAIT ABOUT 15 MS FOR REGISTER
JMP T173A1     /NUMBER NOT FOUND
ISZ REGB       /UPDATE COMPARE REGISTER
JMP T173B1     /TEST FOR NEXT COUNTER PULSE
JMS I NERRR    /CHECK NON-ERROR HANDLER
JMS I ERROR    /ERROR: COUNTER FAILED
4173           /TST173 ERROR MESSAGE
TST173        /SCOPE LOOP

/DOES COUNTER REALLY COUNT ?
/RATE 2, MODE 1
/
TST174,   /AC TO 7777
          /IOT 6133, CLAB
          DCA REGA
          TAD K0200
          TAD K1020
          JMS I XIOTF1
          JMS I XIOTE
          JMP -1
          CLA CLL
          JMS I XIOTG
          DCA REGC
          TAD REGB
          T174B1, /GET RATE + MODE
          DCA SEND /IOT 6132, CLOE
          JMS I XIOTK /IOT 6131, CLSK
          CIA
          TAD REGB
          SNA CLA
          JMP T174A
          ISZ REGC
          JMP T174B
          JMP T174A1
          ISZ REGB
          T174A,   /CLEAR THE AC AND LINK
          JMS I NERRR /IOT 6133, CLAB
          JMS I ERROR /IOT 6133, CLAB
          4174
          TST174 /SAVE OUTPUT FOR ERROR PRINTER
          /IOT 6137, CLCA
          /COMPARE TO THIS REGISTER
          /ARE THEY THE SAME YET ?
          /YES, TEST NEXT NUMBER
          /WAIT ABOUT 15 MS FOR REGISTER
          /NUMBER NOT FOUND
          /UPDATE COMPARE REGISTER
          /TEST FOR NEXT COUNTER PULSE
          /CHECK NON-ERROR HANDLER
          /ERROR: COUNTER FAILED
          /TST174 ERROR MESSAGE
          /SCOPE LOOP

/DOES COUNTER REALLY COUNT ?
/RATE 4, MODE 1
/
TST175,   /AC TO 7777
          /IOT 6133, CLAB
          DCA REGA
          TAD K0400
          TAD K1000
          JMS I XIOTF1
          JMS I XIOTE
          JMP -1
          CLA CLL
          JMS I XIOTG
          DCA REGC
          T175B1, /GET RATE + MODE
          DCA REGC /IOT 6132, CLOE
          TAD REGB /IOT 6131, CLSK
          TAD REGB /CLEAR THE AC AND LINK
          TAD REGB /IOT 6133, CLAB
          TAD REGB
3344      7340
3345      4427
3346      3040
3347      1116
3350      1144
3351      4426
3352      4424
3353      5352
3354      7300
3355      4427
3356      3042
3357      1041

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3360	3070	DCA SEND		/SAVE OUTPUT FOR ERROR PRINTER
3361	4433	JMS I XIOTK		/IOT 6137, CLCA
3362	7041	CIA		
3363	1041	TAD REG8		/COMPARE TO THIS REGISTER
3364	7650	SNA CLA		/ARE THEY THE SAME YET ?
3365	5371	JMP T175A		/YES, TEST NEXT NUMBER
3366	2042	IS2 REGC		
3367	5361	JMP T175B		/WAIT ABOUT 15 MS FOR REGISTER
3370	5374	JMP T175A1		/NUMBER NOT FOUND
3371	2041	IS2 REG8		/UPDATE COMPARE REGISTER
3372	5356	JMP T175B1		/TEST FOR NEXT COUNTER PULSE
3373	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
3374	4473	JMS I ERROR		/ERROR: COUNTER FAILED
3375	4175	4175		/TST175 ERROR MESSAGE
3376	3344	TST175		/SCOPE LOOP
		/		
		/DOES COUNTER REALLY COUNT ?		
		/RATE 2, MODE 2		
		/		
3377	7340	TST176, CLA CLL CMA		/AC TO 7777
3400	4427	JMS I XIOTG		/IOT 6133, CLAB
3401	3040	DCA REGA		
3402	1015	TAD K0200		/GET RATE + MODE
3403	1143	TAD K2000		/IOT 6132, CLOE
3404	4426	JMS I XIOTF1		/CLEAR THE AC AND LINK
3405	7300	CLA CLL		
3406	3042	DCA REGC		
3407	1041	TAD REG8		
3410	3070	DCA SEND		/SAVE OUTPUT FOR ERROR PRINTER
3411	4433	JMS I XIOTK		/IOT 6137, CLCA
3412	7041	CIA		
3413	1041	TAD REG8		/COMPARE TO THIS REGISTER
3414	7650	SNA CLA		/ARE THEY THE SAME YET ?
3415	5221	JMP T176A		/YES, TEST NEXT NUMBER
3416	2042	IS2 REGC		
3417	5211	JMP T176B		/WAIT ABOUT 15 MS FOR REGISTER
3420	5224	JMP T176A1		/NUMBER NOT FOUND
3421	2041	IS2 REG8		/UPDATE COMPARE REGISTER
3422	5205	JMP T176B1		/TEST FOR NEXT COUNTER PULSE
3423	4472	JMS I NERROR		/CHECK NON-ERROR HANDLER
3424	4473	JMS I ERROR		/ERROR: COUNTER FAILED
3425	4176	4176		/TST176 ERROR MESSAGE
3426	3377	TST176		/SCOPE LOOP
		/		
		/DOES COUNTER REALLY COUNT ?		
		/RATE 4, MODE 2		
		/		
3427	7340	TST177, CLA CLL CMA		/AC TO 7777
3430	4427	JMS I XIOTG		/IOT 6133, CLAB
3431	3040	DCA REGA		
3432	1116	TAD K0400		/GET RATE + MODE
3433	1143	TAD K2000		/IOT 6132, CLOE
3434	4426	JMS I XIOTF1		/CLEAR THE AC AND LINK
3435	7300	CLA CLL		
3436	3042	DCA REGC		

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3437      1041      TAD REG8
3440      3070      DCA SEND
3441      4433      JMS I XIOTK
3442      7041      CIA
3443      1041      TAD REG8
3444      7650      SNA CLA
3445      5251      JMP T177A
3446      2042      ISZ REGC
3447      5241      JMP T177B
3448      5254      JMP T177A1
3451      2041      ISZ REG8
3452      5255      JMP T177B1
3453      4472      JMS I NERROR
3454      4473      JMS I ERROR
3455      4177      4177
3456      3427      TST177
/
/DOES COUNTER REALLY COUNT ?
/RATE 4, MODE 3
/
TST200, CLA CLL CMA
3457      7340      JMS I XIOTG
3460      4427      DCA REGA
3461      3040      TAD K2420
3462      1116      TAD K3000
3463      1120      JMS I XIOTF1
3464      4426      CLA CLL
3465      7300      DCA REGC
3466      3042      TAD REG8
3467      1041      DCA SEND
3470      3070      JMS I XIOTK
3471      4433      CIA
3472      7041      TAD REG8
3473      1041      SNA CLA
3474      7650      JMP T200A
3475      5301      ISZ REGC
3476      2042      JMP T200B
3477      5271      JMP T200A1
3500      5304      ISZ REG8
3501      2041      JMP T200B1
3502      5265      JMS I NERROR
3503      4472      JMS I ERROR
3504      4473      4200
3505      4200      TST200
3506      3457
/
/DO IOT'S AFFECT AC ?
/
TST201, CLA CLL CMA
3507      7340      JMS I XIOTG
3510      4427      DCA REGA
3511      3040      6007
3512      6007      TAD K1000
3513      1144      TAD K0200
3514      1015      JMS I XIOTF1
3515      4426      JMS I XIOTE
3516      4424
T177B,
/SAVE OUTPUT FOR ERROR PRINTER
/IOT 6137, CLCA
/COMPARE TO THIS REGISTER
/ARE THEY THE SAME YET ?
/YES, TEST NEXT NUMBER
/WAIT ABOUT 15 MS FOR REGISTER
/NUMBER NOT FOUND
/UPDATE COMPARE REGISTER
/TEST FOR NEXT COUNTER PULSE
/CHECK NON-ERROR HANDLER
/ERROR: COUNTER FAILED
/TST177 ERROR MESSAGE
/SCOPE LOOP

T177A,
/AC TO 7777
/IOT 6133, CLAH
/GET RATE + MODE
/IOT 6132, CLCE
/CLEAR THE AC AND LINK
/SAVE OUTPUT FOR ERROR PRINTER
/IOT 6137, CLCA
/COMPARE TO THIS REGISTER
/ARE THEY THE SAME YET ?
/YES, TEST NEXT NUMBER
/WAIT ABOUT 15 MS FOR REGISTER
/NUMBER NOT FOUND
/UPDATE COMPARE REGISTER
/TEST FOR NEXT COUNTER PULSE
/CHECK NON-ERROR HANDLER
/ERROR: MODE 3, COUNTER FAILED
/TST200 ERROR MESSAGE
/SCOPE LOOP

T200A,
/AC TO 7777
/IOT 6133, CLAB
/PASS COUNT 1
/CAF OR CLEAR THE WORLD
/GET ENABLES
/IOT 6132, CLCE
/IOT 6131, CLSK

T200B1,
/AC TO 7777
/IOT 6133, CLAB
/PASS COUNT 1
/CAF OR CLEAR THE WORLD
/GET ENABLES
/IOT 6132, CLCE
/IOT 6131, CLSK

T200B,
/AC TO 7777
/IOT 6133, CLAB
/PASS COUNT 1
/CAF OR CLEAR THE WORLD
/GET ENABLES
/IOT 6132, CLCE
/IOT 6131, CLSK

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3517 5316 JMP .-1
3520 7340 CLA CLL CMA
3521 4423 JMS I XIOTD
3522 7320 CLA CLL
3523 3070 DCA SEND
3524 1041 TAD REGR
3525 4432 JMS I XIOTJ
3526 7640 SZA CLA
3527 5351 JMP T201A
3530 1041 TAD REGR
3531 4433 JMS I XIOTK
3532 7640 SZA CLA
3533 5351 JMP T201A
3534 1041 TAD REGR
3535 4430 JMS I XIOTH
3536 7640 SZA CLA
3537 5351 JMP T201A
3540 1041 TAD REGR
3541 4431 JMS I XIOTI
3542 7640 SZA CLA
3543 5351 JMP T201A
3544 4424 JMS I XIOTE
3545 5344 JMP .-1
3546 2041 ISZ REGR
3547 5322 JMP T201R
3550 4472 JMS I NERRR
3551 4473 JMS I ERROR
3552 3201 3201
3553 3527 TST201

3554 4570 JMS I XPASS
3555 5463 JMP I XDK8EP

/DOES INPUT 4 CAUSE INT. ROST.
/
/
/DOES INPUT 4 CAUSE INT. ROST.
/

3556 7320 CLA CLL
3557 1112 TAD K7400
3560 3077 DCA LOOP
3561 7340 CLA CLL CMA
3562 3040 DCA REGA
3563 7307 CLA CLL IAC RTL
3564 1142 TAD K0010
3565 4425 JMS I XIOTF
3566 4450 JMS I XPIG02
3567 4472 JMS I NERRR
3570 4473 JMS I ERROR
3571 1602 1602
3572 3561 TST202

/DOES INPUT 2 CAUSE INT. ROST.
/
/DOES INPUT 2 CAUSE INT. ROST.
/

3573 7340 CLA CLL CMA
3574 3040 DCA REGA
3575 7326 CLA CLL CML RTL
3576 1142 TAD K0010

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3577 4425 JMS I XIOTF /IOT 6132, CLOE
3600 4450 JMS I XPIG02 /GO TO PI, PI EXPECTED
3601 4472 JMS I NERROR /CHECK NON-ERROR HANDLER
3602 4473 JMS I ERROR /ERROR: INPUT 2 FAILED
3603 1603 /TST203 ERROR MESSAGE
3604 3573 TST203 /SCOPE LOOP

/DOES INPUT 1 CAUSE INT. ROST.
/DOES INPUT 4 RQST. LAST ?

TST204, CLA CLL CMA /AC TO 7777
DCA REGA
CLA CLL CML RAL /AC TO 0001
TAD K0010 /GET ENABLES
JMS I XIOTF /IOT 6132, CLOE
JMS I XPIG02 /GO TO PI, PI EXPECTED
JMS I NERROR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: INPUT 1 FAILED
1604 /TST204 ERROR MESSAGE
TST204 /SCOPE LOOP

/DOES INPUT 4 RQST. LAST ?
/DOES INPUT 2 RQST. LAST ?

TST205, CLA CLL CMA /AC TO 7777
DCA REGA
CLA CLL IAC RTL /AC TO 0004
TAD K0010 /GET ENABLES
JMS I XIOTF /IOT 6132, CLOE
JMS I XPIG01 /GO TO PI, PI EXPECTED
JMP T205A /NO RQST. FOUND
ISZ REG8 /UPDATE COUNTER
JMP -1 /WAIT 15 MS
JMS I XPIG02 /GO TO PI, PI EXPECTED
JMS I NERROR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: INPUT 4 FAILED
T205A, 1605 /TST205 ERROR MESSAGE
1605 /SCOPE LOOP
TST205

/DOES INPUT 2 RQST. LAST ?
/DOES INPUT 1 RQST. LAST ?

TST206, CLA CLL CMA /AC TO 7777
DCA REGA
CLA CLL IAC RAL /AC TO 0002
TAD K0010 /GET ENABLES
JMS I XIOTF /IOT 6132, CLOE
JMS I XPIG01 /GO TO PI, PI EXPECTED
JMP T206A /NO RQST. FOUND
ISZ REG8 /UPDATE COUNTER
JMP -1 /WAIT 15 MS
JMS I XPIG02 /GO TO PI, PI EXPECTED
JMS I NERROR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: INPUT 2 FAILED
T206A, 1606 /TST206 ERROR MESSAGE
1606 /SCOPE LOOP
TST206

/DOES INPUT 1 RQST. LAST ?
/DOES INPUT 2 RQST. LAST ?

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3653 7340 /
3654 3040 /TST207, CLA CLL CMA /AC TO 7777
3655 7324 DCA REGA
3656 1142 CLA CLL CML RAL /AC TO 0001
3657 4425 TAD K0010 /GET ENABLES
3660 4447 JMS I XIOTF /IOT 6132, CLOE
3661 5266 JMS I XPIG01 /GO TO PI, PI EXPECTED
3662 2041 JMP T207A /NO RQST. FOUND
3663 5262 IS2 REG8 /UPDATE COUNTER
3664 4450 JMP .-1 /WAIT 15 MS
3665 4472 JMS I XPIG02 /GO TO PI, PI EXPECTED
3666 4473 JMS I NERR0R /CHECK NON-ERROR HANDLER
3667 1627 JMS I ERROR /ERROR: INPUT 1 FAILED
3670 3653 TST207 /TST207 ERROR MESSAGE
/SCOPE LOOP

/DOES INPUTS 4,2,1 WITHOUT BIT 8 ?
/
TST210, CLA CLL CMA /AC TO 7777
3671 7340 DCA REGA
3672 3040 CLA CLL IAC RTR /AC TO 4000
3673 7313 TAD K0007
3674 1027 TAD K0600
3675 1147 JMS I XIOTF /IOT 6132, CLOE
3676 4425 JMS I XPIG01 /GO TO PI, NO PI EXPECTED
3677 4447 JMS I NERR0R /CHECK NON-ERROR HANDLER
3700 4472 JMS I ERROR /ERROR:ENABLE BIT 8 FAILED
3701 4473 JMS I ERROR /TST210 ERROR MESSAGE
3702 1210 1212 /SCOPE LOOP
3703 3671 TST210

/DOES INPUT 4 CAUSE SKIP ?
/
TST211, CLA CLL CMA /AC TO 7777
3704 7340 DCA REGA
3705 3040 TAD KTICPS
3706 1113 DCA REGF
3707 3045 CLA CLL IAC RTL /AC TO 0004
3710 7327 JMS I XIOTF /IOT 6132, CLOE
3711 4425 JMS I XIOTE /IOT 6131, CLSK
3712 4424 JMS I SKPWAT /LET'S WAIT FOR A FLAG
3713 4446 JMS I NERR0R /CHECK NON-ERROR HANDLER
3714 4472 JMS I ERROR /ERROR: INPUT 4 OR SKIP FAILED
3715 4473 JMS I ERROR /TST211 ERROR MESSAGE
3716 0611 0611 /SCOPE LOOP
3717 3724 TST211

/DOES INPUT 2 CAUSE SKIP ?
/
TST212, CLA CLL CMA /AC TO 7777
3720 7340 DCA REGA
3721 3040 TAD KTICPS
3722 1113 DCA REGF
3723 3045 CLA CLL CML RTL /AC TO 0002
3724 7326 JMS I XIOTF /IOT 6132, CLOE
3725 4425 JMS I XIOTE /IOT 6131, CLSK
3726 4424 JMS I SKPWAT /LET'S WAIT FOR A FLAG
3727 4446
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3730      4472      JMS I NERROR      /CHECK NON-ERROR HANDLER
3731      4473      JMS I ERROR      /ERROR: INPUT 2 OR SKIP FAILED
3732      0612      0612      /TST212 ERROR MESSAGE
3733      3720      TST212      /SCOPE LOOP

/DOES INPUT 1 CAUSE SKIP ?
/
TST213,    CLA CLL CMA      /AC TO 7777
          DCA REGA
          TAD KTI0PS
          DCA REGF
          CLA CLL IAC      /AC TO 0001
          JMS I XIOTF      /IOT 6132, CLOE
          JMS I XIOTE      /IOT 6131, CLSK
          JMS I SKPMAT      /LET'S WAIT FOR FLAG
          JMS I NERROR      /CHECK NON-ERROR HANDLER
          JMS I ERROR      /ERROR: INPUT 1 OR SKIP FAILED
          0613      /TST213 ERROR MESSAGE
          TST213      /SCOPE LOOP

/DOES INPUT 4 ROST. THEN SKIP AND VICE-VERSA ?
/
TST214,    CLA CLL CMA      /AC TO 7777
          DCA REGA
          CLA CLL IAC RTL /AC TO 0004
          TAD K0010      /GET ENABLES
          JMS I XIOTF      /IOT 6132, CLOE
          JMS I XIOTE      /IOT 6131, CLSK
          JMP .-1
          JMS I XPIG01      /GO TO PI, PI EXPECTED
          JMP T214A      /NO ROST, FOUND
          JMS I XIOTE      /IOT 6131, CLSK
          JMP .-1
          JMS I NERROR      /CHECK NON-ERROR HANDLER
          JMS I ERROR      /ERROR: INPUT 4 SKIP OR INT, ROST, FAILED
          1614      /TST214 ERROR FAILED
          TST214      /SCOPE LOOP

T214A,      JMS I NERROR
          JMS I ERROR
          1614
          TST214

/DOES INPUT 2 SKIP THEN INT, ROST, AND VICE-VERSA ?
/
TST215,    CLA CLL CMA      /AC TO 7777
          DCA REGA
          CLA CLL IAC RAL /AC TO 0002
          TAD K0010      /GET ENABLES
          JMS I XIOTF      /IOT 6132, CLOE
          JMS I XIOTE      /IOT 6131, CLSK
          JMP .-1
          JMS I XPIG01      /GO TO PI, PI EXPECTED
          JMP I XCRS5
          JMS I XIOTE      /IOT 6131, CLSK
          JMP .-1
          JMS I NERROR      /CHECK NON-ERROR HANDLER
          JMS I ERROR      /ERROR: INPUT 2 SKIP OR ROST, FAILED
          1615      /TST215 ERROR MESSAGE
          TST215      /SCOPE LOOP

T215A,      JMS I NERROR
          JMS I ERROR
          1615
          TST215
3767      7340
3770      3040
3771      7305
3772      1142
3773      4425
3774      4424
3775      5374
3776      4447
3777      5575
4000      4424
4001      5200
4002      4472
4003      4473
4004      1615
4005      3767

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/
/DOES INPUT 1 SKIP THEN INT, ROST, AND VICE-VERSA ?
/
TST216, CLA CLL CMA /AC TO 7777
DCA REGA
CLA CLL IAC /AC TO 0001
TAD K0010 /GET ENABLES
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP .-1
JMS I XPIG01 /GO TO PI, PI EXPECTED
JMP T216A
JMS I XIOTE /IOT 6131, CLSK
JMP .-1
JMS I NERROR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: INPUT 1 SKIP OR INT, ROST, FAILED
T216A, 1616 /TST216 ERROR MESSAGE
TST216 /SCOPE LOOP

/DOES CAF CLEAR INPUT 4 INT, ROST, ?
/
TST217, CLA CLL CMA /AC TO 7777
DCA REGA
CLA CLL IAC RTL /AC TO 0004
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP .-1 /WAIT FOR FIRST FLAG
6007 /CAF OR CLEAR THE WORLD
CLA CLL IAC RTL /AC TO 0004
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP .-1 /WAIT FOR SECOND FLAG
6007 /CAF OR CLEAR THE WORLD
CLA CLL IAC RTL
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMS I NERROR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: INPUT 4 SKIP OR ROST, FAILED
0217 /TST217 ERROR MESSAGE
TST217 /SCOPE LOOP

/DOES CAF CLEAR INPUT 2 ROST, ?
/
TST220, CLA CLL CMA /AC TO 7777
DCA REGA
CLA CLL IAC RAL /AC TO 0002
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP .-1 /WAIT FOR FIRST FLAG
6007 /CAF OR CLEAR THE WORLD
CLA CLL IAC RAL /AC TO 0002
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP .-1 /WAIT FOR SECOND FLAG
6007 /CAF OR CLEAR THE WORLD
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/
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4064 7305 CLA CLL IAC RAL /AC TO 0002
4065 4425 JMS I XIOTF /IOT 6132, CLOE
4066 4424 JMS I XIOTE /IOT 6131, CLSK
4067 4472 JMS I NERRR /CHECK NON-ERROR HANDLER
4070 4473 JMS I ERROR /ERROR: INPUT 2 SKIP OR ROST, FAILED
4071 0220 0220 /TST220 ERROR MESSAGE
4072 4050 TST220 /SCOPE LOOP

/DOES CAF CLEAR INTPUT 3 ROST. ?
/
TST221, CLA CLL CMA /AC TO 7777
DCA REGA
CLA CLL IAC /AC TO 0001
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP --1 /WAIT FOR FIRST FLAG
6007 /CAF OR CLEAR THE WORLD
CLA CLL IAC /AC TO 0001
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP --1 /WAIT FOR SECONED FLAG
6007 /CAF OR CLEAR THE WORLD
CLA CLL IAC /IOT 6132, CLOE
JMS I XIOTF /IOT 6131, CLSK
JMS I NERRR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: INPUT 1 SKIP OR ROST, FAILED
0221 /TST221 ERROR MESSAGE
TST221 /SCOPE LOOP

/DOES CLSA READ ROST. INPUT 4 ?
/
TST222, CLA CLL CMA /AC TO 7777
DCA REGA
CLA CLL IAC RTL /AC TO 0004
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP --1 /WAIT FOR FLAG
CMA /AC TO 7773
JMS I XIOTI /IOT 6135, CLSA
JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
JMS I NERRR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: CLSA OR INPUT 4 FAILED
5222 /TST222 ERROR MESSAGE
TST222 /SCOPE LOOP

/DOES CLSA READ ROST. INPUT 2 ?
/
TST223, CLA CLL CMA /AC TO 7777
DCA REGA
CLA CLL IAC RAL /AC TO 0002
JMS I XIOTF /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP --1 /WAIT FOR FLAG
CMA /AC TO 7775

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/
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4142 4431 JMS I XIOTI /IOT 6135, CLSA
4143 4426 JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
4144 4472 JMS I NERROR /CHECK NON-ERROR HANDLER
4145 4473 JMS I ERROR /ERROR: CLSA OR INPUT 2 FAILED
4146 5223 /TST223 ERROR MESSAGE
4147 4133 TST223 /SCOPE LOOP

/DOES CLSA READ RQST. INPUT 1 ?
/
TST224, CLA CLL CMA /AC TO 7777
DCA REGA /AC TO 0001
CLA CLL IAC /IOT 6132, CLOE
JMS I XIOTF /IOT 6131, CLSK
JMS I XIOTE /WAIT FOR FLAG
JMP -1 /AC TO 7776
CMA /IOT 6135, CLSA
JMS I XIOTI /CHECK SEND AND RECEV REGISTERS
JMS I XSNDRV /CHECK NON-ERROR HANDLER
JMS I NERROR /ERROR: CLSA OR INPUT 1 FAILED
JMS I ERROR /TST224 ERROR MESSAGE
5224 /SCOPE LOOP
TST224

/DOES CLSA CLEAR INPUT 4 RQST. ?
/
TST225, CLA CLL CMA /AC TO 7777
DCA REGA /AC TO 0004
CLA CLL IAC RTL /IOT 6132, CLOE
JMS I XIOTF1 /IOT 6131, CLSK
JMS I XIOTE /WAIT FOR FIRST FLAG
JMP -1 /IOT 6135, CLSA
JMS I XIOTI /IOT 6131, CLSK
JMS I XIOTE /WAIT FOR SECOND FLAG
JMP -1 /IOT 6135, CLSA
JMS I XIOTI /IOT 6131, CLSK
JMS I XIOTE /CHECK NON-ERROR HANDLER
JMS I NERROR /ERROR: CLSA OR INPUT 1 FAILED
JMS I ERROR /TST225 ERROR MESSAGE
0225 /SCOPE LOOP
TST225

/DOES CLSA CLEAR INPUT 2 RQST. ?
/
TST226, CLA CLL CMA /AC TO 7777
DCA REGA /AC TO 0002
CLA CLL IAC RAL /IOT 6132, CLOE
JMS I XIOTF /IOT 6131, CLSK
JMS I XIOTE /WAIT FOR FIRST FLAG
JMP -1 /IOT 6135, CLSA
JMS I XIOTI /IOT 6131, CLSK
JMS I XIOTE /WAIT FOR SECOND FLAG
JMP -1 /IOT 6135, CLSA
JMS I XIOTI /IOT 6131, CLSK
JMS I XIOTE /CHECK NON-ERROR HANDLER
JMS I NERROR

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/
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4220      4473      JMS I ERROR      /ERROR: CLSA OR INPUT 2 FAILED
4221      0226      0226      /TST226 ERROR MESSAGE
4222      4224      TST226      /SCOPE LOOP

/DOES CLSA CLEAR INPUT 4 RQST. ?
/
TST227,    CLA CLL CMA      /AC TO 7777
          DCA REGA
          CLA CLL IAC
          JMS I XIOTF      /AC TO 0001
          JMS I XIOTE      /IOT 6132, CLOE
          JMP -1           /IOT 6131, CLSK
          JMS I XIOTI      /WAIT FOR FIRST FLAG
          JMS I XIOTE      /IOT 6135, CLSA
          JMP -1           /IOT 6131, CLSK
          JMS I XIOTI      /WAIT FOR SECOND FLAG
          JMS I XIOTE      /IOT 6135, CLSA
          JMS I XIOTE      /IOT 6131, CLSK
          JMS I NERROR      /CHECK NON-ERROR HANDLER
          JMS I ERROR      /ERROR: CLSA OR INPUT 1 FAILED
          0227      /TST227 ERROR MESSAGE
          TST227      /SCOPE LOOP

/DOES CLSA READ INPUT 4,2,1 ?
/
TST230,    CLA CLL CMA      /AC TO 7777
          DCA REGA
          TAD K0007      /GET ENABLES
          JMS I XIOTF      /IOT 6132, CLOE
          NOP
          ISZ REGB
          JMP -2
          JMS I XIOTE      /WAIT FOR ALL
          JMP -1           /IOT 6131, CLOE
          CLA CLL CMA      /WAIT FOR FLAGS
          JMS I XIOTI      /AC TO 7777
          JMS I XSNDRV      /IOT 6135, CLSA
          SKP CLA          /CHECK SEND AND RECEV REGISTERS
          JMP T230A        /ERROR, STATUS REGISTER
          DCA SEND        /SAVE OUTPUT FOR ERROR PRINTER
          CLA CLL CMA      /AC TO 7777
          JMS I XIOTI      /IOT 6135, CLSA
          SNA CLA          /WAS STATUS ALL 0'S ?
          JMS I NERROR      /CHECK NON-ERROR HANDLER
          JMS I ERROR      /ERROR: CLSA OR INPUTS 1,2,3 FAILED
          5230      /TST230 ERROR MESSAGE
          TST230      /SCOPE LOOP

/DOES INPUT 4 CLEAR BIT 7 ?
/
TST231,    CLA CLL CMA
          DCA REGA
          CLA CLL IAC RTL /AC TO 0004
          DCA SEND
          TAD SEND
          TAD K0020      /GET ENABLES
          4270      7340
          4271      3040
          4272      7307
          4273      3070
          4274      1070
          4275      1140

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4276 JMS I XIOTF1 /IOT 6132, CLOE
4277 JMS I XIOTE /IOT 6131, CLSK
4300 JMP .-1 /WAIT FOR FLAG
4301 CLA CLL CMA /AC TO 7777
4302 JMS I XIOTH /IOT 6134, CLEN
4303 JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
4304 JMS I NERROR /CHECK NON-ERROR HANDLER
4305 JMS I ERROR /ERROR:BIT 7 OR INPUT 4 FAILED
4306 4631 /TST231 ERROR MESSAGE
4307 TST231 /SCOPE LOOP

/DOES INPUT 2 CLEAR BIT 7 ?
/
TST232, CLA CLL CMA
DCA REGA
CLA CLL IAC RAL /AC TO 0002
DCA SEND /SAVE OUTPUT FOR ERROR PRINTER
TAD SEND
TAD K0020
JMS I XIOTF1 /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP .-1 /WAIT FOR FLAG
CLA CLL CMA /IOT 6134, CLEN
JMS I XIOTH /CHECK SEND AND RECEV REGISTERS
JMS I XSNDRV /CHECK NON-ERROR HANDLER
JMS I NERROR /ERROR: BIT 7 OR INPUT 2 FAILED
JMS I ERROR /TST232 ERROR MESSAGE
4632 /TST232 ERROR MESSAGE
TST232 /SCOPE LOOP

/DOES INPUT 1 CLEAR BIT 7 ?
/
TST233, CLA CLL CMA /AC TO 7777
DCA REGA /AC TO 0001
CLA CLL IAC /SAVE OUTPUT FOR ERROR PRINTER
DCA SEND
TAD SEND
TAD K0020
JMS I XIOTF1 /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP .-1 /WAIT FOR FLAG
CLA CLL CMA /AC TO 7777
JMS I XIOTH /IOT 6134, CLEN
JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
JMS I NERROR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: BIT 7 OR INPUT 1 FAILED
4633 /TST233 ERROR MESSAGE
TST233 /SCOPE LOOP

/DOES INPUT 4,2,1 GENERATE CLR CNT ?
/MODE 3, RATE 0
/
TST234, CLA CLL CMA /AC TO 7777
DCA REGA /GET AC NUMBER
TAD K2525

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 4353 4427
 4354 7307
 4355 1120
 4356 4426
 4357 4424
 4360 5357
 4361 7300
 4362 3070
 4363 7340
 4364 4433
 4365 7650
 4366 4472
 4367 4473
 4370 4234
 4371 4350

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JMS I XIOTG /IOT 6133, CLAB
 CLA CLL IAC RTL /AC TO 0004
 TAD K3000 /GET ENABLES
 JMS I XIOTF1 /IOT 6132, CLOE
 JMS I XIOTE /IOT 6131, CLSK
 JMP -1 /WAIT FOR FLAG
 CLA CLL /SAVE OUTPUT FOR ERROR PRINTER
 DCA SEND /AC TO 7777
 CLA CLL CMA /IOT 6137, CLCA
 JMS I XIOTK /WAS COUNTER ALL 0'S ?
 SNA CLA /CHECK NON-ERROR HANDLER
 JMS I NERROR /ERROR: CLR CNT FAILED
 JMS I ERROR /TST234 ERROR MESSAGE
 4234 /SCOPE LOOP
 TST234

/
 /DOES INPUT 4,2,1 CAUSE CLR CNT ?
 /MODE 3, RATE 0
 /

TST235, CLA CLL CMA /AC TO 7777
 DCA REGA /GET AC NUMBER
 TAD K252 /IOT 6133, CLAB
 JMS I XIOTG /AC TO 0002
 CLA CLL IAC RAL /GET ENABLES
 TAD K3000 /IOT 6132, CLOE
 JMS I XIOTF1 /IOT 6131, CLSK
 JMS I XIOTE /WAIT FOR FLAG
 JMP -1 /SAVE OUTPUT FOR ERROR PRINTER
 CLA CLL /AC TO 7777
 DCA SEND /IOT 6137, CLCA
 CLA CLL CMA /WAS COUNTER ALL 0'S ?
 JMS I XIOTK /CHECK NON-ERROR HANDLER
 SNA CLA /ERROR: CLR CNT FAILED ?
 JMS I NERROR /TST235 ERROR MESSAGE
 JMS I ERROR 4235
 4235 /SCOPE LOOP
 TST235

/
 /DOES INPUT 4,2,1 TRANSFER COUNTER TO BUFFER ?
 /

TST236, CLA CLL CMA /AC TO 7777
 DCA REGA /GET AC NUMBER
 TAD K2525 /IOT 6133, CLAB
 JMS I XIOTG /CAF OR CLEAR THE WORLD
 6007 /AC TO 0001
 CLA CLL IAC /GET ENABLES
 TAD K3000 /IOT 6132, CLOE
 JMS I XIOTF1 /IOT 6131, CLSK
 JMS I XIOTE /WAIT FOR FLAG
 JMP -1 /AC TO 7777
 CLA CLL CMA /IOT 6136, CLBA
 JMS I XIOTJ /CHECK SEND AND RECEV REGISTERS
 JMS I XSNDRV /CHECK NON-ERROR HANDLER
 JMS I NERROR /ERROR: COUNTER TO BUFFER FAILED
 JMS I ERROR 4413
 4413 4372

4414 7340
 4415 3040
 4416 1016
 4417 4427
 4420 6027
 4421 7301
 4422 1120
 4423 4426
 4424 4424
 4425 5224
 4426 7340
 4427 4432
 4430 4456
 4431 4472
 4432 4473

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4433 3636 /TST236 ERROR MESSAGE
4434 4414 /SCOPE LOOP

/DOES INPUT 4,2,1 TRANSFER COUNTER TO BUFFER ?
/
TST237, CLA CLL CMA /AC TO 7777
DCA REGA
TAD K252
JMS I XIOTG
6007
CLA CLL IAC
TAD K3000
JMS I XIOTF1
JMS I XIOTE
JMP -1
CLA CLL CMA
JMS I XIOTJ
JMS I XSNDRV
JMS I NERRR
JMS I ERROR
JMS I ERROR
3637
TST237
/SCOPE LOOP

/DOES INPUT 4,2,1 GENERATE CLR CNT ?
/
/MODE 2, RATE 0
/
TST240, CLA CLL CMA /AC TO 7777
DCA REGA
TAD K2525
JMS I XIOTG
6007
CLA CLL IAC RTL /AC TO 0004
TAD K2000
JMS I XIOTF1
JMS I XIOTE
JMP -1
CLA CLL CMA
JMS I XIOTK
JMS I XSNDRV
JMS I NERRR
JMS I ERROR
JMS I ERROR
4240
TST240
/SCOPE LOOP

/DOES INPUT 4,2,1 CAUSE CLR CNT ?
/
/MODE 2, RATE 0
/
TST241, CLA CLL CMA
DCA REGA
TAD K252
JMS I XIOTG
6007
CLA CLL IAC RAL
TAD K2000

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/
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4506      4426      JMS I XIOTF1      /IOT 6132, CLOE
4507      4424      JMS I XIOTE      /IOT 6131, CLSK
4510      5307      JMP .-1      /WAIT FOR FLAG
4511      7340      CLA CLL CMA      /AC TO 7777
4512      4433      JMS I XIOTK      /IOT 6137, CLCA
4513      4456      JMS I XSNDRV      /CHECK SEND AND RECEV REGISTERS
4514      4472      JMS I NERRR      /CHECK NON-ERROR HANDLER
4515      4473      JMS I ERROR      /ERROR: CLR CNT FAILED, MODE 2
4516      4241      /TST241 ERROR MESSAGE
4517      4477      TST241      /SCOPE LOOP

/DOES COUNTER TRANSFER TO BUFFER ?
/MODE 2, RATE 0
/
TST242, CLA CLL CMA      /AC TO 7777
DCA REGA      /GET AC NUMBER
TAD K2525      /IOT 6133, CLAH
JMS I XIOTG      /CAF OR CLEAR THE WORLD
6007
CLA CLL IAC RTL      /GET ENABLES
TAD K2000      /IOT 6132, CLOE
JMS I XIOTF1      /IOT 6131, CLSK
JMS I XIOTE      /WAIT FOR FLAG
JMP .-1      /AC TO 7777
CLA CLL CMA      /IOT 6136, CLBA
JMS I XIOTJ      /CHECK SEND AND RECEV REGISTERS
JMS I XSNDRV      /CHECK NON-ERROR HANDLER
JMS I NERRR      /ERROR: COUNTER TO BUFFER FAILED
JMS I ERROR      /TST242 ERROR MESSAGE
3642
TST242      /SCOPE LOOP

/DOES COUNTER TRANSFER TO BUFFER ?
/MODE 2, RATE 0
/
TST243, CLA CLL CMA      /AC TO 7777
DCA REGA      /GET AC NUMBER
TAD K2525      /IOT 6133, CLAB
JMS I XIOTG      /CAF OR CLEAR THE WORLD
6007
CLA CLL IAC RAL      /AC TO 0002
TAD K2000      /GET ENABLES
JMS I XIOTF1      /IOT 6132, CLOE
JMS I XIOTE      /IOT 6131, CLSK
JMP .-1      /WAIT FOR FLAG
CLA CLL CMA      /IOT 6136, CLBA
JMS I XIOTJ      /CHECK SEND AND RECEV REGISTERS
JMS I XSNDRV      /CHECK NON-ERROR HANDLER
JMS I NERRR      /ERROR: COUNTER TO BUFFER FAILED
JMS I ERROR      /TST243 ERROR MESSAGE
3643
TST243      /SCOPE LOOP

/DOES INPUT 4,2,1 AFFECT MODE 0 ?
/

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4562 7340 TST244, CLA CLL CMA
4563 3040 DCA REGA
4564 1016 TAD K2525
4565 4027 JMS I XIOTG
4566 6027 6007
4567 7327 CLA CLL IAC RTL
4570 4426 JMS I XIOTF1
4571 4424 JMS I XIOTE
4572 5371 JMP .-1
4573 7340 CLA CLL CMA
4574 4433 JMS I XIOTK
4575 4456 JMS I XSNDRV
4576 4472 JMS I NERROR
4577 4473 JMS I ERROR
4600 4244
4601 4502 TST244
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/DOES INPUT 4,2,1 AFFECT MODE 0 ?
/
TST245, CLA CLL CMA
4602 7340 DCA REGA
4603 3040
4604 1017 TAD K5252
4605 4427 JMS I XIOTG
4606 7321 CLA CLL IAC
4607 4426 JMS I XIOTF1
4610 4424 JMS I XIOTE
4611 5210 JMP .-1
4612 7340 CLA CLL CMA
4613 4432 JMS I XIOTJ
4614 4456 JMS I XSNDRV
4615 4472 JMS I NERROR
4616 4473 JMS I ERROR
4617 3645 3645
4620 4602 TST245
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/DOES INPUT 4,2,1 AFFECT MODE 1 ?
/
TST246, CLA CLL CMA
4621 7340 DCA REGA
4622 3040
4623 1016 TAD K2525
4624 4427 JMS I XIOTG
4625 6027 6007
4626 3070 DCA SEND
4627 7321 CLA CLL IAC
4630 1144 TAD K1000
4631 4426 JMS I XIOTF1
4632 4424 JMS I XIOTE
4633 5232 JMP .-1
4634 7340 CLA CLL CMA
4635 4432 JMS I XIOTJ
4636 7650 SNA CLA
4637 4472 JMS I NERROR
4640 4473 JMS I ERROR
4641 4246 4246
4642 4621 TST246
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/GET AC NUMBER
/IOT 6133, CLAB
/CAF OR CLEAR THE WORLD
/AC TO 0004
/IOT 6132, CLOE
/IOT 6131, CLSK
/WAIT FOR FLAG
/AC TO 7777
/IOT 6137, CLCA
/CHECK SEND AND RECEV REGISTERS
/CHECK NON-ERROR HANDLER
/ERROR: MODE 0 FAILED
/TST 244 ERROR MESSAGE
/SCOPE LOOP

/AC TO 7777
/GET AC NUMBER
/IOT 6133, CLAB
/AC TO 0001
/IOT 6132, CLOE
/IOT 6131, CLSK
/WAIT FOR FLAG
/AC TO 7777
/IOT 6136, CLBA
/CHECK SEND RECEV REGISTERS
/CHECK NON-ERROR HANDLER
/ERROR: MODE 0 FAILED
/TST245 ERROR MESSAGE
/SCOPE LOOP

/AC TO 7777
/GET AC NUMBER
/IOT 6133, CLAB
/CAF OR CLEAR THE WORLD
/SAVE OUTPUT FOR ERROR PRINTER
/AC TO 0001
/GET ENABLES
/IOT 6132, CLOE
/IOT 6131, CLOE
/WAIT FOR FLAG
/AC TO 7777
/IOT 6136, CLBA
/WAS BUFFER STILL ALL 0'S ?
/CHECK NON-ERROR HANDLER
/ERROR: MODE 1 FAILED
/TST246 ERROR MESSAGE
/SCOPE LOOP
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4643 7340
4644 3040
4645 1017
4646 4427
4647 7307
4650 1144
4651 4426
4652 4424
4653 5252
4654 7340
4655 4432
4656 4426
4657 4472
4660 4473
4661 3647
4662 4643

/DOES INPUT 4,2,1 AFFECT MODE 1 ?
/
TST247, CLA CLL CMA /AC TO 7777
DCA REGA
TAD K252 /GET AC NUMBER
JMS I XIOTG /IOT 6133, CLAB
CLA CLL IAC RTL /AC TO 0004
TAD K1000
JMS I XIOTF1 /IOT 6132, CLOE
JMS I XIOTE /IOT 6131, CLSK
JMP -1 /WAIT FOR FLAG
CLA CLL CMA /AC TO 7777
JMS I XIOTJ /IOT 6136, CLBA
JMS I XSNDRV /CHECK SEND AND RECEV REGISTERS
JMS I NERROR /CHECK NON-ERROR HANDLER
JMS I ERROR /ERROR: MODE 1 FAILED
3647 /TST247 ERROR MESSAGE
TST247 /SCOPE LOOP

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/DOES CLSA READ INPUTS 4,2,1 ?
/
TST250, CLA CLL CMA /AC TO 7777
DCA REGA
TAD K0007 /GET ENABLES
JMS I XIOTF1 /IOT 6132, CLOE
NOP
ISZ REG8
JMP -2 /WAIT FOR ALL
JMS I XIOTE /IOT 6131, CLSK
JMP -1 /IOT 6130, CLZE
JMS I XIOTD /CLEAR THE AC AND LINK
CLA CLL /SAVE OUTPUT FOR ERROR PRINTER
DCA SEND /AC TO 7777
CLA CLL CMA /IOT 6135, CLSA
JMS I XIOTI /WAS STATUS ALL 0'S ?
SNA CLA /CHECK NON-ERROR HANDLER
JMS I NERROR /ERROR: INPUT 4,2,1 OR STATUS FAILED
JMS I ERROR /TEST250 ERROR MESSAGE
5250 /SCOPE LOOP
TST250

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/DOES CLSA READ STATUS REGISTER ?
/
TST251, CLA CLL CMA /AC TO 7777
DCA REGA
TAD K0007 /GET ENABLES
JMS I XIOTF /IOT 6132, CLOE
NOP
ISZ REG8
JMP -2 /WAIT FOR FLAGS
JMS I XIOTE /IOT 6131, CLSK
JMP -1 /AC TO 7777
CLA CLL CMA /IOT 6135, CLSA
JMS I XIOTI

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4706 7340
4707 3040
4710 1007
4711 4425
4712 7000
4713 2041
4714 5312
4715 4424
4716 5315
4717 7340
4720 4431

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4721	4456	JMS I XSNDRV		/CHECK SEND AND RECEV REGISTERS
4722	4472	JMS I NERRR		/CHECK NON-ERROR HANDLER
4723	4473	JMS I ERROR		/ERROR: CLSA OK STATUS REGISTER
4724	5251	TST251		/TST251 ERROR MESSAGE
4725	4706			/SCOPE LOOP
4726	7300	CLA CLL		
4727	2077	ISZ LOOP		
4730	5464	JMP I XMITT		/DO TEST 4096 TIMES
4731	4570	JMS I XPASS		/TYPE PASS COMPLETE
4732	5465	JMP I XMITT1		/CONTINUE TESTING
		/NON-ERROR HANDLER FOR PROGRAM		
5000	5000	*5000		
5001	0000	NERRR,		/CAF OR CLEAR THE WORLD
5002	6007	ISZ NERRR		
5003	2200	ISZ NERRR		
5004	2040	ISZ REGA		
5005	5215	JMP OUT		
5006	4460	JMS I XCLREG		/CLEAR ALL REGISTERS
5007	7604	LAS		
5010	0137	AND K0040		/IS IT LOOP ON NON-
5011	7640	S2A CLA		/FAILING TEST.
5012	5215	JMP OUT		
5013	2200	ISZ NERRR		
5014	5600	JMP I NERRR		/TO NEXT TEST
5015	1600	OUT,		
5016	3220	TAD I NERRR		
5017	5620	DCA ERRO		
		JMP I ERRO		
		/ERROR HANDLER FOR PROGRAM		
5020	0000	ERRO,		/CAF OR CLEAR THE WORLD
5021	6007	LAS		
5022	7604	RTL		
5023	7006	SMA CLA		/CHECK SWR2 FOR INH. PRINT
5024	7720	JMS I XSORT		/GET ERROR MESSAGE
5025	4503	JMS I XBELL		/RING BELL
5026	4510	JMS I XCLREG		
5027	4460	ISZ ERRO		
5030	2220	LAS		
5031	7604	AND K0200		
5032	0015	SNA CLA		/CHECK SWR4 FOR INH. HLT
5033	7650	HLT		/MONITOR ERROR HALT, READ TYPEOUT
5034	7402	EHLT1,		/AND REFERENCE LISTING.
5035	7604	LAS		
5036	0013	AND K0100		
5037	7640	S2A CLA		/CHECK SWR5 FOR SCOPE LOOP
5040	5243	JMP IN		
5041	2220	ISZ ERRO		

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5042	5620		JMP I ERRO	/ENTER SCOPE LOOP
5043	1620	/	TAD I ERRO	
5044	3200	IN,	DCA NERRO	
5045	5600		JMP I NERRO	
5046	0000	/	BELL,	
5047	7604		0000	
5050	0116		LAS	
5051	7640		AND K0400	
5052	5646		SZA CLA	
5053	1026		JMP I BELL	
5054	4527		TAD K0207	
5055	5646		JMS I XTYPE	
			JMP I BELL	
5056	0020	/	TYPE,	
5057	6046		0000	
5060	6041		TLS	
5061	5260		TSF	
5062	7200		JMP -1	
5063	6042		CLA	
5064	5656		TCF	
			JMP I TYPE	
5065	0020	/	CLRREG, 0000	/CLEAR THE AC AND LINK
5066	7300		CLA CLL	
5067	3041		DCA REGB	
5070	3042		DCA REGC	
5071	3043		DCA REGD	
5072	3070		DCA SEND	
5073	3071		DCA RECEV	
5074	7604		LAS	
5075	0117		AND K0000	
5076	7650		SNA CLA	
5077	7340		CLA CLL CMA	
5100	3040		DCA REGA	
5101	5665		JMP I CLRREG	
5102	0020	/	IOTA,	/FIELD SERVICE CHANGE
5103	6131		0000	
5104	5702		6131	
5105	2302		JMP I IOTA	
5106	5702		ISZ IOTA	
			JMP I IOTA	
5107	0020	/	IOTB,	/FIELD SERVICE CHANGE
5110	6132		0000	
5111	5707		6132	
5112	2307		JMP I IOTB	
5113	5707		ISZ IOTB	
			JMP I IOTB	
5114	0020	/	IOTC,	/FIELD SERVICE CHANGE
5115	6133		0000	
5116	5714		6133	
5117	2314		JMP I IOTC	
5120	5714		ISZ IOTC	
			JMP I IOTC	

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5121	0000	IOTD,	0000	
5122	3070		DCA SEND	/SAVE OUTPUT FOR ERROR PRINTER
5123	1070		TAD SEND	
5124	6130		6130	/FIELD SERVICE CHANGE
5125	5721		JMP I IOTD	
5126	7422	EHLT2,	HLT	/SKIP TRAP, CLDE
5127	0000	/		
5130	6131	IOTE,	0000	
5131	5727		6131	/FIELD SERVICE CHANGE
5132	2527		JMP I IOTE	
5133	5727		ISZ IOTE	
			JMP I IOTE	
5134	0000	/		
5135	3070	IOTF,	0000	
5136	1070		DCA SEND	/SAVE OUTPUT FOR ERROR PRINTER
5137	6132		TAD SEND	
5140	5734		6132	/FIELD SERVICE CHANGE
5141	7422	EHLT3,	JMP I IOTF	
		/	HLT	/SKIP TRAP, CLOE
5142	0000	IOTF1,	0000	
5143	6132		6132	/FIELD SERVICE CHANGE
5144	5742		JMP I IOTF1	
5145	7422	EHLT4,	HLT	/SKIP TRAP, CLOE
		/		
5146	0000	IOTG,	0000	
5147	3070		DCA SEND	/SAVE OUTPUT FOR ERROR PRINTER
5150	1070		TAD SEND	
5151	6133		6133	/FIELD SERVICE CHANGE
5152	5746		JMP I IOTG	
5153	7422	EHLT5,	HLT	/SKIP TRAP, CLAB
		/		
5154	0000	IOTH,	0000	
5155	6134		6134	/FIELD SERVICE CHANGE
5156	7410		SKP	
5157	7422	EHLT6,	HLT	/SKIP TRAP, CLEN
5160	3071		DCA RECEV	/SAVE OUTPUT FOR ERROR PRINTER
5161	1071		TAD RECEV	
5162	5754		JMP I IOTH	
5163	0000	/		
5164	6135	IOTI,	0000	
5165	7410		6135	/FIELD SERVICE CHANGE
5166	7422		SKP	
5167	3071	EHLT7,	HLT	/SKIP TRAP, CLSA
5170	1071		DCA RECEV	/SAVE OUTPUT FOR ERROR PRINTER
5171	5763		TAD RECEV	
			JMP I IOTI	
5200	0000	/		
5201	6136	*5200		
5202	7410	/		
5203	7422	IOTJ,	0000	
5204	3071		6136	/FIELD SERVICE CHANGE
			SKP	
		EHLT10,	HLT	/SKIP TRAP, CLBA
			DCA RECEV	/SAVE OUTPUT FOR ERROR PRINTER

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5205	1071		TAD RECEV	
5206	5600		JMP I 107J	
5207	0000	/	10TK, 0000	/FIELD SERVICE CHENGE
5210	6137		6137	
5211	7410		SKP	/SKIP TRAP, CLCA
5212	7402		EH111, HLT	/SAVE OUTPUT FOR ERROR PRINTER
5213	3071		DCA RECEV	
5214	1071		TAD RECEV	
5215	5607		JMP I 10TK	
5216	0000	/	SNDRV, 0000	
5217	7041		CIA	
5220	1072		TAD SEND	
5221	7640		SZA CLA	/WAS SEND AND RECEV THE SAME ?
5222	2216		ISZ SNDRV	
5223	5616		JMP I SNDRV	
5224	0000	/	RANDOM, 0000	
5225	1044		TAD REGE	
5226	7024		RAL	
5227	7430		SZL	
5230	1410		TAD I 10	
5231	3044		DCA REGE	
5232	1044		TAD REGE	
5233	5624		JMP I RANDOM	
5234	0000		0000	
5235	7300		CLA CLL	/CLEAR THE AC AND LINK
5236	1254		TAD PRET5	
5237	3002		DCA 2	/SET FOR PI RETURN
5240	6001		ION	
5241	7300		CLA CLL	/CLEAR THE AC AND LINK
5242	1076		TAD KREGC	
5243	3042		DCA REGC	
5244	4433		JMS I XIOTK	
5245	2042		ISZ REGC	/READ THE COUNTER
5246	5245		JMP .-1	
5247	2043		ISZ REGD	
5250	5241		JMP .-7	
5251	2254		ISZ PIG05	
5252	6002		PIRET5, IOF	/DISABLE PROGRAM INTERRUPT
5253	5634		JMP I PIG05	
5254	5252	/	PIRET5, PIRET5	
5255	0000	/	PIG01, 0000	
5256	7300		CLA CLL	/CLEAR THE AC AND LINK
5257	1267		TAD PRET1	
5260	3002		DCA 2	/SET FOR PI RETURN
5261	6001		ION	/ENABLE PROGRAM INTERRUPT
5262	4454		JMS I XISZ	
5263	7410		SKP	
5264	2255		PIRET1, ISZ PIG01	
5265	6002		IOF	/DISABLE PROGRAM INTERRUPT
5266	5655		JMP I PIG01	

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5267	5264	/		
		PRET1, PIRET1		
5270	0000	/		
5271	7300	PIG02, 0000		/CLEAR THE AC AND LINK
5272	1301	CLA CLL		
5273	3002	TAD PRET2		/SET FOR PI RETURN
5274	6001	DCA 2		
5275	4001	ION		/WAIT
5276	2270	JMS I XIS2		
5277	6002	ISZ PIG02		
5300	5670	PIRET2, IOF		
		JMP I PIG02		
5301	5277	/		
		PRET2, PIRET2		
5302	0000	/		
5303	4422	SYNC, 0000		
5304	5303	JMS I XIOTC		
5305	4422	JMP -1		
5306	5305	JMS I XIOTC		
5307	5702	JMP -1		
		JMP I SYNC		
5310	0000	/		
5311	7300	ISZLOP, 0000		
5312	1113	CLA CLL		
5313	3049	TAD XIOTCPS		
5314	7001	DCA REGF		
5315	7000	IAC		
5316	2043	NOP		
5317	5314	ISZ REGD		
5320	2045	JMP -3		
5321	5314	ISZ REGF		
5322	5710	JMP -5		
		JMP I ISZLOP		
5323	0000	/		
5324	7300	PIG03, 0000		/CLEAR THE AC AND LINK
5325	1335	CLA CLL		
5326	3002	TAD PRETC		
5327	6001	DCA 2		
5330	7000	ION		
5331	7410	NOP		
5332	2323	SKP		
5333	6002	ISZ PIG03		
5334	5723	IOF		
		JMP I PIG03		
5335	5332	/		
		PRETC, RETC		
5336	0000	/		
5337	7300	PIG04, 0000		/CLEAR THE AC AND LINK
5340	1347	CLA CLL		
5341	3002	TAD PRETD		
5342	6001	DCA 2		
5343	7000	ION		
5344	2336	NOP		
5345	6002	ISZ PIG04		
		IOF		
		RETD,		

5346	5736	JMP I PIG04
5347	5345	PRETD, RETD
5350	0000	IOTS,
5351	6132	
5352	6134	
5353	6132	
5354	6134	
5355	6132	
5356	6134	
5357	5750	JMP I IOTS
5360	0000	
5361	6133	IOTS1,
5362	6136	
5363	6133	
5364	6136	
5365	6133	
5366	6136	
5367	5760	JMP I IOTS1
5370	0000	
5371	6133	IOTS2,
5372	6137	
5373	6133	
5374	6137	
5375	6133	
5376	6137	
5377	5770	JMP I IOTS2
5400	5400	
5401	6134	IOTS3,
5402	7040	
5403	6130	
5404	7040	
5405	6134	
5406	7040	
5407	6130	
5410	7040	
5411	6134	
5412	5600	JMP I IOTS3
5413	0000	
5414	7624	CLOCK,
5415	0027	
5416	3075	
5417	5613	
5420	0020	
5421	0000	
5422	0000	
5423	0000	
5424	0000	
5425	0000	
5426	0000	
5427	0000	
5428	0000	
5429	0000	
5430	0000	
5431	0000	
5432	0000	
5433	0000	
5434	0000	
5435	0000	
5436	0000	
5437	0000	
5438	0000	
5439	0000	
5440	0000	
5441	0000	
5442	0000	
5443	0000	
5444	0000	
5445	0000	
5446	0000	
5447	0000	
5448	0000	
5449	0000	
5450	0000	
5451	0000	
5452	0000	
5453	0000	
5454	0000	
5455	0000	
5456	0000	
5457	0000	
5458	0000	
5459	0000	
5460	0000	
5461	0000	
5462	0000	
5463	0000	
5464	0000	
5465	0000	
5466	0000	
5467	0000	
5468	0000	
5469	0000	
5470	0000	
5471	0000	
5472	0000	
5473	0000	
5474	0000	
5475	0000	
5476	0000	
5477	0000	
5478	0000	
5479	0000	
5480	0000	
5481	0000	
5482	0000	
5483	0000	
5484	0000	
5485	0000	
5486	0000	
5487	0000	
5488	0000	
5489	0000	
5490	0000	
5491	0000	
5492	0000	
5493	0000	
5494	0000	
5495	0000	
5496	0000	
5497	0000	
5498	0000	
5499	0000	
5500	0000	

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5421 7006      RTL
5422 7006      RTL
5423 3041      DCA REG8
5424 1130      TAD K7774
5425 3042      DCA REGC
5426 1041      TAD REG8
5427 0027      AND K0007
5430 1123      TAD K0260
5431 4527      JMS I XTYPE
5432 1041      TAD REG8
5433 7006      RTL
5434 7004      RAL
5435 3041      DCA REG8
5436 2042      ISZ REGC
5437 5226      JMP I-11
5440 5620      JMP I OCTEL

/ROUTINE FOR CRLF
/
CRLF, 0000
CLA CLL
TAD K0215
JMS I XTYPE
TAD K0212
JMS I XTYPE
JMP I CRLF

/ROUTINE TO TYPE CLOCK
/
POPR, 0000
CLA CLL
TAD KTADCK
TAD CKLOCKS
DCA I+1
TAD KTADCK
JMS I XOCTEL
JMS I XPRINT
FMES
JMP I POPR

KTADCK, TAD CLKNO
/
CLKNO, 0001
0050
0100
0120
0500
5000

/ROUTINE TO SORT ERROR MESSAGES
/
SORT, 0000
CLA CLL
JMS I XCRLF
TAD I ERROR
DCA REGE

5471 0000
5472 7300
5473 4521
5474 1473
5475 3044
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5476	4525	JMS I XMESS	/GO PRINT TEST + ADDRESS
5477	1444	TAD I REGE	
5500	7012	RTR	
5501	7012	RTR	
5502	7012	RTR	
5503	7012	RTR	
5504	0127	AND K0017	/MOVE IT TO BITS 8-11
5505	3044	DCA REGE	/MASK 8-11
5506	7300	CLA CLL	/SAVE POINTER
5507	1044	TAD REGE	/CLEAR THE AC AND LINK
5510	1926	TAD KTADM	/GET POINTER
5511	3312	DCA .+1	
5512	1926	TAD KTADM	/MODIFIED BY TEST
5513	3316	DCA .+3	/STORE MESSAGE POINTER
5514	4501	JMS I XCRLF	/CRLF
5515	4506	JMS I XPRINT	/PRINT MESSAGE
5516	0000		/MODIFIED MESSAGE POINTER
5517	7300	CLA CLL	
5520	1044	TAD REGE	/GET MESSAGE POINTER
5521	1132	TAD K7772	/IS IT GREATER THAN
5522	7620	SNL CLA	
5523	5671	JMP I SORY	
5524	4502	JMS I XREG	
5525	5671	JMP I SORY	
5526	1927	KTADM, TAD KTMX	
5527	6107	MES1	
5530	6131	MES2	
5531	6152	MES3	
5532	6202	MES4	
5533	6231	MES5	
5534	6256	MES6	
5535	6303	MES7	
5536	6324	MES8	
5537	6353	MES9	
5540	6402	MES10	
5541	6431	MES11	
5542	0000		/ROUTINE TO PRINT TEST + ADDRESS
5543	7300	CLA CLL	/CLEAR THE AC AND LINK
5544	4501	JMS I XCRLF	/CRLF
5545	4506	JMS I XPRINT	/GO PRINT TEST
5546	6046	TMES	
5547	1473	TAD I ERROR	/GET ERROR MESSAGE
5550	3043	DCA REGD	/STORE MESSAGE POINTER
5551	1443	TAD I REGD	
5552	0136	AND K0377	/MASK 4-11
5553	4504	JMS I XOCTEL	/GO PRINT NUMBER
5554	2043	IS2 REGD	/UPDATE POINTER
5555	4506	JMS I XPRINT	/GO PRINT STARTING ADDRESS
5556	6051	AMES	
5557	1443	TAD I REGD	

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5560	4524	JMS I XOCTEL	/GO PRINT NUMBER	
5561	7300	CLA CLL	/CLEAR THE AC AND LINK	
5562	5742	JMP I MESS		
		/ROUTINE TO PRINT AC		
		/		
5563	0000	PREG, 0000		
5564	4521	JMS I XCRLF	/CRLF	
5565	4526	JMS I XPRINT	/GO PRINT MESSAGE	
5566	6067	GMES		
5567	1070	TAD SEND	/GET GOOD AC	
5570	4524	JMS I XOCTEL	/PRINT IT	
5571	4526	JMS I XPRINT	/PRINT BAD AC	
5572	6077	BMES		
5573	1071	TAD RECEV	/GET BAD AC	
5574	4524	JMS I XOCTEL	/PRINT IT	
5575	7300	CLA CLL	/CLEAR THE AC AND LINK	
5576	5763	JMP I PREG		
		/		
	5600	*5600		
		/		
5600	0020	SETO, 0000		
5601	1120	TAD JMP12	/GET JMP I 2	
5602	3021	DCA I	/SET FOR PI RETURN	
5603	5600	JMP I SETO		
		/ROUTINE TO TYPE LISTING		
		/ENTER WITH JMS +1 EQUAL TO START OF LIST		
		/		
5604	0020	PRINT, 0000		
5605	7300	CLA CLL	/CLEAR THE AC AND LINK	
5606	1624	TAD I PRINT		
5607	2204	ISZ PRINT	/SET FOR RETURN +1	
5610	3041	DCA REG8	/SAVE THE POINTER	
5611	1441	TAD I REG8	/GET THE CHARACTER	
5612	0012	AND K7700	/MASK BITS 0-5	
5613	7450	SNA	/END OF MESSAGE	
5614	5240	JMP EXIT	/YES, EXIT	
5615	7520	SMA	/IS AC MINUS	
5616	7020	CML	/NO, SET THE LINK	
5617	7021	IAC		
5620	7012	RTR		
5621	7012	RTR		
5622	7012	RTR		
5623	4507	JMS I XTYPE	/PRINT THE CHARACTER	
5624	1441	TAD I REG8	/GET THE WORD	
5625	0133	AND K0077	/MASK BITS 6-11	
5626	7450	SNA	/END OF MESSAGE	
5627	5240	JMP EXIT	/YES, EXIT	
5630	1125	TAD K3740	/NO, ADD A CONSTANT	
5631	7520	SMA		
5632	1124	TAD K4100		
5633	1126	TAD K0240		
5634	4527	JMS I XTYPE	/TYPE THE CHARACTER	
5635	2041	ISZ REG8	/UPDATE WORD LIST	

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5636	7320		CLA CLL		/CLEAR THE AC AND LINK
5637	5211		JMP PRINT+5		
5640	7320		CLA CLL		/CLEAR THE AC AND LINK
5641	5624		JMP I PRINT		/YES EXIT
			EXIT,		
			/ROUTINE TO WAIT FOR OVERFLOWS		
			XWAIT,		
5642	0000		DCA SAVAC		/SAVE THE AC
5643	3011		CLA CLL CMA RAL		
5644	7344		TAD XWAIT		
5645	1242		DCA XWAIT		/SET FOR RETURN ADDRESS
5646	3242		ISZ REGB		
5647	2041		JMP RETURN		
5650	5256		ISZ REGF		
5651	2045		JMP RETURN		
5652	5256		CLA CLL CML IAC RAL		
5653	7325		TAD XWAIT		
5654	1242		DCA XWAIT		/UPDATE FOR ERROR RETURN
5655	3242		RETURN, TAD SAVAC		
5656	1011		JMP I XWAIT		
5657	5642				
5660	0000		SWLAS,		
5661	7624		LAS		
5662	0142		AND K0010		/CHECK FOR EXTERNAL CLOCK SCOPE LOOP
5663	7640		SZA CLA		/ENTER SCOPE LOOP
5664	5325		JMP CLKIN		
5665	7624		LAS		
5666	0140		AND K0020		/CHECK FOR EXTERNAL PULSE SCOPE LOOP
5667	7640		SZA CLA		
5670	5313		JMP EXTER		/ENTER SCOPE LOOP
5671	7340		CLA CLL CMA		/AC TO 7777
5672	3113		DCA K11CPS		
5673	7624		LAS		
5674	0114		AND K6007		
5675	7640		SZA CLA		
5676	5321		JMP +3		
5677	1111		TAD KPRMTI		
5700	3113		DCA K11CPS		
5701	7624		LAS		/GET HIS SWITCHES
5702	7024		RAL		/GET BIT 1
5703	7710		SPA CLA		
5704	5660		JMP I SWLAS		/TEST SCHMITT
5705	2260		ISZ SWLAS		
5706	7624		LAS		/GET HIS SWITCHES
5707	7710		SPA CLA		
5710	5660		JMP I SWLAS		/TEST DK8-EP
5711	2260		ISZ SWLAS		
5712	5660		JMP I SWLAS		/TEST DK8-EA OR DK8-EC
5713	7340		EXTER,		
5714	4427		CLA CLL CMA		
5715	7320		JMS I X10TG		/IOT 6133, CLAB
5716	1137		CLA CLL		
			TAD K0040		

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5717	1147		TAD K0600	/GET ENABLES
5720	4425		JMS I XIOTF	/IOT 6132, CLOE
5721	4424		JMS I XIOTE	/IOT 6131, CLSK
5722	5321		JMP .-1	/WAIT FOR OVERFLOW
5723	6007		6007	/CAF OR CLEAR THE WORLD
5724	5313		JMP EXTER	/CONTINUE WITH SCOPE LOOP
5725	7340	/	CLA CLL CMA	/AC TO 7777
5726	4427	CLKIN,	JMS I XIOTG	/IOT 6133, CLAB
5727	7300		CLA CLL	
5730	1013		TAD K0100	/GET ENABLES
5731	4426		JMS I XIOTF1	/IOT 6132, CLOE
5732	4424		JMS I XIOTE	/IOT 6131, CLSK
5733	5332		JMP .-1	/WAIT FOR OPERATOR
5734	6007		6007	/CAF OR CLEAR THE WORLD
5735	1006		TAD K0207	
5736	4527		JMS I XTYPE	/TTY SIGNAL
5737	5325		JMP CLKIN	/LOOP
5740	0000	/	PASS,	
5741	4501		JMS I XCRLF	/CRLF
5742	4506		JMS I XPRINT	/PRINT MESSAGE
5743	6014		PHES	
5744	6007		6007	
5745	5740		JMP I PASS	
5746	0000	/	GTAD,	
5747	1075		TAD CLOCKS	/GET SELECTED CLOCK
5750	1354		TAD CLTAD	
5751	3746		DCA I GTAD	
5752	2346		ISZ GTAD	
5753	5746		JMP I GTAD	
5754	5755	/	CLTAD,	
5755	6000		CLTAD +1	
5756	1612		6000	
5757	4776		1612	
5760	5367		4776	
5761	7306		5367	
5762	7747		7306	
5763	4000		7747	
5764	1527		4000	
5765	4552		1527	
5766	5217		4552	
5767	7276		5217	
5770	7741		7276	
			7741	
5771	0000	/	TIMCLK,	
5772	7624		0000	
5773	0114		LAS	
5774	7650		AND K6007	
5775	1166		SNA CLA	
5776	1012		TAD PATCH	
5777	5771		TAD K7700	
			JMP I TIMCLK	

DKMES, TEXT ?DK8E CLOCKS DIAGNOSTIC?

PMES, TEXT ?DK8E PASS COMPLETE?

FMES, TEXT ? CPS CLOCK SELECTED BY OPERATOR?

TMES, TEXT ?TEST ?

AMES, TEXT ? FAILED, STARTING ADDRESS ?

PAL10	V141	6000	0413
		6001	7005
		6002	4003
		6003	1417
		6004	0313
		6005	2340
		6006	0411
		6007	0107
		6010	1617
		6011	2324
		6012	1103
		6013	0000
		6014	0413
		6015	7005
		6016	4020
		6017	0103
		6020	2340
		6021	0317
		6022	1520
		6023	1405
		6024	2405
		6025	0020
		6026	4003
		6027	2023
		6030	4003
		6031	1417
		6032	0313
		6033	4023
		6034	0514
		6035	0503
		6036	2405
		6037	0440
		6040	0201
		6041	4017
		6042	2005
		6043	2201
		6044	2417
		6045	2200
		6046	2405
		6047	2524
		6050	4000
		6051	4006
		6052	0111
		6053	1405
		6054	0454
		6055	4023
		6056	2401
		6057	2224
		6060	1116
		6061	0740
		6062	0104
		6063	0422
		6064	0523
		6065	2340
		6066	0000

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6067 2410
 6070 0540
 6071 0717
 6072 1704
 6073 4021
 6074 0340
 6075 7540
 6076 0000
 6077 4021
 6100 1624
 6101 4022
 6102 0124
 6103 4021
 6104 0340
 6105 7540
 6106 0000
 6107 0314
 6110 1703
 6111 1340
 6112 2313
 6113 1120
 6114 4026
 6115 0111
 6116 1425
 6117 0424
 6120 4016
 6121 1740
 6122 2313
 6123 1120
 6124 4025
 6125 3020
 6126 0523
 6127 2425
 6130 0420
 6131 0314
 6132 1723
 6133 1340
 6134 2313
 6135 1120
 6136 4026
 6137 0111
 6140 1425
 6141 0424
 6142 4023
 6143 1311
 6144 2040
 6145 0530
 6146 2025
 6147 0324
 6150 0524
 6151 0020
 6152 2022
 6153 1727
 6154 2221
 6155 1540

GMES, TEXT ?THE GOOD AC = ?

BMES, TEXT ? AND BAD AC = ?

MES1, TEXT ?CLOCK SKIP FAILED, NO SKIP EXPECTED?

MES2, TEXT ?CLOCK SKIP FAILED, SKIP EXPECTED?

MES3, TEXT ?PROGRAM INTERRUPT FAILED, NO INTERRUPT EXPECTED?

6136 1116
6137 2405
6160 2222
6161 2520
6162 2440
6163 0621
6164 1114
6165 0504
6166 5440
6167 1617
6170 4011
6171 1624
6172 0522
6173 2225
6174 2024
6175 4025
6176 3020
6177 0523
6200 2405
6201 0400
6202 2022
6203 1707
6204 2201
6205 1540
6206 1116
6207 2405
6210 2222
6211 2520
6212 2440
6213 0621
6214 1114
6215 0504
6216 5440
6217 1116
6220 2405
6221 2222
6222 2520
6223 2440
6224 0530
6225 2025
6226 0324
6227 0504
6230 0000
6231 0314
6232 1703
6233 1340
6234 1725
6235 2420
6236 2524
6237 4026
6240 0111
6241 1425
6242 0454
6243 4023
6244 1417

MES4, TEXT ?PROGRAM INTERRUPT FAILED, INTERRUPT EXPECTED?

MES5, TEXT ?CLOCK OUTPUT FAILED, CLOCK FREQUENCY FAST?

PAL10 V141 6245 0313
 6246 4006
 6247 2205
 6250 2125
 6251 0516
 6252 0331
 6253 4006
 6254 0123
 6255 2420
 6256 0314
 6257 1703
 6260 1340
 6261 1725
 6262 2420
 6263 2524
 6264 4006
 6265 0111
 6266 1405
 6267 0454
 6270 4003
 6271 1417
 6272 0313
 6273 4006
 6274 2205
 6275 2125
 6276 0516
 6277 0331
 6300 4023
 6301 1417
 6302 2700
 6303 2410
 6304 0540
 6305 0103
 6306 4027
 6307 0123
 6310 4003
 6311 1001
 6312 1607
 6313 0504
 6314 4002
 6315 3140
 6316 0140
 6317 0314
 6320 1703
 6321 1340
 6322 1117
 6323 2400
 6324 0314
 6325 1703
 6326 1340
 6327 0225
 6330 0606
 6331 0522
 6332 4022
 6333 0507

MES6, TEXT ?CLOCK OUTPUT FAILED, CLOCK FREQUENCY SLOW?

MES7, TEXT ?THE AC WAS CHANGED BY A CLOCK IOT?

MES8, TEXT ?CLOCK BUFFER REGISTER AND AC TRANSFER FAILED?

PAL10 V141 6334 1123
6335 2405
6336 2240
6337 0116
6340 0440
6341 0103
6342 4024
6343 2201
6344 1623
6345 0605
6346 2240
6347 0601
6350 1114
6351 0504
6352 0000
6353 0314
6354 1703
6355 1340
6356 0317
6357 2516
6360 2405
6361 2240
6362 2205
6363 0711
6364 2324
6365 0522
6366 4001
6367 1604
6370 4001
6371 0340
6372 2422
6373 0116
6374 2306
6375 0522
6376 4006
6377 0111
6400 1405
6401 0400
6402 0314
6403 1703
6404 1340
6405 0516
6406 0102
6407 1405
6410 4022
6411 0507
6412 1123
6413 2405
6414 2240
6415 0116
6416 0440
6417 0103
6420 4024
6421 2201
6422 1623

MES9, TEXT ?CLOCK COUNTER REGISTER AND AC TRANSFER FAILED?

MES10, TEXT ?CLOCK ENABLE REGISTER AND AC TRANSFER FAILED?

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6423 0605
6424 2240
6425 0601
6426 1114
6427 0504
6430 0000
6431 0314
6432 1703
6433 1340
6434 2324
6435 0124
6436 2523
6437 4022
6440 0507
6441 1123
6442 2405
6443 2240
6444 0116
6445 0440
6446 0103
6447 4024
6450 2201
6451 1623
6452 0605
6453 2240
6454 0601
6455 1114
6456 0504
6457 0000

MES11, TEXT ?CLOCK STATUS REGISTER AND AC TRANSFER FAILED?

[illegible]

AMES	6051	K0017	0127	LOOP	0077	SETO	5600
AUTO10	0010	K0020	0140	MES1	6107	SKPWAT	0046
BEGIN	0200	K0040	0137	MES10	6402	SNDRV	5216
BELL	5046	K0077	0133	MES11	6431	SORT	5471
BGNEAC	0215	K0100	0013	MES2	6131	SWLAS	5660
BMES	6077	K0200	0015	MES3	6152	SYNC	5302
CLKIN	5725	K0207	0006	MES4	6202	T113A	1655
CLKNO	5463	K0212	0135	MES5	6231	T113H	1646
CLOCK	5413	K0215	0134	MES6	6256	T114A	1673
CLOCKS	0075	K0240	0126	MES7	6303	T114H	1664
CLPREG	5065	K0260	0123	MES8	6324	T11A	0354
CLTAD	5754	K0300	0145	MES9	6353	T120A	1751
CLLF	5441	K0377	0136	MESS	5542	T121A	1766
DMES	6000	K0400	0116	NERRO	5000	T122A	2014
EHLT1	5034	K0500	0146	NERRO	0072	T122H	1775
EHLT10	5203	K0600	0147	OCTEL	5420	T123A	2043
EHLT11	5212	K0700	0150	OUT	5015	T123H	2024
EHLT2	5126	K1000	0144	OVER2	0061	T124A	2072
EHLT3	5141	K2000	0143	OVER2A	0062	T124P	2053
EHLT4	5145	K2525	0016	PASS	5740	T125A	2120
EHLT5	5153	K3000	0120	PATCH	0166	T125P	2102
EHLT6	5157	K3740	0125	PIG01	5255	T126A	2147
EHLT7	5166	K4000	0014	PIG02	5270	T126P	2125
ERR0	5020	K4100	0124	PIG03	5323	T127A	2200
ERROR	0073	K5000	0121	PIG04	5336	T12A	0366
EXIT	5640	K5252	0017	PIG05	5234	T130A	2220
EXTER	5713	K6000	0117	PIRET1	5264	T133A	2306
FMS	6026	K6007	0114	PIRET2	5277	T133E	2273
GMES	6067	K7000	0141	PIRET5	5252	T147A	2555
GTAD	5746	K7400	0112	PMES	6014	T147B	2537
IN	5043	K7700	0012	POPR	5450	T150A	2603
IOTA	5102	K7770	0122	PREG	5563	T150P	2565
IOTB	5127	K7772	0132	PRET1	5267	T151A	2631
IOTC	5114	K7773	0131	PRET2	5301	T151P	2613
IOTD	5141	K7774	0130	PRET5	5254	T152A	2657
IOTE	5127	KPRMT1	0111	PRETC	5335	T152P	2641
IOTF	5134	KREGC	0076	PRETD	5347	T153A	2705
IOTF1	5142	KT1CPS	0113	PRINT	5604	T153E	2667
IOTG	5146	KTA	0151	RANDOM	5224	T154A	2733
IOTH	5154	KTA1	0152	RANDY	0055	T154B	2715
IOTI	5163	KTADCK	5462	RECEV	0071	T172A	3254
IOTJ	5200	KTADM	5526	REGA	0040	T172A1	3257
IOTK	5207	KT8	0153	REGB	0041	T172B	3244
IOTS	5350	KT81	0154	REGC	0042	T172P1	3240
IOTS1	5360	KTC	0155	REGD	0043	T173A	3303
IOTS2	5370	KTC1	0156	REGE	0044	T173A1	3306
IOTS3	5400	KTC2	0157	REGF	0045	T173B	3273
ISELOP	5310	KTD	0160	RETC	5332	T173B1	3267
JMPI2	0100	KTD1	0101	RETD	5345	T174A	3336
K0006	0115	KTE	0162	RETURN	5656	T174A1	3341
K0007	0007	KTE1	0163	SAVAC	0011	T174B	3326
K0010	0142	KTMX	5527	SEND	0070	T174B1	3323

PAL10	V141	9-0CT-71	15144	PAGE 1-77
T175A	3371	TST110	1605	TST167
T175A1	3374	TST111	1616	TST17
T175B	3361	TST112	1631	TST170
T175B1	3356	TST113	1642	TST171
T176A	3421	TST114	1660	TST172
T176A1	3424	TST115	1676	TST173
T176B	3411	TST116	1707	TST174
T176B1	3405	TST117	1722	TST175
T177A	3451	TST118	1735	TST176
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T200A1	3504	TST123	1821	TST181
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T200B1	3465	TST125	1855	TST183
T201A	3551	TST126	1872	TST184
T201B	3522	TST127	1889	TST185
T205A	3632	TST128	1906	TST186
T206A	3650	TST129	1923	TST187
T207A	3666	TST130	1940	TST188
T214A	3764	TST131	1957	TST189
T215A	4003	TST132	1974	TST190
T216A	4022	TST133	1991	TST191
T22A	4073	TST134	2008	TST192
T230A	4265	TST135	2025	TST193
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T45B	0763	TST137	2059	TST195
T46A	1014	TST138	2076	TST196
T46B	1025	TST139	2093	TST197
T47A	1033	TST140	2110	TST198
T47B	1024	TST141	2127	TST199
T50A	1052	TST142	2144	TST200
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T70A	1332	TST144	2178	TST202
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PAL10	V141
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TST77	1451
TYPE	5056
XBELL	0110
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XCLOCK	0060
XCRLF	0101
XCRS1	0171
XCRS2	0172
XCRS3	0173
XCRS4	0174
XCRS5	0175
XDK8EP	0063
XGETM	0167
XGTAD	0067
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XIOTC	0022
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XIOTE	0024
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XIOTI	0031
XIOTJ	0032
XIOTK	0033
XIOTS	0034
XIOTS1	0035
XIOTS2	0036
XIOTS3	0037
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XLAS	0066
XMESS	0105
XMITT	0064
XMITT1	0065
XOCIEL	0104
XOPR	0165
XPASS	0170
XPI001	0047
XPI002	0050
XPI003	0051
XPI004	0052
XPI005	0053
XPRINT	0106
XREG	0102
XSETO	0164
XSNDRV	0056
XSORT	0103
XSYNC	0057
XTYPE	0107
XWAIT	5642

ERRORS DETECTED: 0
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RUN-TIME: 39 SECONDS
3K CORE USED